EXAM PACK

EEE455

VLSI Design and Testing



Exam Pack

Session 2018–19 *Solutions and formula sheet of the 4th Year 1st Semester Final Examination-2022 and Term Tests 1 and 2*

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Question Paper Solutions 2018–19 — pages 3--84 — 82 pages

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Prepared by Nur Mohammad

November 2026

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Question Paper Solutions 2018–19

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PAST PAPER SOLUTIONS

EEE455

VLSI Design and Testing

Question Paper Solutions

Session 2018–19

Final Examination-2022 (Q1–Q6) and Term Tests 1 and 2 (Q7, Q8)

QUESTION MAP

Q1	Q2	Q3	Q4	Q5	Q6
25 marks	25 marks	25 marks	25 marks	25 marks	25 marks

Q7	Q8
20 marks	25 marks

EACH ANSWER



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November 2026

BACK OF THE COVER

The paper at a glance

3 h	100	4 of 6	45 min
Total time	Total marks	Questions to answer	Per question

Time = marks × 1.8 min (180 min for 100 marks); a part with its own time (a term test) uses that. Tick the questions you will answer; each sub-part links to its solution.

Q	Sub-part	Marks	Time	Pick
Part - A · (Answer any two of the following questions)				
Q1	(a) Effect of Adding O2 and H2 on Etching	5	9 min	☐
	(b) Silicon Oxidation Model and Its Limiting Cases	5	9 min	
	(c) Sustaining a Plasma and Plasma Etching Chemistry	7	13 min	
	(d) VHDL Code for the Boolean Expression	8	14 min	
Q2	(a) Threshold Voltage without Body Effect	6	11 min	☐
	(b) Moore's Law and MOSFET Scaling	6	11 min	
	(c) Saturated-Load NMOS Inverter	5	9 min	
	(d) DC Characteristics of the CMOS Inverter	8	14 min	
Q3	(a) VHDL and Its Advantages	2	4 min	☐
	(b) Elmore Delays of a 3-Input NOR Gate	6	11 min	
	(c) Logical Effort of the Path in Figure 3c	10	18 min	
	(d) Number of Stages for Minimum Delay	7	13 min	
Part - B · (Answer any two of the following questions)				
Q4	(a) Why NMOS Is Not Used as a Pull-Up Network	5	9 min	☐
	(b) Sources of Power Dissipation	5	9 min	
	(c) Activity Factors of Figure 4.1	7	13 min	
	(d) Switching and Static Power of a System-on-Chip			
Q5	(a) Stick Diagram of a 3-Input NAND	8	14 min	☐
	(b) Advantages of CMOS	5	9 min	
	(c) Max-Delay and Min-Delay Constraints	6	11 min	
	(d) Flip-Flop with Synchronous Reset	6	11 min	
Q6	(a) Principle of the Dynamic Inverter	5	9 min	☐
	(b) Full Adder Design	6	11 min	
	(c) PLA and PAL	7	13 min	
	(d) DRAM Write and Read	7	13 min	
Term Test 1 · 20 marks · 40 min				
Q7	(a) Steps of Photolithography	8	16 min	☐
	(b) Silicon Oxidation Model in the Term Test	6	12 min	
	(c) Vapor Phase Epitaxy	6	12 min	
Term Test 2 · 20 marks · 40 min				
Q8	(a) Threshold Voltage with Body Effect	5	10 min	☐
	(b) DIBL and Gate Leakage	5	10 min	
	(c) CMOS Gate for the Carry Function	5	10 min	
	(d) Minimum Delay and Gate Sizes of the Term-Test Circuit			
	(e) Stage Effort for the Ideal Number of Stages	5	10 min	

Formulas follow Weste and Harris, CMOS VLSI Design (delay, logical effort, power, leakage, scaling, timing, circuit families, adders, PLA, DRAM, VHDL), Kang and Leblebici, CMOS Digital Integrated Circuits (threshold voltage, saturated-load inverter, inverter DC equations), and Plummer, Deal and Griffin, Silicon VLSI Technology (oxidation, epitaxy, lithography, plasma etching).

Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

4th Year 1st Semester Final Examination-2022

Course Code: EEE 455 Session: 2018-19

Course Title: VLSI Design and Testing

Total Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part - A

(Answer any two of the following questions)

1. (a) What are the effects of O_2 and H_2 adding on etching? 5
   (b) Discuss the silicon oxidation model mentioning its limiting cases. 5
   (c) Explain how does the plasma sustain. Also explain the chemistries involved in plasma etching. 7
   (d) Write the VHDL code to implement the following Boolean expression: $F = AB + \overline{A}BC + BC$. 8

2. (a) Find threshold voltage of MOSFET without body effect. 6
   (b) State Moore's law. Explain how different parameters of MOSFET is scaled. 6
   (c) Explain operation of saturated load NMOS inverter. What are the disadvantages of this configuration? 5
   (d) Find DC characteristics of CMOS inverter using V-I characteristics curve. 8

3. (a) What is VHDL? Write down the advantages of VHDL. 2
   (b) Sketch a 3-input NOR gate with transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model. Assume that every source or drain has fully contacted diffusion when making your estimate of capacitance. 6
   (c) For the circuit in figure 3c find-
      i) Logical effort, G
      ii) Branching effort, B
      iii) Electrical effort, E
      iv) Path effort, F
      v) Stage effort, f
      vi) Parasitic delay, P
      vii) Minimum path delay, D
      viii) Size of the gate. 10
   (d) Find the number of stages for minimum delay. 7

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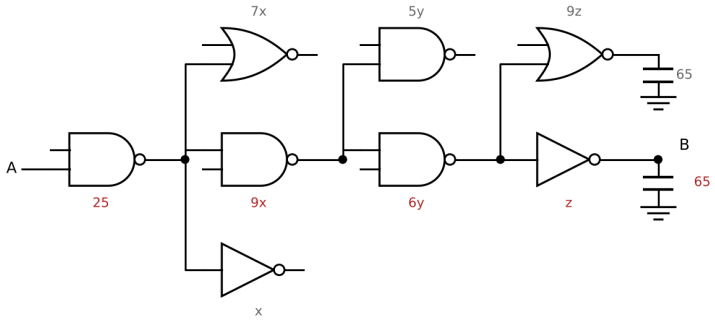


Figure: 3c of the paper (redrawn); the dashed path from A to B on the paper runs through the gates marked 25, 9x, 6y and z

Part - B

(Answer any two of the following questions)

4. (a) Explain why NMOS network is not used as pull-up network. 5

   (b) What are the sources of power dissipation in CMOS logic circuit? 5

   (c) What is activity factor? Figure 4.1 shows a 4-input AND gate built using a tree and a chain of gates. Determine the activity factors at each node in the circuit assuming the input probabilities PA = PB = PC = PD = 0.5. 7

   (d) A digital system-on-chip in a 1 V 65 nm process (with 50 nm drawn channel lengths and $\lambda = 25$ nm) has 1 billion transistors, of which 50 million are in logic gates and the remainder in memory arrays. The average logic transistor width is 12 λ and the average memory transistor width is 4 λ. The memory arrays are divided into banks and only the necessary bank is activated so the memory activity factor is 0.02. The static CMOS logic gates have an average activity factor of 0.1. Assume each transistor contributes 1 fF/Rm [*sic*: should be fF/µm] of gate capacitance and 0.8 fF/µm of diffusion capacitance. Neglect wire capacitance for now (though it could account for a large fraction of total power). **Estimate the switching power when operating at 1 GHz.** If Subthreshold leakage for OFF devices is 100 nA/µm for low-threshold devices and 10 nA/µm for high-threshold devices. Gate leakage is 5 nA/µm. Junction leakage is negligible. Memories use low-leakage devices everywhere. Logic uses low-leakage devices in all but 5% of the paths that are most critical for performance. **Estimate the static power consumption.** 8

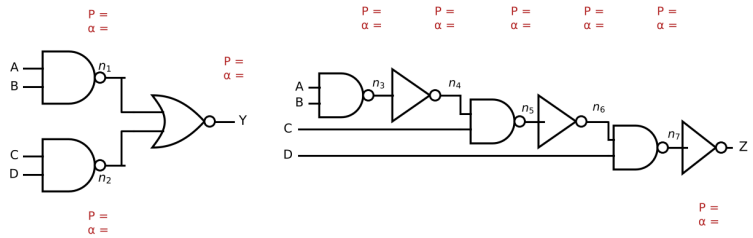


Figure: 4.1 of the paper (redrawn): the 4-input AND as a tree and as a chain

5. (a) Draw the stick diagram of the following Boolean function: $f = \overline{A.B.C}$ with their symbolic diagram and truth table. 8

   (b) Write down the advantages of CMOS. 5

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(c) Explain max-delay constraint and min-delay constraint in sequential circuits. 6

(d) Sketch logic circuit of flip-flop with synchronous reset and explain its operation. 6

6. (a) Explain the principle of dynamic inverter. 5

(b) Design and implement full adder logic circuit. 6

(c) Define PLA and PAL. Sketch a 3-input and 2-output PLA implementing the following logic expressions- $s = a\overline{b}\,\overline{c} + \overline{a}b\overline{c} + \overline{a}\,\overline{b}c + abc$, $c = ab + bc + ac$ 7

(d) Explain how data is written and read in a DRAM sub-array. 7

Term Test 1 (its questions 1 to 3 are 7(a) to 7(c))

[4th Year 1st Semester Term Test-01, Set - A, Full Marks: 20, Time: 40 minutes]

7. (a) Briefly describe steps of photo-lithography. 8

(b) Explain the silicon oxidation model mentioning its limiting cases. 6

(c) Briefly explain vapor phase epitaxy. 6

Term Test 2 (its questions 1 to 5 are 8(a) to 8(e))

[4th Year 1st Semester Term Test-02, Set - B, Answer any four out of five questions, Full Marks: 20, Time: 40 minutes]

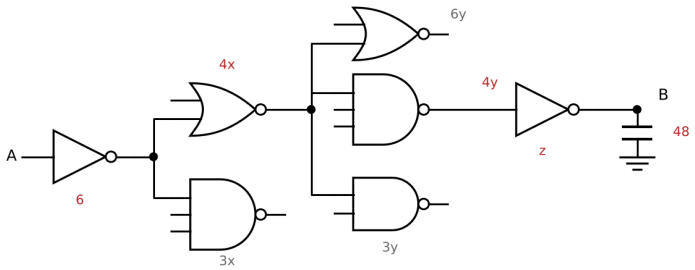
8. (a) Find the threshold voltage of MOSFET considering the body effect. 5

(b) Briefly explain the following terms: i) Drain induced barrier lowering, ii) Gate leakage. 5

(c) Implement the following expression using CMOS logic: $Y = AB + C(A + B)$. 5

(d) Find minimum number delay [*sic*: should be "minimum delay"] for the logic circuit below. Also find the gate size to achieve this delay. 5

(e) Find the stage effort for the ideal number of stages. 5



The figure of Term Test 2, question 4 (redrawn): inverter 6, then NOR2 4x and NAND3 3x, then NOR2 6y, NAND3 4y and NAND2 3y, then inverter z and the load 48; A and B are faint marks on the paper

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1. (a) What are the effects of O_2 and H_2 adding on etching? Marks: 5
   (b) Discuss the silicon oxidation model mentioning its limiting cases. Marks: 5
   (c) Explain how does the plasma sustain. Also explain the chemistries involved in plasma etching. Marks: 7
   (d) Write the VHDL code to implement the following Boolean expression: $F = AB + \overline{A}BC + BC$. Marks: 8

1.(a) What are the effects of O_2 and H_2 adding on etching? Marks: 5

1. Theory

1. What is etching? (Core Idea)

A chip is built layer by layer. After a layer (oxide, nitride, polysilicon, metal) is put on the whole wafer, photolithography covers the parts to keep with a patterned **photoresist**. **Etching** then removes the layer wherever it is not covered. The resist is the stencil; etching is the cutting.

Two numbers describe an etch:

- **Etch rate:** how fast a layer is removed (nm/min).
- **Selectivity:** the etch rate of the layer we want to remove divided by the etch rate of the layer below it. A selectivity of 20 means that while 200 nm of oxide is removed, only 10 nm of the silicon below is lost.

2. Plasma etching with CF_4

The wafer sits in a low-pressure chamber filled with CF_4 gas. A radio-frequency voltage turns the gas into a **plasma**: free electrons hit the molecules and break them into reactive pieces, called **radicals**, such as F and CF_3:

$$CF_4 + e^- \rightarrow CF_3 + F + e^-$$

A fluorine atom reaches the surface and reacts. The product SiF_4 is a gas, so it leaves the surface and is pumped away:

$$Si + 4F \rightarrow SiF_4 \uparrow \qquad SiO_2 + 4F \rightarrow SiF_4 \uparrow + O_2$$

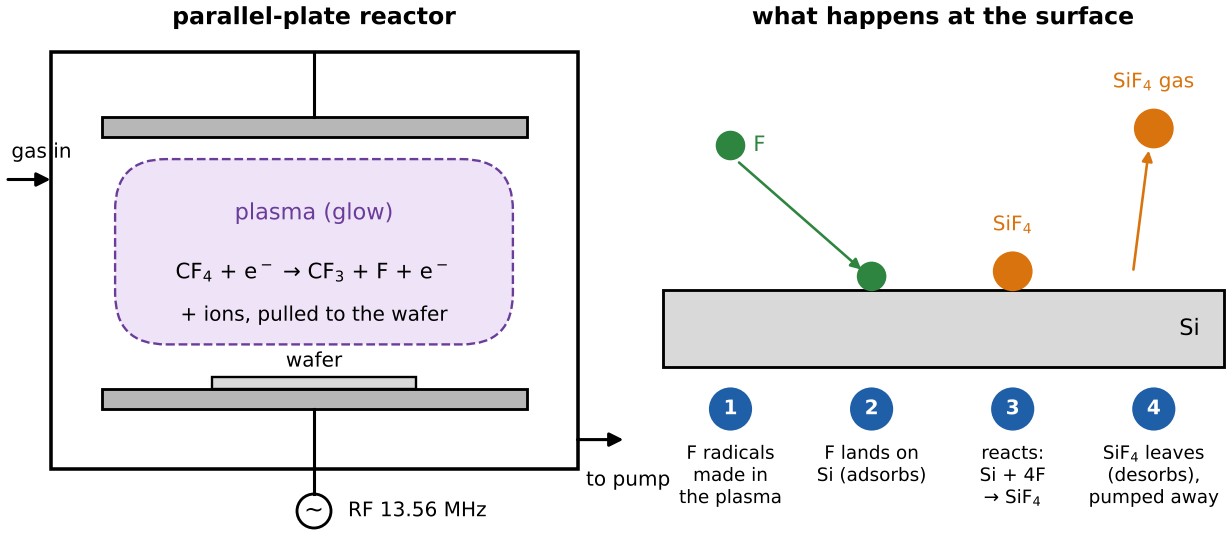


Figure 1: A parallel-plate plasma etcher and the four steps of the chemical etch at the silicon surface (reused from the 2019-20 paper's solution)

Reading the figure. On the left, gas enters, the RF supply keeps a plasma glowing above the wafer, and the products are pumped away. On the right, an F radical is made, lands on the silicon, four F

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atoms join one Si atom, and the SiF_4 leaves as a gas.

3. Why does the etch rate depend on what else is in the gas?

The etch rate of silicon depends mainly on **how many free F atoms** are in the plasma. In pure CF_4, many F atoms join a CF_3 fragment again ($F + CF_3 \rightarrow CF_4$), so few are left. Anything that removes the CF_x fragments raises the F count; anything that removes F lowers it. O_2 does the first, H_2 does the second.

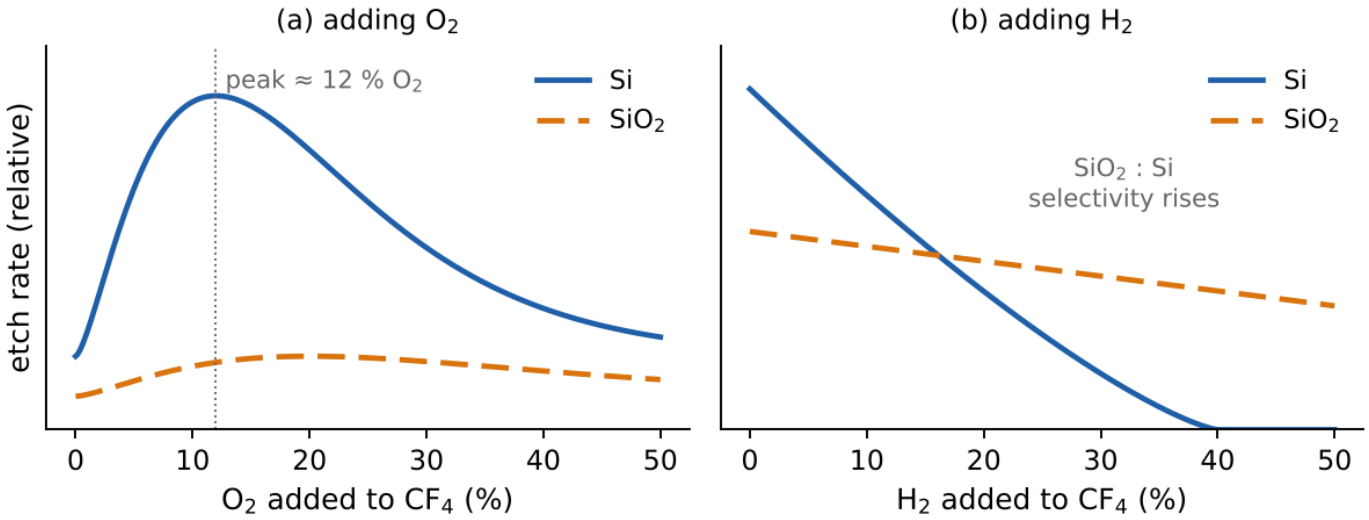


Figure 2: Etch rates of Si and SiO2 in a CF4 plasma: (a) with O2 added, (b) with H2 added (shapes after the classic measurements reported in Plummer; the numbers are relative; reused from the 2019-20 paper's solution)

Reading the figure. In (a), the silicon etch rate (solid blue curve) rises steeply as O_2 is added, peaks near 12 % O_2 and then falls; the oxide rate (dashed orange curve) changes much less, with a small peak near 20 %. In (b), the silicon rate falls quickly as H_2 is added and is almost zero at about 40 % H_2, while the oxide rate stays nearly the same.

2. Answer

Plasma etching in CF_4 works through free fluorine: the plasma makes F by $CF_4 + e^- \rightarrow CF_3 + F + e^-$, and F etches silicon as $Si + 4F \rightarrow SiF_4 \uparrow$. Added gases change the amount of free F.

Effect of adding O_2:

- O atoms react with the CF_x fragments, for example $O + CF_3 \rightarrow COF_2 + F$. Fewer F atoms recombine into CF_4, so more free F is left and the Si etch rate **rises**, to a maximum near 12 % O_2. The SiO_2 rate peaks later, near 20 %.
- Beyond the peak, extra O_2 only **dilutes** the F, and O atoms adsorb on the silicon and oxidise it, which blocks F. The rate **falls**.

Effect of adding H_2:

- H atoms capture fluorine: $H + F \rightarrow HF$. Free F drops, so the **Si etch rate falls** quickly, to almost zero near 40 % H_2.
- The plasma becomes rich in carbon, which forms a thin fluorocarbon polymer on the silicon and protects it. On SiO_2, the oxygen released from the oxide removes the carbon as CO and CO_2, so the **SiO_2 rate stays almost constant**.
- Result: the **SiO_2:Si selectivity rises** strongly. This is used to etch contact holes in oxide that must stop on silicon.

3. Keyword Explanation

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1. Radical

A neutral fragment of a molecule with an unpaired electron, such as F or CF_3. It is chemically very active, so it does most of the chemical etching.

2. Selectivity

The ratio of the etch rates of two materials, for example SiO_2:Si = 20:1. High selectivity lets an etch stop on the layer below.

4. Notes and Remarks

- The curves follow the shapes of the classic measurements by Mogab et al. (O_2 in CF_4) and Ephrath (H_2 in CF_4) as shown in Plummer, Deal and Griffin; the exact percentages depend on the reactor, so quote them as "about".
- Also solved in 1(a) of the 2019-20 paper, as extra reading.

1.(b) Discuss the silicon oxidation model mentioning its limiting cases. Marks: 5

1. Theory

1. What is thermal oxidation? (Core Idea)

Silicon heated to 800–1200 °C in oxygen or steam grows a layer of silicon dioxide, the gate insulator and isolation of the MOS transistor:

$$Si + O_2 \rightarrow SiO_2 \qquad Si + 2H_2O \rightarrow SiO_2 + 2H_2$$

The first is **dry** oxidation (slow, best quality, used for gate oxides); the second is **wet** oxidation (fast, used for thick oxides).

The important fact is *where* the oxide grows. The reaction does not happen on top of the oxide. The oxygen must travel through the oxide already grown and react at the **Si/SiO_2 interface**, which moves down into the wafer. So the thicker the oxide, the longer the oxygen's journey, and the slower the growth. The **Deal-Grove model** turns this idea into an equation.

2. The three steps (fluxes)

A **flux** F is the number of oxidant molecules crossing a unit area each second. Three steps happen in series.

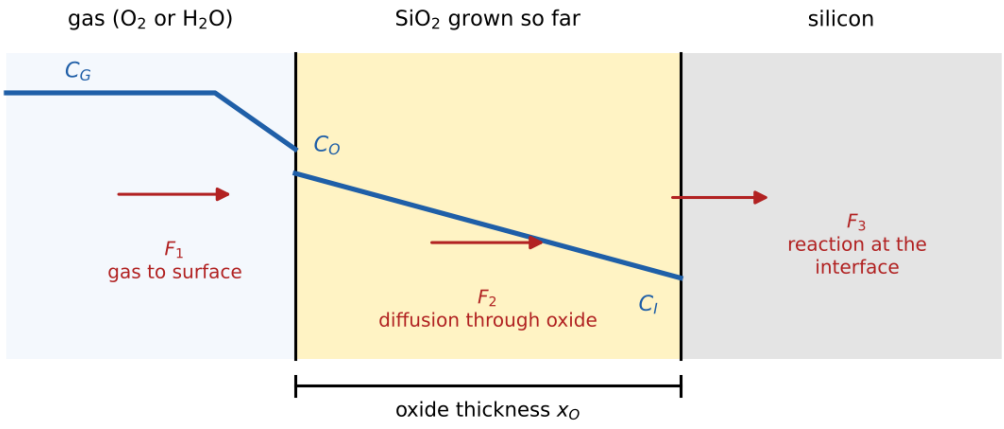


Figure 3: Oxidant concentration from the gas through the oxide to the silicon, and the three fluxes of the Deal-Grove model

Reading the figure. The solid blue line is the concentration of oxidant. It is highest in the gas (C_G), drops at the surface to C_O just inside the oxide, falls steadily across the oxide, and has its lowest value C_I at the silicon. A slope in concentration is what drives the diffusion; the red arrows are the three fluxes.

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- **Step 1, gas to oxide surface:** proportional to how far the surface is below its equilibrium concentration C^*:

$$F_1 = h\,(C^* - C_O)$$

where h is the gas-phase mass-transfer coefficient and C^* is the solubility of the oxidant in the oxide (the concentration it would reach with no reaction below).

- **Step 2, diffusion through the oxide:** Fick's first law with a straight-line profile across thickness x_O:

$$F_2 = D\,\frac{C_O - C_I}{x_O}$$

where D is the diffusivity of the oxidant in SiO_2.

- **Step 3, reaction at the interface:** proportional to the oxidant that arrives:

$$F_3 = k_S\,C_I$$

where k_S is the interface reaction rate constant.

3. Deriving the growth law, step by step

Step 1: equal fluxes. In steady state nothing piles up, so $F_1 = F_2 = F_3 = F$.
Step 2: find C_I. Set $F_2 = F_3$ and solve for C_O:

$$D\,\frac{C_O - C_I}{x_O} = k_S C_I \;\Rightarrow\; C_O = C_I\left(1 + \frac{k_S x_O}{D}\right)$$

Set $F_1 = F_3$ and put this C_O in:

$$h\left[C^* - C_I\left(1 + \frac{k_S x_O}{D}\right)\right] = k_S C_I \;\Rightarrow\; C_I = \frac{C^*}{1 + \dfrac{k_S}{h} + \dfrac{k_S x_O}{D}}$$

Step 3: the flux. Since $F = k_S C_I$,

$$F = \frac{k_S C^*}{1 + \dfrac{k_S}{h} + \dfrac{k_S x_O}{D}}$$

Step 4: flux to growth speed. Each unit volume of SiO_2 holds N_1 oxidant molecules (2.2×10^{22} cm^{-3} for O_2, 4.4×10^{22} cm^{-3} for H_2O). So the oxide thickens at

$$\frac{dx_O}{dt} = \frac{F}{N_1} = \frac{k_S C^*/N_1}{1 + \dfrac{k_S}{h} + \dfrac{k_S x_O}{D}}$$

Step 5: separate and integrate. Move the denominator to the left and dt to the right:

$$\left(1 + \frac{k_S}{h} + \frac{k_S x_O}{D}\right) dx_O = \frac{k_S C^*}{N_1}\,dt$$

Divide by $k_S/(2D)$, which turns the coefficient of $x_O\,dx_O$ into 2:

$$\left[2D\left(\frac{1}{k_S} + \frac{1}{h}\right) + 2x_O\right] dx_O = \frac{2DC^*}{N_1}\,dt$$

Name the two constants:

$$A = 2D\left(\frac{1}{k_S} + \frac{1}{h}\right), \qquad B = \frac{2DC^*}{N_1}$$

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so $(A + 2x_O)\,dx_O = B\,dt$. Integrate from thickness x_i (any oxide already present) at $t = 0$ to x_O at time t:

$$Ax_O + x_O^2 - Ax_i - x_i^2 = Bt$$

Define $\tau = (x_i^2 + Ax_i)/B$, the time the starting oxide would have taken to grow. Then

$$x_O^2 + Ax_O = B(t + \tau)$$

This is the **linear-parabolic growth law**. Solving the quadratic for x_O (take the positive root):

$$x_O = \frac{A}{2}\left[\sqrt{1 + \frac{t+\tau}{A^2/4B}} - 1\right]$$

4. The two limits

- **Thin oxide (short time):** x_O^2 is much smaller than Ax_O. Then $Ax_O \approx B(t + \tau)$, so $x_O \approx (B/A)(t + \tau)$: **linear growth**. Physically the oxide is so thin that diffusion is easy, and the **interface reaction** sets the speed. With $h \gg k_S$, $B/A \approx k_S C^*/N_1$.
- **Thick oxide (long time):** x_O^2 is much larger than Ax_O. Then $x_O^2 \approx B(t+\tau)$, so $x_O \approx \sqrt{B(t + \tau)}$: **parabolic growth**. Physically the oxide is thick, so **diffusion** through it sets the speed, and $B = 2DC^*/N_1$ holds only diffusion terms.

That is why B/A is called the **linear rate constant** and B the **parabolic rate constant**.

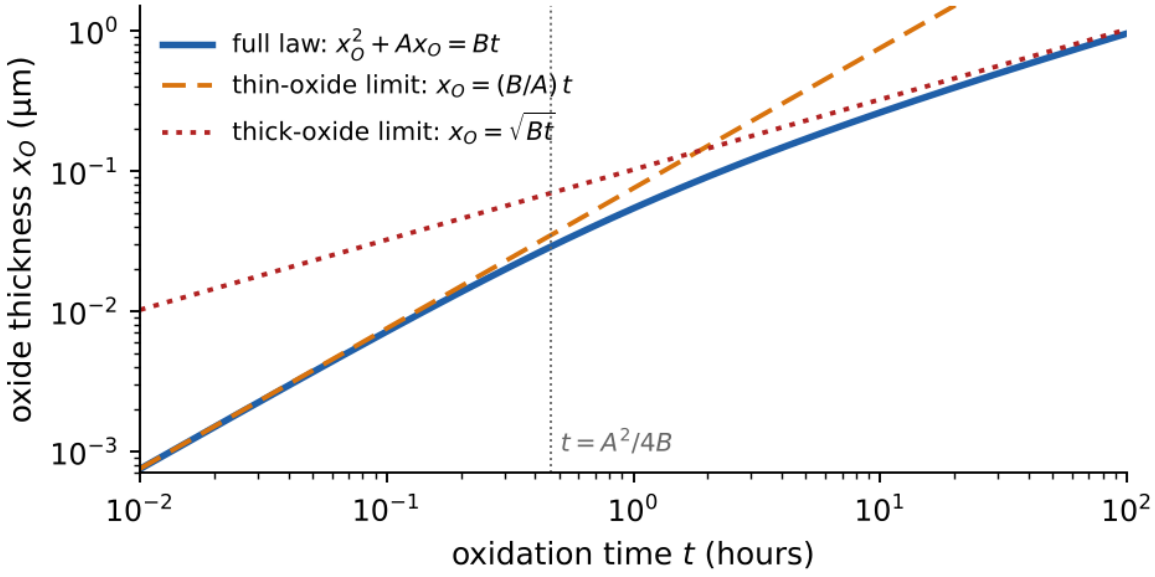


Figure 4: Oxide thickness against time for dry O2 at 1000 °C on (111) silicon on log axes, with the linear and parabolic limits

Reading the figure. The numbers use Plummer's Table 6-2 at 1000 °C: $B = 0.0104$ µm^2/h and $B/A = 0.0752$ µm/h, so $A = 0.139$ µm. On log axes the two limits are straight lines. The full law (solid blue) follows the dashed orange line for short times and the dotted red line for long times. The change-over is near $t = A^2/4B = 0.46$ h. After 1 hour the full law gives 0.054 µm; after 20 hours, 0.39 µm: twenty times the time gives only about seven times the oxide.

2. Answer

In the **Deal-Grove model**, oxidant reaches the silicon through three steps in series: transport from the gas to the oxide surface ($F_1 = h(C^* - C_O)$), diffusion across the oxide ($F_2 = D(C_O - C_I)/x_O$), and reaction at the Si/SiO_2 interface ($F_3 = k_S C_I$).
In steady state $F_1 = F_2 = F_3$. Eliminating C_O and C_I:

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$$F = \frac{k_S C^*}{1 + k_S/h + k_S x_O/D}$$

The growth speed is $dx_O/dt = F/N_1$, where N_1 is the number of oxidant molecules per unit volume of oxide. Integrating from $x_O = x_i$ at $t = 0$:

$$x_O^2 + Ax_O = B(t + \tau)$$

$$A = 2D\left(\frac{1}{k_S} + \frac{1}{h}\right), \quad B = \frac{2DC^*}{N_1}, \quad \tau = \frac{x_i^2 + Ax_i}{B}$$

Limiting cases:

1. **Thin oxide, short time (reaction limited):** $x_O^2 \ll Ax_O$, so $x_O \approx \frac{B}{A}(t + \tau)$, linear in time. $B/A \approx k_S C^*/N_1$ is the linear rate constant.
2. **Thick oxide, long time (diffusion limited):** $x_O^2 \gg Ax_O$, so $x_O \approx \sqrt{B(t + \tau)}$, parabolic in time. $B = 2DC^*/N_1$ is the parabolic rate constant.

3. Keyword Explanation

1. Flux

The number of molecules crossing unit area per unit time.

2. Reaction limited and diffusion limited

The slowest of the series steps sets the overall speed. If it is the interface reaction, the growth is reaction limited; if it is diffusion through the oxide, it is diffusion limited.

4. Notes and Remarks

- h is very large, so the gas step is rarely the limit; Plummer drops k_S/h in most expressions.
- The model fails for dry oxides thinner than about 20 nm, which grow faster than it predicts. τ is sometimes used to fit this range.
- B and B/A follow Arrhenius laws, $C\exp(-E/kT)$. From Plummer's Table 6-2, for dry O_2 on (111) Si: $B = 7.72\times10^2\exp(-1.23\text{ eV}/kT)$ μm^2/h and $B/A = 6.23\times10^6\exp(-2.0\text{ eV}/kT)$ μm/h. For (100) silicon the B/A values are divided by 1.68.

1.(c) Explain how does the plasma sustain. Also explain the chemistries involved in plasma etching. Marks: 7

1. Theory

1. What is a plasma? (Core Idea)

A **plasma** is a gas in which some atoms have lost electrons, so it holds free electrons, positive ions and neutral atoms and radicals. Overall it is neutral. In an etcher the gas is at low pressure (about 1 mtorr to 1 torr) and only a small fraction of it is ionised.

A plasma is not a stable thing. Electrons and ions are lost all the time to the walls, and they recombine. So a plasma lasts only if **new electron-ion pairs are made as fast as they are lost**. The question "how does the plasma sustain?" asks what makes them.

2. The sustaining loop

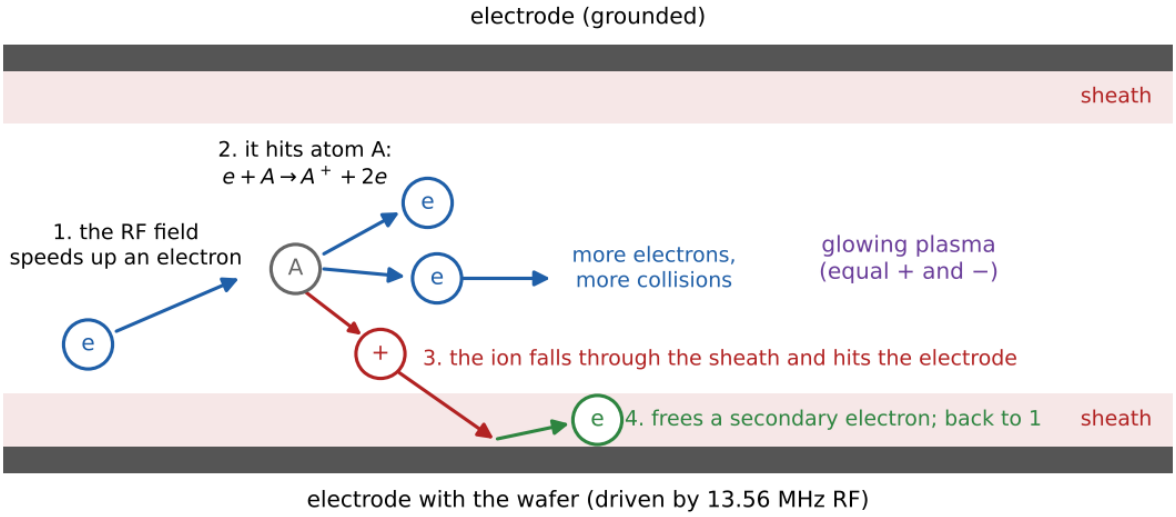


Figure 5: The loop that sustains a plasma: the RF field accelerates electrons, they ionise atoms, ions hit the electrode and free more electrons

Reading the figure. Follow the numbers. (1) The RF electric field speeds up a free electron. (2) The fast electron hits a neutral atom and knocks out another electron, so one electron has become two. (3) The positive ion is pulled to the electrode, across the dark layer called the **sheath**, and hits it hard. (4) The impact frees a **secondary electron**, which starts again at step 1.

3. Why the electrons, and why RF?

Electrons are thousands of times lighter than ions, so they respond to the field almost at once and reach energies of a few eV, while the heavy ions hardly move in one RF cycle. Three electron collisions matter:

- **Ionisation** (makes the plasma): $e^- + \mathrm{Ar} \rightarrow \mathrm{Ar}^+ + 2e^-$
- **Dissociation** (makes the radicals): $e^- + \mathrm{CF_4} \rightarrow \mathrm{CF_3} + \mathrm{F} + e^-$
- **Excitation** (makes the glow): an electron lifts an atom to a higher state, and it emits light as it falls back.

The usual supply is 13.56 MHz. Because it alternates, charge never has to flow through the insulating wafer, so oxides and resists can sit on the electrode.

4. The sheath

Electrons are lost to the electrodes much faster than ions, because they are so much faster. The plasma therefore charges positive with respect to the electrodes, until the field holds most electrons back. This region of strong field next to each electrode is the **sheath**. It has few electrons, so little excitation, and it looks dark. Ions fall through it and strike the wafer **vertically**, with energy. That is the key to anisotropic etching.

5. The four steps of a plasma etch

For chemistry to etch: (1) the plasma makes a reactive species; (2) the species reaches and adsorbs on the surface; (3) it reacts with the film; (4) the product is **volatile** (a gas), desorbs and is pumped away. If step 4 fails, the surface is blocked. So the gas is chosen by its volatile product.

2. Answer

How the plasma sustains. An RF voltage (usually 13.56 MHz) across two electrodes accelerates the free electrons in the low-pressure gas. When an electron has gained enough energy it **ionises** a neutral atom, $e^- + A \rightarrow A^+ + 2e^-$, so the number of electrons multiplies. The ions are drawn across the sheath to the electrodes, and their impact releases **secondary electrons**, which are accelerated in turn.

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Electrons and ions are lost to the walls and by recombination. The plasma sustains itself when the rate of ionisation equals the rate of loss, and the applied power keeps supplying the energy. The electrons, being light, lose to the electrodes faster than the ions, so the plasma stands at a positive potential and a **sheath** forms at each electrode. Electron collisions also dissociate molecules into radicals and excite atoms, giving the glow.

Chemistries in plasma etching. Etching takes four steps: the plasma makes reactive radicals and ions; they reach the surface; they react with the film; the **volatile** product leaves and is pumped out.

$$CF_4 + e^- \rightarrow CF_3 + F + e^-$$

$$Si + 4F \rightarrow SiF_4 \uparrow \qquad SiO_2 + 4F \rightarrow SiF_4 \uparrow + O_2$$

Layer	Typical gases	Volatile product
Si, poly-Si	CF_4/O_2, SF_6, Cl_2, HBr	SiF_4, $SiCl_4$, $SiBr_4$
SiO_2	CF_4/H_2, CHF_3, C_2F_6	SiF_4, CO, CO_2
Si_3N_4	CF_4/O_2, CHF_3	SiF_4, N_2
Al	Cl_2, BCl_3	$AlCl_3$
Photoresist	O_2	CO, CO_2, H_2O

Table 1: Plasma-etch chemistries

Three mechanisms act together:

- **Chemical etching:** radicals react with the film. It is selective and isotropic.
- **Physical etching (sputtering):** ions from the sheath knock atoms off. It is directional but poorly selective.
- **Ion-enhanced etching (RIE):** ions break bonds or remove protective films only on the bottom of a feature, and radicals do the chemistry. The result is much faster than either alone, directional and acceptably selective.

3. Keyword Explanation

1. Secondary electron

An electron knocked out of an electrode surface by the impact of an ion.

2. Volatile

Easily turned to gas at the process temperature, so the product can leave the surface and be pumped away.

4. Notes and Remarks

- Plummer's Section 10.2.2 gives the system and the two reasons plasma replaced wet etching: very reactive species are made at low temperature, and ions make directional etching possible.
- If one electrode is smaller, a larger voltage appears across its sheath. Putting the wafer on the smaller electrode raises the ion energy there.

1.(d) Write the VHDL code to implement the following Boolean expression: $F = AB + \overline{A}BC + BC$. Marks: 8

1. Theory

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1. What is VHDL? (Core Idea)

VHDL is a hardware description language: a text language that describes a digital circuit. A synthesiser turns the text into gates; a simulator lets us test it before anything is built. Every VHDL description has two parts:

- an **entity**, which lists the ports (the pins seen from outside);
- an **architecture**, which says what is inside.

2. The operators

In a concurrent signal assignment, `<=` means "this output is always driven by this expression". The logic operators are `and`, `or`, `not`, `nand`, `nor`, `xor`. In VHDL, `and` and `or` have **equal precedence**, and mixing them without brackets is an error, so each product must be put in brackets.

3. Simplifying first

Take BC common from the last two terms:

$$F = AB + \overline{A}BC + BC = AB + BC(\overline{A} + 1)$$

Since $\overline{A} + 1 = 1$:

$$F = AB + BC = B(A + C)$$

A	**B**	**C**	AB	$\overline{A}BC$	BC	**F**	$B(A+C)$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
0	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	0	1	1	1

Table 2: Truth table of F

The last two columns agree in every row.

2. Answer

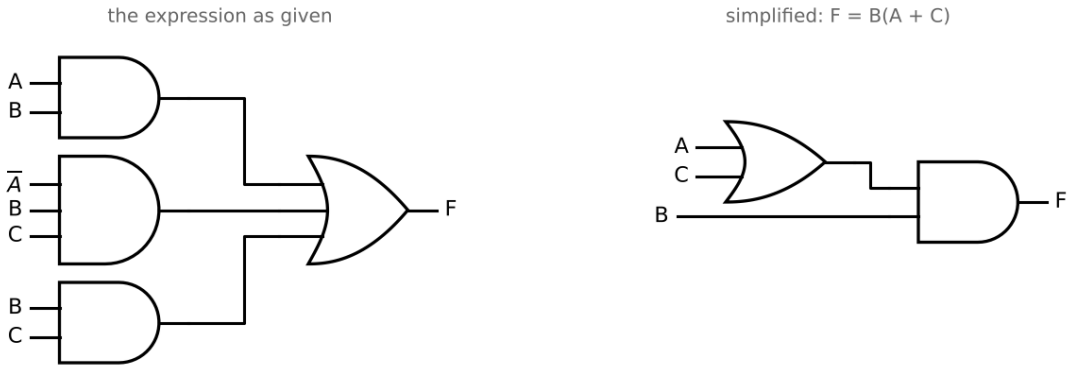


Figure 6: The expression as given and its simplified form

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VHDL code of the expression as given (dataflow style):

```vhdl
library IEEE;
use IEEE.STD_LOGIC_1164.all;

entity boolean_f is
  port(A, B, C: in  STD_LOGIC;
       F:       out STD_LOGIC);
end boolean_f;

architecture dataflow of boolean_f is
begin
  F <= (A and B) or ((not A) and B and C) or (B and C);
end dataflow;
```

Listing 1: Listing

How it works: the `library` and `use` lines bring in the type `STD_LOGIC`. The entity declares three inputs and one output. The architecture has one concurrent assignment: whenever A, B or C changes, F is recomputed.

Simplified form: $F = AB + \overline{A}BC + BC = AB + BC(\overline{A} + 1) = B(A + C)$. The assignment can then be written with two gates instead of five:

```vhdl
  F <= B and (A or C);
```

Listing 2: Listing

3. Keyword Explanation

1. Entity

The outside view of a design unit: its name and its ports.

2. Architecture

The inside of a design unit: how the outputs depend on the inputs.

4. Notes and Remarks

- The same logic in structural form instantiates `and2` and `or2` components; a synthesiser gives the same gates, so dataflow is shorter.
- The code style follows Weste and Harris, Appendix A, which shows VHDL beside SystemVerilog.

5. Final Concise Answer

(a) Adding O_2 to CF_4 removes CF_x ($O + CF_3 \rightarrow COF_2 + F$), so free F and the Si rate rise to a peak near 12 % O_2, then fall by dilution and surface oxidation. Adding H_2 ties F up as HF and leaves polymer on Si, so the Si rate falls (near zero at about 40 %) while SiO_2 etches on: SiO_2:Si selectivity rises.

(b) Deal-Grove: $F_1 = h(C^* - C_O)$, $F_2 = D(C_O - C_I)/x_O$, $F_3 = k_S C_I$, equal in steady state, give $x_O^2 + Ax_O = B(t + \tau)$ with $A = 2D(1/k_S + 1/h)$, $B = 2DC^*/N_1$. Thin oxide: $x_O \approx (B/A)(t + \tau)$, reaction limited. Thick oxide: $x_O \approx \sqrt{B(t + \tau)}$, diffusion limited.

(c) RF-accelerated electrons ionise atoms ($e + A \rightarrow A^+ + 2e$), ions hit the electrodes and free secondary electrons; the plasma lasts while ionisation balances loss. Etch chemistry: radicals such as F form volatile products (SiF_4, $AlCl_3$, CO_2); ions crossing the sheath make it directional.

(d) Entity with inputs A, B, C and output F; the architecture assigns the expression to F with one concurrent assignment. Since $F = B(A + C)$, it can be written `F <= B and (A or C);`.

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2. (a) Find threshold voltage of MOSFET without body effect. Marks: 6
   (b) State Moore's law. Explain how different parameters of MOSFET is scaled. Marks: 6
   (c) Explain operation of saturated load NMOS inverter. What are the disadvantages of this configuration? Marks: 5
   (d) Find DC characteristics of CMOS inverter using V-I characteristics curve. Marks: 8

2.(a) Find threshold voltage of MOSFET without body effect. Marks: 6

1. Theory

1. What is the threshold voltage? (Core Idea)

An nMOS transistor has a p-type body between its two n^+ regions. With no channel, no current flows. As the gate goes positive, it first pushes holes away, leaving a **depletion region** of fixed negative ions. Raised further, it attracts electrons to the surface until they form a thin n-type **inversion layer**, the channel. The **threshold voltage** V_{T0} is the gate voltage at which this happens, with the source and body at the same potential. "Without body effect" means $V_{SB} = 0$.

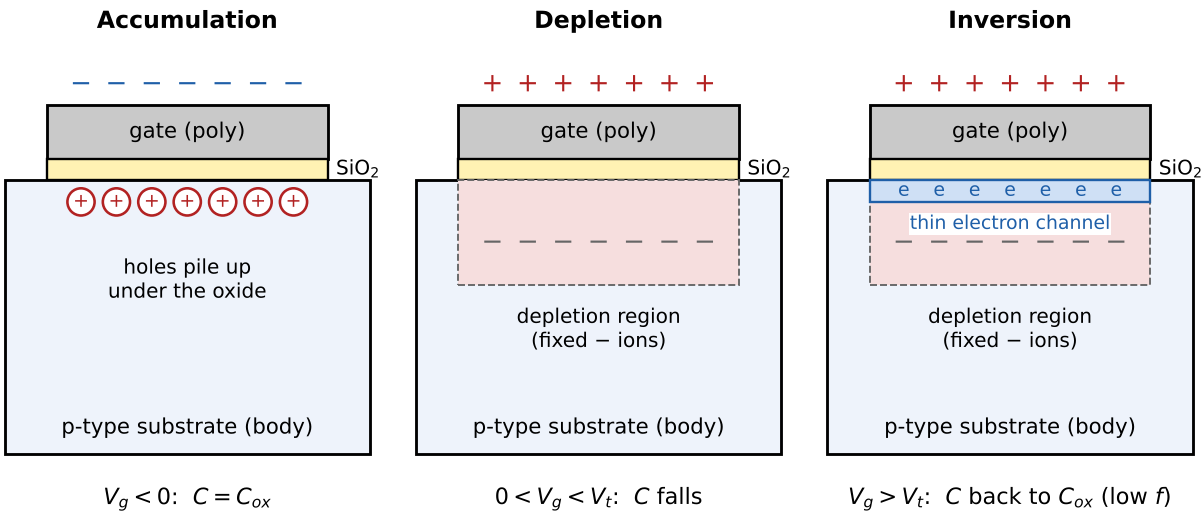


Figure 7: Accumulation, depletion and inversion under the gate (reused from the 2020-21 paper's solution)

Reading the figure. With a negative gate, holes pile up (accumulation). With a small positive gate, holes are pushed down, leaving fixed negative ions (depletion). Past the threshold, a thin layer of electrons forms (inversion).

2. The Fermi potential and the inversion condition

The doping sets how far the Fermi level sits from the middle of the bandgap. The **Fermi potential** of a p-type body is

$$\phi_F = \frac{kT}{q} \ln \frac{n_i}{N_A}$$

which is negative for p-type. Here n_i is the intrinsic carrier density and N_A is the doping. The surface is called **strongly inverted** when it is as n-type as the bulk is p-type. That needs the bands to bend by twice this amount, so the surface potential must change by $-2\phi_F$.

3. The depletion charge

Find how much charge the depletion layer holds at inversion. With depth x_d, Poisson's equation gives $d^2\phi/dx^2 = qN_A/\varepsilon_{Si}$, and integrating twice from the bulk to the surface gives the band bending across the depletion layer as $qN_Ax_d^2/2\varepsilon_{Si}$. Setting this equal to $|2\phi_F|$ and solving for the depth:

$$x_d = \sqrt{\frac{2\varepsilon_{Si}\,|2\phi_F|}{qN_A}}$$

and the fixed charge per unit area is the ion density times the depth:

$$Q_{B0} = -qN_Ax_d = -\sqrt{2qN_A\varepsilon_{Si}\,|2\phi_F|}$$

4. Adding up the gate voltage

The gate voltage has four jobs, listed by Kang and Leblebici:

1. **The built-in difference.** The gate and the body are different materials, so a voltage ϕ_{GC} already exists: $\phi_{GC} = \phi_F(\text{substrate}) - \phi_M$ for a metal gate, and $\phi_F(\text{substrate}) - \phi_F(\text{gate})$ for a polysilicon gate.
2. **Bending the bands**, by $-2\phi_F$.
3. **Holding the depletion charge.** This charge sits across the oxide capacitor C_{ox} and needs a voltage $-Q_{B0}/C_{ox}$.
4. **Cancelling the oxide charge.** A fixed positive charge Q_{ox} at the interface helps the gate, so it subtracts: $-Q_{ox}/C_{ox}$.

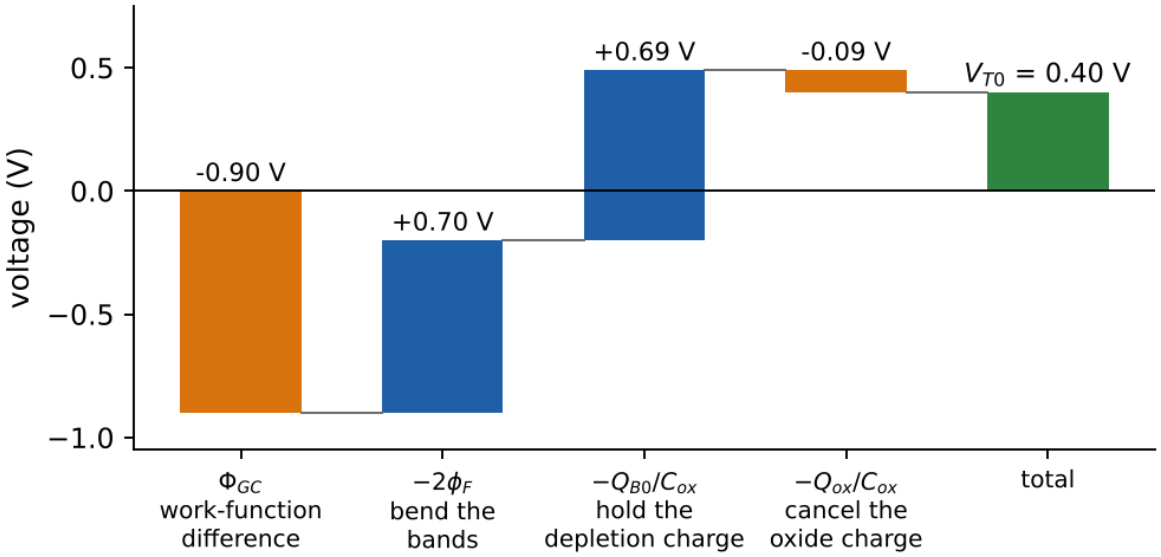


Figure 8: The threshold voltage built up from its four parts, for the values of the worked example below

Reading the figure. Read the bars from left to right, each starting where the last one ended. Starting at 0: the work-function difference goes to −0.90 V, band bending adds 0.70 V, the depletion charge adds 0.69 V, and the oxide charge takes off 0.09 V. The last bar is the total, 0.40 V.

2. Answer

Assumptions: n-channel MOSFET on a p-type body, $V_{SB} = 0$. The threshold voltage is the gate voltage at which the surface inverts. It is the sum of four components.

1. Work-function difference between gate and channel:

$$\phi_{GC} = \phi_F(\text{substrate}) - \phi_M \text{ (metal gate)}, \qquad \phi_{GC} = \phi_F(\text{substrate}) - \phi_F(\text{gate}) \text{ (poly gate)}$$

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2. Gate voltage to change the surface potential by the amount needed for inversion: $-2\phi_F$, where

$$\phi_F(\text{substrate}) = \frac{kT}{q}\ln\frac{n_i}{N_A}$$

3. Gate voltage to offset the depletion charge: $-Q_{B0}/C_{ox}$, where

$$Q_{B0} = -\sqrt{2qN_A\varepsilon_{Si}\,|-2\phi_F|}, \qquad C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$

4. Gate voltage to offset the fixed oxide charge: $-Q_{ox}/C_{ox}$.
Adding them gives the threshold voltage for zero substrate bias:

$$\boxed{V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}}$$

Illustration (Kang and Leblebici, Example 3.2; the paper gives no values): n^+ polysilicon gate, $N_A = 10^{16}$ cm^{-3}, $t_{ox} = 500$ Å, $N_{ox} = 4\times10^{10}$ cm^{-2}, $n_i = 1.45\times10^{10}$ cm^{-3}, $kT/q = 0.026$ V.

$$\phi_F(\text{substrate}) = 0.026\ln\frac{1.45\times10^{10}}{10^{16}} = -0.35\text{ V}$$

The heavily doped gate is degenerate, so $\phi_F(\text{gate}) \approx 0.55$ V, and

$$\phi_{GC} = -0.35 - 0.55 = -0.90\text{ V}$$

$$Q_{B0} = -\sqrt{2(1.6\times10^{-19})(10^{16})(11.7)(8.85\times10^{-14})(0.70)} = -4.82\times10^{-8}\text{ C/cm}^2$$

$$C_{ox} = \frac{3.97\times8.85\times10^{-14}}{500\times10^{-8}} = 7.03\times10^{-8}\text{ F/cm}^2$$

$$Q_{ox} = (1.6\times10^{-19})(4\times10^{10}) = 6.4\times10^{-9}\text{ C/cm}^2$$

$$V_{T0} = -0.90 + 0.70 + 0.69 - 0.09 = 0.40\text{ V}$$

$$\boxed{V_{T0} = 0.40\text{ V}}$$

3. Keyword Explanation

1. Strong inversion

The state in which the electron density at the surface equals the hole density of the bulk.

2. Fermi potential

The distance, in volts, between the Fermi level and the middle of the bandgap. It measures the doping.

4. Notes and Remarks

- For a p-channel device the same formula holds with signs changed: ϕ_F and Q_{B0} are positive and V_{T0} is usually negative.
- Real thresholds are set by an ion implant into the channel, which adds a term $-qN_I/C_{ox}$ for an implanted dose N_I.
- The same device with $V_{SB} \neq 0$ is solved in 8(a).
- Written with exact arithmetic, the sum is 0.394 V; Kang rounds each term and quotes 0.40 V.

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2.(b) State Moore's law. Explain how different parameters of MOSFET is scaled. Marks: 6

1. Theory

1. Moore's law (Core Idea)

In 1965 Gordon Moore noticed that the number of transistors on the most economical chip had doubled every year, and predicted it would go on doing so. In 1975 he revised the rate to a doubling about **every two years**. The industry then used the prediction as a target, and it held for about five decades.

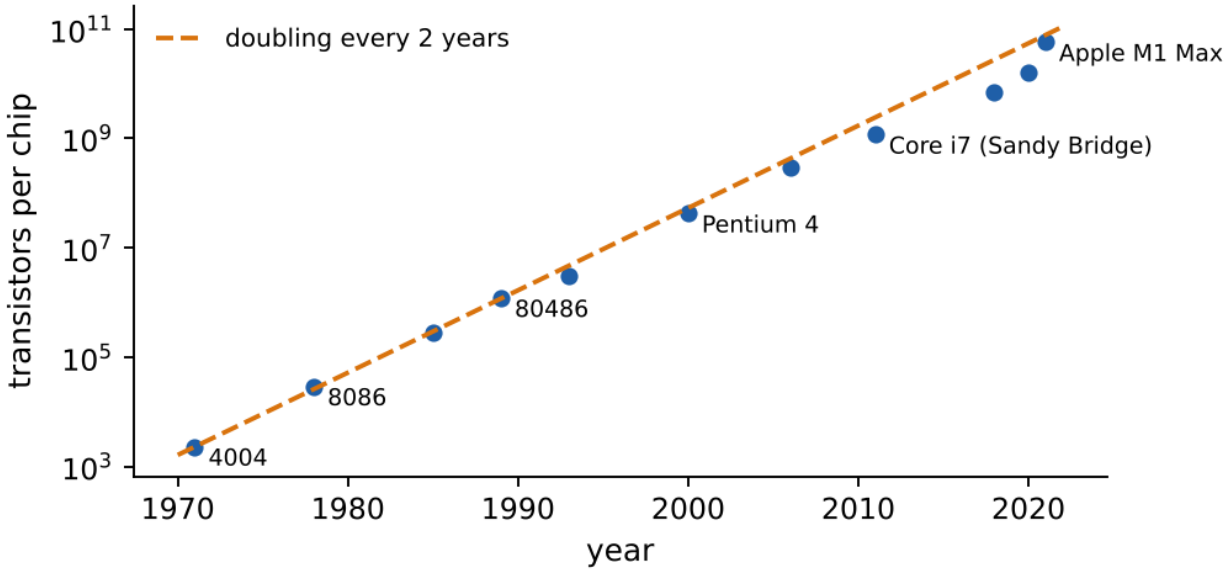


Figure 9: Transistors per chip against year, on a log scale, with the line for doubling every two years (reused from the 2019-20 paper's solution)

Reading the figure. On a log scale, steady doubling is a straight line (dashed). The chips follow it from the Intel 4004 (2300 transistors, 1971) to the Apple M1 Max (57 billion, 2021).

2. What does "scaling" mean?

Each new process generation makes transistors smaller. If every length shrinks by a factor S (about $\sqrt{2} = 1.4$ per generation), each transistor takes $1/S^2$ of the area, so twice as many fit on the same chip. How we change the voltages and doping decides whether the smaller transistor works well.

- **Constant-field (Dennard) scaling:** the voltages also shrink by S, so the electric fields stay the same.
- **Constant-voltage scaling:** V_{DD} stays, so the fields grow by S.

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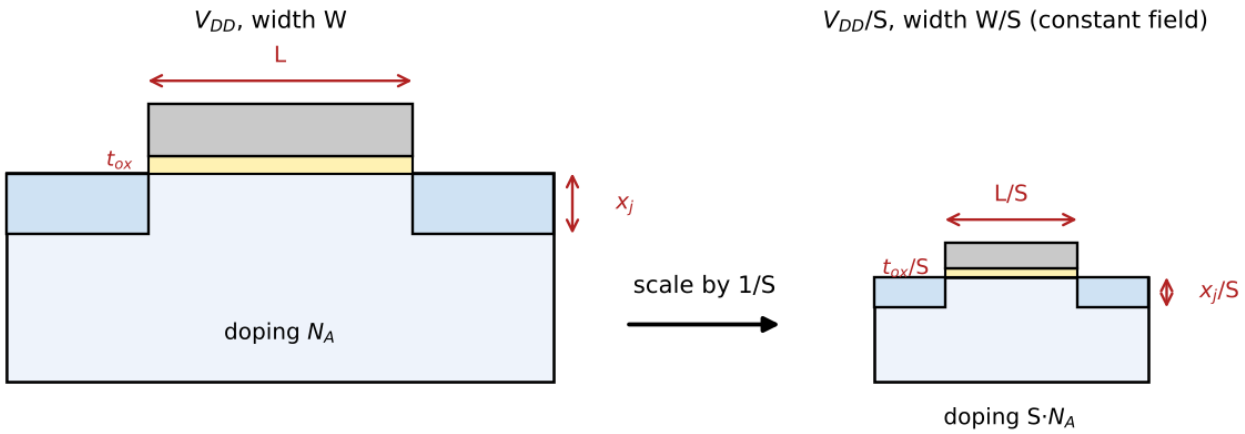


Figure 10: A transistor before and after scaling every dimension by 1/S, drawn for S = 2 (reused from the 2019-20 paper's solution)

Reading the figure. Every length (L, x_j, t_{ox}, W) shrinks by $1/S$. The body doping rises by S to keep the depletion regions in proportion, and the supply falls to V_{DD}/S.

3. Working out each parameter (constant-field)

Start from the saturation current $I = \frac{\beta}{2}(V - V_t)^2$ with $\beta = \mu C_{ox} W/L$.

- $C_{ox} = \varepsilon_{ox}/t_{ox}$: t_{ox} falls by S, so C_{ox} rises by S.
- $\beta = \mu C_{ox} W/L$: W/L is unchanged, so β rises by S.
- $I \propto \beta V^2$: $S \times (1/S)^2 = 1/S$.
- Gate capacitance $C = C_{ox} WL$: $S \times (1/S) \times (1/S) = 1/S$.
- Gate delay $t = CV/I$: $(1/S)(1/S)/(1/S) = 1/S$.
- Power per gate $P = IV$: $(1/S)(1/S) = 1/S^2$.
- Power density P/area: $(1/S^2)/(1/S^2) = 1$.

2. Answer

Moore's law: the number of transistors on an integrated circuit doubles about every two years (Moore, 1965: every year; revised in 1975), at about constant cost per chip.

Scaling: each generation reduces the dimensions of the MOSFET by a factor $S > 1$ (typically $\sqrt{2}$). The voltages and doping are changed by plan. Two classic ways:

Parameter	Constant-field scaling	Constant-voltage scaling
Length L, width W, oxide t_{ox}, junction depth x_j	$1/S$	$1/S$
Supply V_{DD}, threshold V_t	$1/S$	1
Body doping N_A	S	S^2
C_{ox}, $\beta = \mu C_{ox} W/L$	S	S
Current per transistor $I \propto \beta V^2$	$1/S$	S
Gate capacitance $C \propto C_{ox} WL$	$1/S$	$1/S$
Gate delay CV/I	$1/S$	$1/S^2$
Power per gate IV	$1/S^2$	S
Switching energy CV^2	$1/S^3$	$1/S$
Transistors per area	S^2	S^2

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Parameter	Constant-field scaling	Constant-voltage scaling
Power density	1	S^3

Table 3: How the parameters scale (factor by which each changes)

Interpretation: constant-field scaling makes gates S times faster and S^2 times denser, with constant power per area. Constant-voltage scaling gives faster gates but the power density rises as S^3, and the high fields damage the devices. In practice V_{DD} was scaled more slowly than the lengths, and since about 2005 it has hardly scaled at all, because V_t cannot fall further without large leakage.

3. Keyword Explanation

1. Scaling factor S

The ratio of old to new dimensions; about 1.4 per technology node (for example 90 nm to 65 nm).

2. Dennard scaling

Another name for constant-field scaling, after R. Dennard's 1974 paper.

4. Notes and Remarks

- Wires scale differently: Weste and Harris's Table 7.5 gives resistance per length rising by S^2, so a local wire's RC delay stays about constant instead of falling, and global wires get slower.
- Also discussed in 2(c) and 4(a) of the 2019-20 paper, as extra reading.

2.(c) Explain operation of saturated load NMOS inverter. What are the disadvantages of this configuration? Marks: 5

1. Theory

1. What is an nMOS inverter? (Core Idea)

An inverter needs something to pull the output up and something to pull it down. In early chips only nMOS transistors were available. An nMOS **driver** pulls the output down when the input is high, and the pull-up is a **load**. A resistor would work but takes a huge area, so a second nMOS transistor is used as the load.

2. Why "saturated"?

The load's gate is tied to its drain, and both go to V_{DD}. So $V_{GS} = V_{DS}$ for the load. A transistor is in saturation when $V_{DS} \geq V_{GS} - V_T$. Here that reads $V_{DS} \geq V_{DS} - V_T$, which is **always true**. So whenever the load conducts, it is saturated.

3. The two output levels

Use the current equations (Kang's $k = \mu C_{ox} W/L$):

- Load (saturated): $I_L = \frac{k_L}{2}(V_{DD} - V_{out} - V_{TL})^2$
- Driver in saturation: $I_D = \frac{k_D}{2}(V_{in} - V_T)^2$
- Driver in the linear region: $I_D = \frac{k_D}{2}[2(V_{in} - V_T)V_{out} - V_{out}^2]$

Output high. With $V_{in} < V_T$ the driver is off, so the current must be zero. From the load equation, $I_L = 0$ needs $V_{DD} - V_{out} - V_{TL} = 0$:

$$V_{OH} = V_{DD} - V_{TL}$$

The output **cannot reach** V_{DD}: as it nears $V_{DD} - V_{TL}$, the load's V_{GS} falls to V_{TL} and it switches off. The load's source is above the body, so the body effect raises V_{TL} and makes this worse.

Output low. With $V_{in} = V_{OH}$ the driver is in the linear region. Set $I_D = I_L$:

$$k_D\left[2(V_{OH}-V_T)V_{OL}-V_{OL}^2\right]=k_L\left(V_{OH}-V_{OL}\right)^2$$

Both transistors are on, so the output is a voltage divider and V_{OL} is above 0. A small V_{OL} needs a large **ratio** $k_R = k_D/k_L$. With $V_{DD} = 5$ V, $V_T = 1$ V, $k_R = 9$ and the body effect neglected: $V_{OH} = 4$ V, and the equation gives a quadratic $10V_{OL}^2 - 62V_{OL} + 16 = 0$ whose smaller root is $V_{OL} = 0.27$ V.

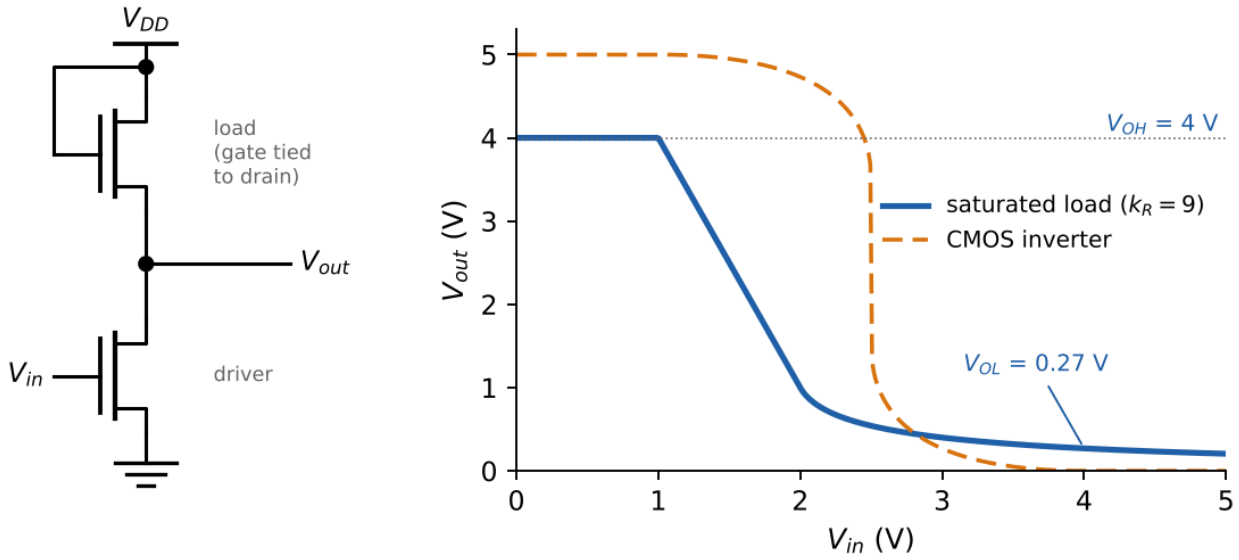


Figure 11: The saturated-load inverter and its transfer curve (VDD 5 V, VT 1 V, kR 9, no body effect) against a CMOS inverter

Reading the figure. The solid blue curve starts at 4 V, not 5 V. It falls in a straight line from $V_{in} = 1$ V with slope $-\sqrt{k_R} = -3$, because both transistors are saturated, and then bends into a long tail that never reaches 0. The dashed orange CMOS curve runs from 5 V to 0 V with a sharp step.

2. Answer

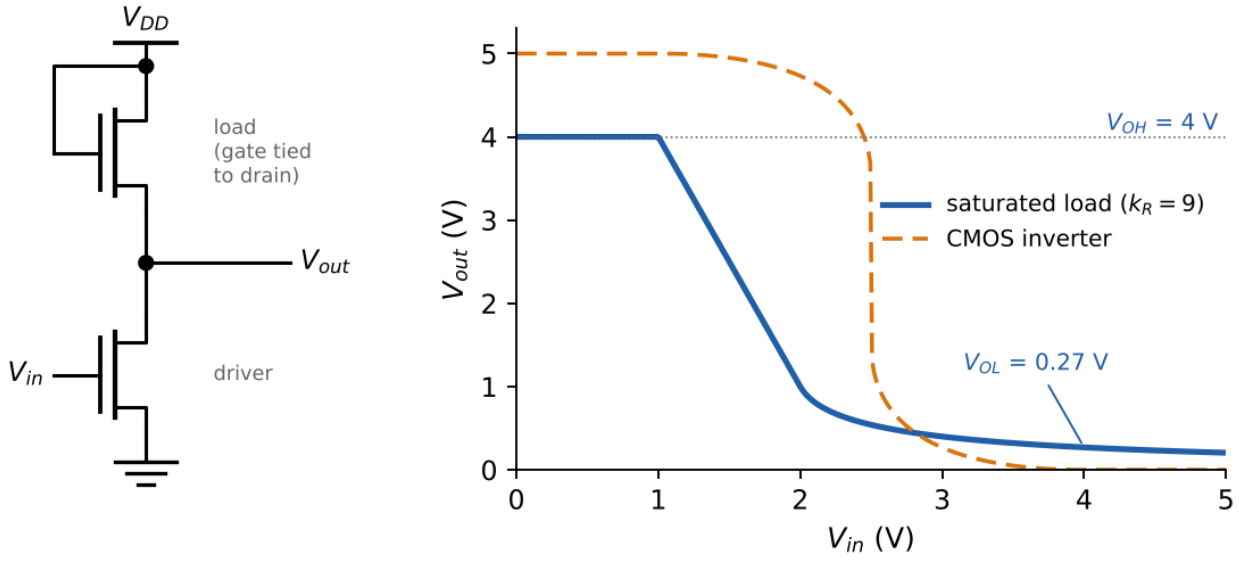


Figure 12: Circuit of the saturated enhancement-load inverter and its transfer curve

Circuit: an enhancement nMOS **driver** from the output to ground, with the input on its gate, and an enhancement nMOS **load** from V_{DD} to the output with its gate tied to its drain. Since $V_{GS} = V_{DS}$,

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the load is always saturated while it conducts.

Operation:

- $V_{in} < V_T$ **(input low):** the driver is off, and no DC current flows. The load pulls the output up until its V_{GS} falls to V_{TL}, so $V_{OH} = V_{DD} - V_{TL}$.
- V_{in} **just above** V_T**:** the driver and load are both saturated, so $k_D(V_{in} - V_T)^2 = k_L(V_{OH} - V_{out})^2$, which gives a straight line of slope $-\sqrt{k_R}$, $k_R = k_D/k_L$.
- V_{in} **high:** the driver is in the linear region. Both transistors are on, and the output is set by their ratio:

$$k_R\left[2(V_{OH} - V_T)V_{OL} - V_{OL}^2\right] = (V_{OH} - V_{OL})^2$$

For $V_{DD} = 5$ V, $V_T = 1$ V, $k_R = 9$ (illustrative): $V_{OH} = 4$ V, $V_{OL} = 0.27$ V.

Disadvantages:

1. V_{OH} **is below** V_{DD} by V_{TL}, made worse by the body effect, so the output swing and high noise margin are reduced.
2. **Ratioed logic:** V_{OL} is not 0 and depends on k_R; a low V_{OL} needs a large driver and so more area.
3. **Static power:** current flows from V_{DD} to ground whenever the output is low.
4. **Slow rise:** the load's current falls towards zero as the output rises.
5. **Poor transition:** the slope $-\sqrt{k_R}$ is small, so the gain and noise margins are small.

3. Keyword Explanation

1. Ratioed logic

Logic whose output level depends on the size ratio of its transistors.

2. Body effect

The rise in threshold voltage when the source is above the body; see 8(a).

4. Notes and Remarks

- Kang's Section 5.3 also shows the linear enhancement load (gate on a second, higher supply), which reaches V_{DD} but needs two supplies, and the depletion load, which solved most of these faults and needs one supply.
- With the same numbers, the static current in the low state is $\frac{k_L}{2}(V_{OH} - V_{OL})^2 = 139$ µA for $k_L = 20$ µA/V^2 (an illustrative value), so 696 µW per gate at 5 V. This is why large chips moved to CMOS.

2.(d) Find DC characteristics of CMOS inverter using V-I characteristics curve. Marks: 8

1. Theory

1. The CMOS inverter (Core Idea)

A CMOS inverter is one pMOS from V_{DD} to the output and one nMOS from the output to ground, with both gates tied to the input. When the input is low, the pMOS is ON and the nMOS OFF, so the output is pulled to V_{DD}; when the input is high, the reverse. Its **DC characteristic**, or **voltage transfer characteristic (VTC)**, is the graph of V_{out} against V_{in} when the input changes slowly.

2. The I-V equations

Long-channel equations for the nMOS, with $\beta = \mu C_{ox} W/L$:

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$$I_{Dn} = \begin{cases} 0 & V_{GS} < V_{tn} \text{ (cutoff)} \\ \beta_n \left(V_{GS} - V_{tn} - \dfrac{V_{DS}}{2} \right) V_{DS} & V_{DS} < V_{GS} - V_{tn} \text{ (linear)} \\ \dfrac{\beta_n}{2} (V_{GS} - V_{tn})^2 & V_{DS} \geq V_{GS} - V_{tn} \text{ (saturation)} \end{cases}$$

The pMOS has the same form with V_{SG}, V_{SD} and $|V_{tp}|$. In the inverter:

- nMOS: $V_{GS} = V_{in}$, $V_{DS} = V_{out}$.
- pMOS: $V_{SG} = V_{DD} - V_{in}$, $V_{SD} = V_{DD} - V_{out}$.

3. The key fact: the two currents are equal

The next gate draws no DC current, so whatever flows down through the pMOS flows on through the nMOS:

$$I_{Dn} = |I_{Dp}|$$

For each V_{in} the output settles where the nMOS curve and the pMOS curve cross.

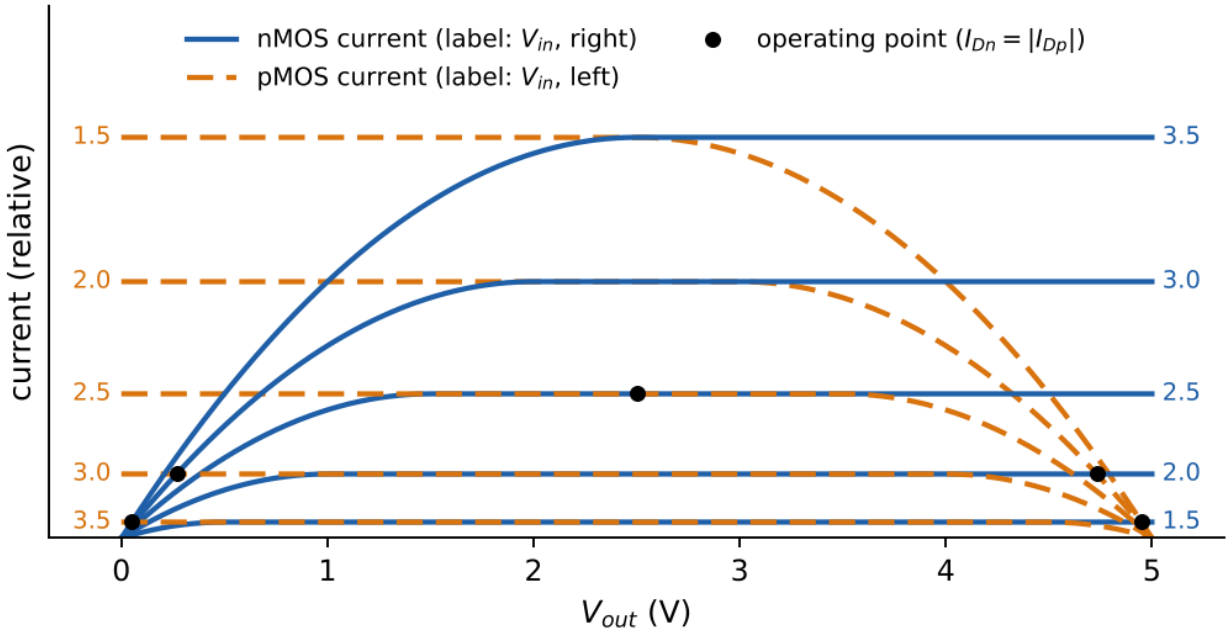


Figure 13: nMOS (blue) and pMOS (orange) currents against Vout for Vin from 1.5 V to 3.5 V (VDD = 5 V, Vt = 1 V, equal beta); each black dot is where the two currents are equal (reused from the 2019-20 paper's solution)

Reading the figure. For $V_{in} = 1.5$ V the nMOS curve is low and the pMOS curve is high, so they cross near $V_{out} = 5$ V. As V_{in} rises, the nMOS curves rise and the pMOS curves fall, so the crossing moves left. At 2.5 V both are flat and equal and the crossing is at the middle. At 3.5 V it is near 0 V. Plotting each dot's V_{out} against its V_{in} gives the VTC.

4. Solving each region (symmetric inverter)

Take $\beta_n = \beta_p = \beta$ and $V_{tn} = |V_{tp}| = V_t$, and write $a = V_{DD} - V_{in} - V_t$ and $b = V_{in} - V_t$.

Region B (nMOS saturated, pMOS linear). Let $u = V_{DD} - V_{out}$. Equating currents:

$$\frac{\beta}{2} b^2 = \beta \left(a - \frac{u}{2} \right) u$$

Multiply by $2/\beta$ and rearrange:

$$u^2 - 2au + b^2 = 0 \;\Rightarrow\; u = a - \sqrt{a^2 - b^2}$$

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The smaller root is the one with $u < a$, which the linear region needs. Now $a^2 - b^2 = (a-b)(a+b)$, with $a - b = V_{DD} - 2V_{in}$ and $a + b = V_{DD} - 2V_t$. So

$$V_{out} = V_{DD} - u = V_{in} + V_t + \sqrt{(V_{DD} - 2V_{in})(V_{DD} - 2V_t)}$$

Region D (nMOS linear, pMOS saturated). By the mirror symmetry of the circuit:

$$V_{out} = V_{in} - V_t - \sqrt{(2V_{in} - V_{DD})(V_{DD} - 2V_t)}$$

Check: at $V_{in} = V_t$ in region B, the root is $V_{DD} - 2V_t$, so $V_{out} = 2V_t + V_{DD} - 2V_t = V_{DD}$, joining region A. For $V_{DD} = 5$ V, $V_t = 1$ V: $V_{in} = 1.5$ V gives 4.95 V, 2 V gives 4.73 V, 3 V gives 0.27 V, 3.5 V gives 0.05 V.

2. Answer

Method: for each input, set $I_{Dn}(V_{in}, V_{out}) = |I_{Dp}|(V_{in}, V_{out})$ and solve for V_{out}. Graphically this is the intersection of the nMOS and pMOS I-V curves, with one pair of curves for each V_{in}:

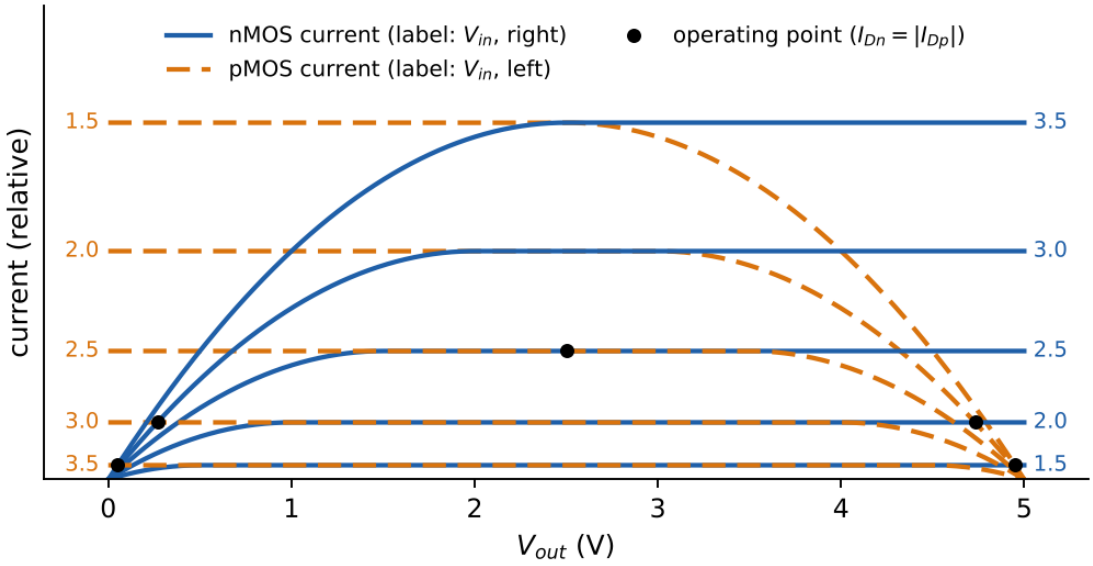


Figure 14: The nMOS and pMOS curves for each input; the dots are the operating points

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Region	V_{in}	nMOS	pMOS	V_{out}
A	$0 \le V_{in} < V_{tn}$	cutoff	linear	V_{DD}
B	$V_{tn} \le V_{in} < V_M$	saturation	linear	slightly below V_{DD}
C	$V_{in} = V_M$	saturation	saturation	falls steeply
D	$V_M < V_{in} \le V_{DD} - \lvert V_{tp}\rvert$	linear	saturation	slightly above 0
E	$V_{in} > V_{DD} - \lvert V_{tp}\rvert$	linear	cutoff	0

Table 4: Operating regions of the CMOS inverter

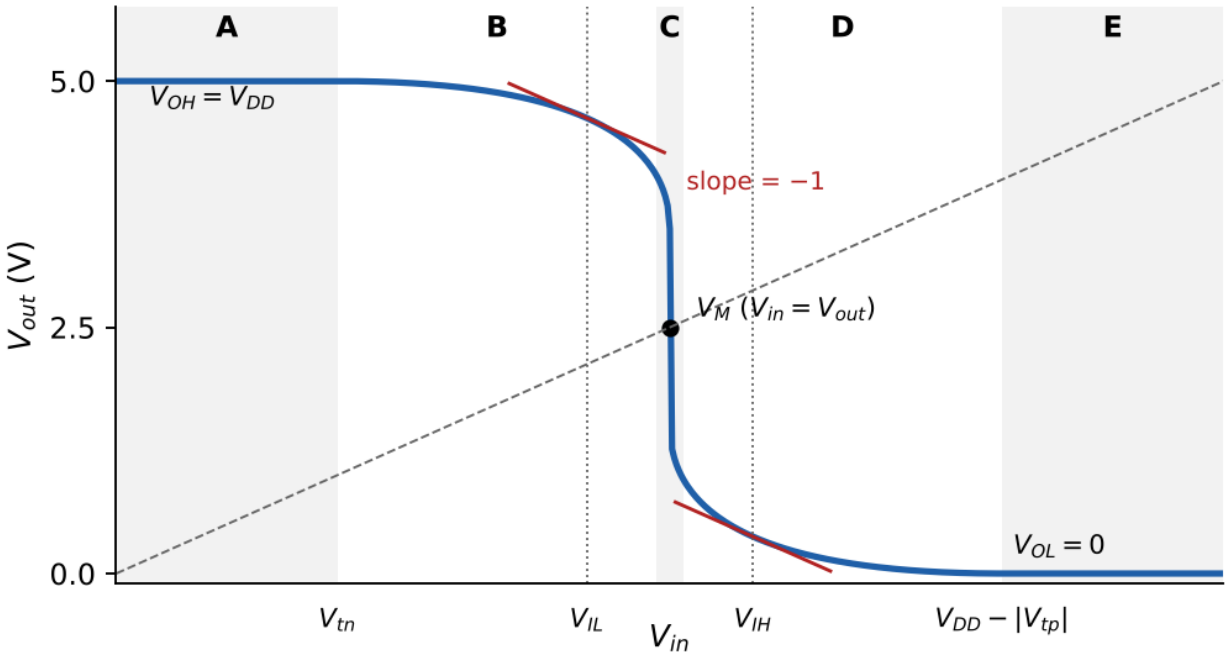


Figure 15: VTC of the CMOS inverter with its five regions (VDD = 5 V, Vt = 1 V, beta_n = beta_p; reused from the 2019-20 paper's solution)

Output levels: in A one transistor is off and no current flows, so $V_{OH} = V_{DD}$; in E, $V_{OL} = 0$. The inverter is **ratioless**: full swing whatever the sizes.

Regions B and D, for $\beta_n = \beta_p$, $V_{tn} = |V_{tp}| = V_t$:

$$V_{out} = V_{in} + V_t + \sqrt{(V_{DD} - 2V_{in})(V_{DD} - 2V_t)} \quad \text{(B)}$$

$$V_{out} = V_{in} - V_t - \sqrt{(2V_{in} - V_{DD})(V_{DD} - 2V_t)} \quad \text{(D)}$$

Switching threshold V_M (region C, $V_{in} = V_{out} = V_M$, both saturated):

$$\frac{\beta_n}{2}(V_M - V_{tn})^2 = \frac{\beta_p}{2}(V_{DD} - V_M - |V_{tp}|)^2$$

Take the square root of both sides and let $r = \sqrt{\beta_p/\beta_n}$:

$$V_M - V_{tn} = r\,(V_{DD} - V_M - |V_{tp}|) \;\Rightarrow\; V_M = \frac{V_{tn} + r\,(V_{DD} - |V_{tp}|)}{1 + r}$$

For $\beta_n = \beta_p$ ($W_p \approx 2W_n$) and $V_{tn} = |V_{tp}|$: $V_M = V_{DD}/2$.

Noise-margin points (slope $dV_{out}/dV_{in} = -1$; Kang and Leblebici). For the symmetric inverter:

$$V_{IL} = \frac{3V_{DD} + 2V_t}{8}, \qquad V_{IH} = \frac{5V_{DD} - 2V_t}{8}$$

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Example

($V_{DD} = 5$ V, $V_t = 1$ V, $\beta_n = \beta_p$): $V_M = 2.5$ V, $V_{IL} = 2.125$ V, $V_{IH} = 2.875$ V, $NM_L = V_{IL} - V_{OL} = 2.125$ V, $NM_H = V_{OH} - V_{IH} = 2.125$ V.

3. Keyword Explanation

1. Switching threshold V_M

The input voltage at which $V_{out} = V_{in}$; it sits in the middle of the steep part of the VTC.

2. Noise margin

How much noise an input can carry and still be read correctly: $NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$.

4. Notes and Remarks

- With ideal long-channel transistors the VTC is exactly vertical at V_M, because both currents are flat in saturation; with channel-length modulation the slope is large but finite.
- A larger β_p/β_n moves V_M up.
- $V_{DD} = 5$ V and $V_t = 1$ V are illustrative values; the paper gives none.
- Also solved in 3(a) of the 2019-20 paper, as extra reading.

5. Final Concise Answer

(a) $V_{T0} = \phi_{GC} - 2\phi_F - Q_{B0}/C_{ox} - Q_{ox}/C_{ox}$, with $Q_{B0} = -\sqrt{2qN_A\varepsilon_{Si}|2\phi_F|}$. For $N_A = 10^{16}$ cm^{-3}, $t_{ox} = 500$ Å, $N_{ox} = 4 \times 10^{10}$ cm^{-2}: $V_{T0} = 0.40$ V.
(b) Moore's law: the transistor count doubles about every two years. Constant-field scaling: lengths and voltages $1/S$, doping S; current $1/S$, capacitance $1/S$, delay $1/S$, power per gate $1/S^2$, power density 1. Constant-voltage scaling: delay $1/S^2$, power density S^3.
(c) Load nMOS with gate on drain, always saturated. $V_{OH} = V_{DD} - V_{TL}$; V_{OL} is set by the ratio k_R (0.27 V for 5 V, 1 V, $k_R = 9$). Faults: reduced V_{OH}, ratioed V_{OL}, static power, slow rise, small gain.
(d) Equate I_{Dn} and $|I_{Dp}|$: five regions A to E; $V_{OH} = V_{DD}$, $V_{OL} = 0$; $V_M = [V_{tn} + r(V_{DD} - |V_{tp}|)]/(1+r)$ with $r = \sqrt{\beta_p/\beta_n}$, $V_{DD}/2$ when symmetric; $V_{IL} = (3V_{DD} + 2V_t)/8$, $V_{IH} = (5V_{DD} - 2V_t)/8$.

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3. (a) What is VHDL? Write down the advantages of VHDL. Marks: 2

(b) Sketch a 3-input NOR gate with transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model. Assume that every source or drain has fully contacted diffusion when making your estimate of capacitance. Marks: 6

(c) For the circuit in figure 3c find-

i) Logical effort, G
ii) Branching effort, B
iii) Electrical effort, E
iv) Path effort, F
v) Stage effort, f
vi) Parasitic delay, P
vii) Minimum path delay, D
viii) Size of the gate. Marks: 10

(d) Find the number of stages for minimum delay. Marks: 7

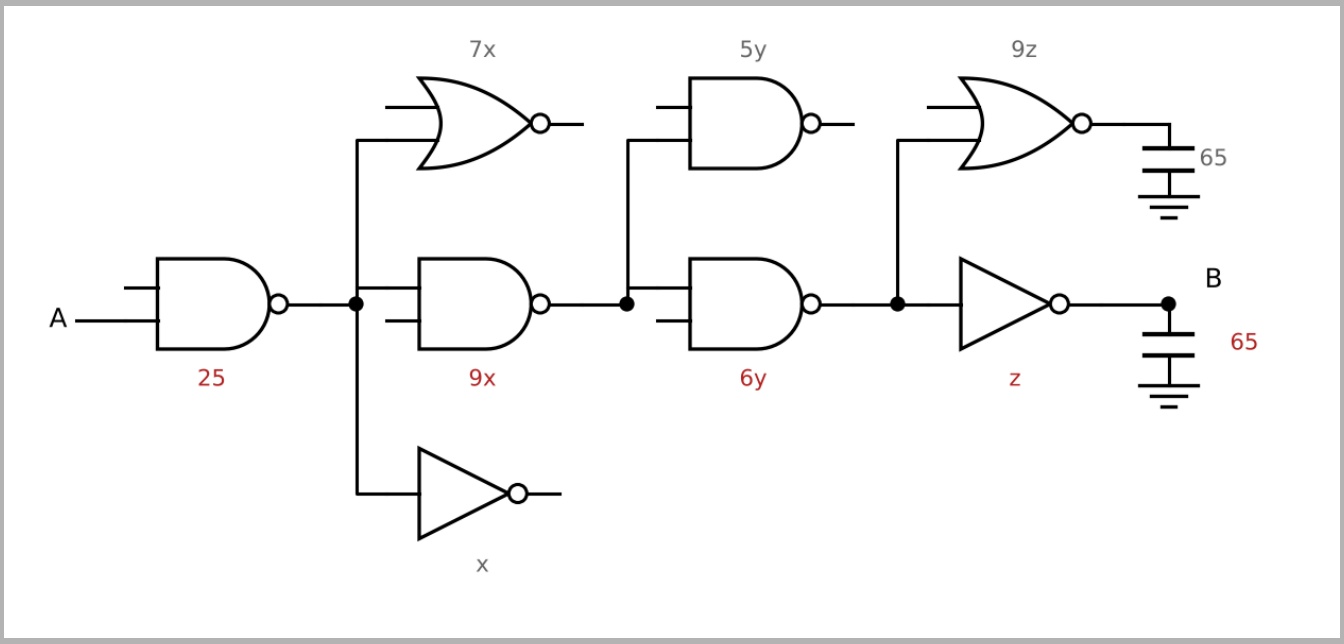


Figure 16: Figure: 3c of the paper (redrawn); the dashed path from A to B on the paper runs through the gates marked 25, 9x, 6y and z

3.(a) What is VHDL? Write down the advantages of VHDL. Marks: 2

1. Theory

1. Why describe hardware in text? (Core Idea)

A modern chip has billions of transistors, so nobody can draw it by hand. Designers write what the circuit should do in a **hardware description language (HDL)**. Tools then simulate the text to check it, and **synthesise** it into gates. VHDL and Verilog are the two main HDLs.

The name VHDL stands for **VHSIC Hardware Description Language**, and VHSIC means Very High Speed Integrated Circuit. It came from a US Department of Defense programme and became IEEE Standard 1076.

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2. Answer

VHDL (VHSIC Hardware Description Language) is an IEEE-standard language (IEEE 1076) that describes the structure and behaviour of digital hardware, so that it can be simulated and synthesised into gates.

Advantages:

- It is a **standard**, so a design moves between tools and vendors (FPGA or ASIC) without being rewritten.
- It supports **several levels** of description: behavioural, dataflow and structural.
- A design can be **simulated and verified** before fabrication, with testbenches written in the same language.
- **Synthesis** tools turn it into gates automatically, which shortens design time.
- Designs are **reusable**, and hierarchy (entities and components) makes large designs manageable.
- **Strong typing** catches many errors early, and the text documents the design.

3. Keyword Explanation

1. Synthesis

The automatic translation of HDL text into a netlist of gates.

4. Notes and Remarks

- The marks (2) need only a definition and a few points.

3.(b) Sketch a 3-input NOR gate with transistor widths chosen to achieve effective rise and fall resistances equal to a unit inverter. Compute the rising and falling propagation delays of the NOR gate driving h identical NOR gates using the Elmore delay model. Assume that every source or drain has fully contacted diffusion when making your estimate of capacitance. Marks: 6

1. Theory

1. A transistor as a resistor (Core Idea)

For delay estimates, an ON transistor is a resistor, and each terminal is a capacitor.

- A **unit nMOS** has width 1 and resistance R.
- A **unit pMOS** has width 2 and also resistance R, because holes are about half as mobile as electrons.
- The **unit inverter** is one of each. Resistance falls as width rises: width k times unit gives R/k.
- Every transistor has gate capacitance C per unit width, and diffusion capacitance C per unit width at each source and at each drain.

2. Choosing the widths

The aim is the same worst-case resistance R as the unit inverter.

- **Pull-up (pMOS):** the output is 1 only when A, B and C are all 0, so three pMOS are in **series**. Their resistances add. Each must be $R/3$, and a pMOS of resistance $R/3$ is 3 times the unit pMOS: width $3 \times 2 = 6$.
- **Pull-down (nMOS):** any one input high pulls the output low, so three nMOS are in **parallel**. In the worst case only one is ON, and it alone must give R: width 1.

3. The Elmore delay

A chain of resistors charges several capacitors. The **Elmore delay** is an estimate: multiply each capacitor by the total resistance between it and the source, and add:

$$t_{pd} = \sum_i R_{i,\text{to source}}\, C_i$$

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4. "Fully contacted diffusion"

The question says every source or drain has its own contacted diffusion. So each source or drain of a transistor of width W counts as the full WC, even where two series transistors meet. That node carries two terminals. (A shared, uncontacted node would be cheaper; see the Notes.)

2. Answer

Sizing: three series pMOS, each of width 6 (resistance $R/3$ each, R in series); three parallel nMOS, each of width 1 (resistance R when one is ON).

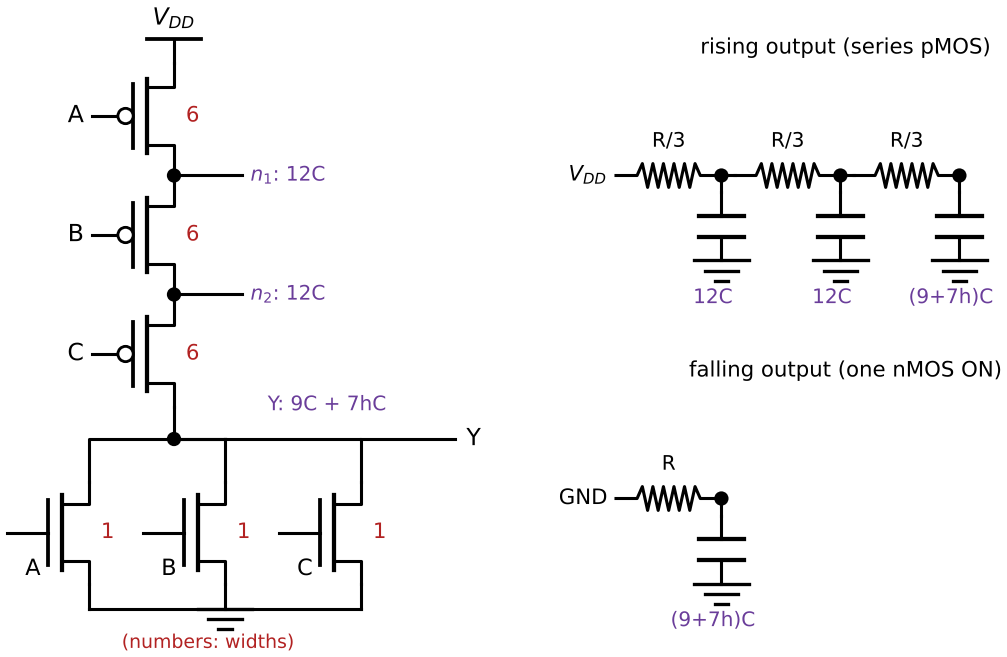


Figure 17: 3-input NOR with widths and node capacitances, and the Elmore equivalents for the rising and falling outputs

Capacitances, in units of C:

- Input capacitance of one input: $6 + 1 = 7C$. So each of the h loads adds $7C$.
- Node n_1 (between pMOS A and B): two diffusions of width 6, $6 + 6 = 12C$.
- Node n_2 (between pMOS B and C): $6 + 6 = 12C$.
- Output Y: one pMOS drain (6) and three nMOS drains $(1 + 1 + 1)$, so $9C$ of diffusion, plus the load: $(9 + 7h)C$.
- Sources on V_{DD} and ground have both plates at a fixed voltage, so they do not count.

Rising delay (worst case: all three pMOS are ON at the end, and n_1, n_2 and Y all start low, so all must be charged through the chain from V_{DD}):

$$t_{pdr} = \left(\frac{R}{3}\right)(12C) + \left(\frac{R}{3} + \frac{R}{3}\right)(12C) + \left(\frac{R}{3} + \frac{R}{3} + \frac{R}{3}\right)(9 + 7h)C$$

$$= 4RC + 8RC + (9 + 7h)RC$$

$$\boxed{t_{pdr} = (21 + 7h)\,RC}$$

Falling delay (worst case: only one nMOS is ON, resistance R, and it discharges the output node):

$$t_{pdf} = R\,(9 + 7h)C$$

$$\boxed{t_{pdf} = (9 + 7h)\,RC}$$

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Final Answer: with $\tau = 3RC$, $t_{pdr} = (7 + 2.33h)\tau$ and $t_{pdf} = (3 + 2.33h)\tau$. The effort term $7h$ is the same in both; the extra $12RC$ in the rise comes from the internal nodes of the series stack.

3. Keyword Explanation

1. Worst-case delay

The longest delay over all input patterns; the propagation delay uses it.

2. Effective resistance

The constant resistance that gives the same delay as the transistor's real non-linear current.

4. Notes and Remarks

- **Better capacitance estimate:** Weste and Harris's Exercise 4.2 says a node between two series transistors with no contact needs only half the capacitance. Then n_1 and n_2 are $6C$ each, and $t_{pdr} = 2+4+9+7h = (15+7h)RC$. The falling delay does not change. The question says "fully contacted", so the Answer keeps $12C$.
- The question follows Exercise 4.1 of Weste and Harris (which asks the same for a 2-input NOR).
- A logical-effort check: $g = 7/3$ for a NOR3, since $C_{in} = 7$ against 3 for an inverter. The effort term in units of τ is $gh = 7h/3 = 2.33h$, as above.

3.(c) For the circuit in figure 3c find- i) Logical effort, G ii) Branching effort, B iii) Electrical effort, E iv) Path effort, F v) Stage effort, f vi) Parasitic delay, P vii) Minimum path delay, D viii) Size of the gate. Marks: 10

1. Theory

1. The delay of one gate (Core Idea)

Logical effort writes the delay of a gate, in units of $\tau = 3RC$, as

$$d = gh + p$$

- g is the **logical effort**: how much worse the gate drives than an inverter of the same input capacitance.
- $h = C_{out}/C_{in}$ is the **electrical effort** or fanout.
- p is the **parasitic delay**, from its own diffusion capacitance.
- $f = gh$ is the **stage effort**.

Gate	g	p
Inverter	1	1
NAND2	4/3	2
NOR2	5/3	2

Table 5: Logical effort and parasitic delay of the gates in the figure

Why $g = 4/3$ for a NAND2: series nMOS of width 2 and parallel pMOS of width 2 give $C_{in} = 4$, against 3 for a unit inverter. **Why $g = 5/3$ for a NOR2:** series pMOS of width 4 and parallel nMOS of width 1 give $C_{in} = 5$.

2. A whole path

For a path of N stages:

- **Path logical effort:** $G = \prod g_i$.
- **Path electrical effort:** $H = C_{out(path)}/C_{in(path)}$ (the paper calls it E).
- **Branching effort:** at a node where the signal splits, part of the drive goes off the path. $b = (C_{on} + C_{off})/C_{on}$, and $B = \prod b_i$.

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- **Path effort:** $F = GBH$. It equals the product of the stage efforts, because the C ratios cancel.
- **Parasitic delay:** $P = \sum p_i$.

3. Why equal stage efforts?

The delay is $D = \sum f_i + P$, and the product of the f_i is fixed at F. A sum with a fixed product is smallest when all terms are equal. So the best is

$$\hat{f} = F^{1/N}, \qquad \hat{D} = NF^{1/N} + P$$

4. Finding the sizes

Work backwards from the load. Each stage has $C_{in} = g\,C_{out}/\hat{f}$, where C_{out} includes all that the stage drives, on and off the path.

2. Answer

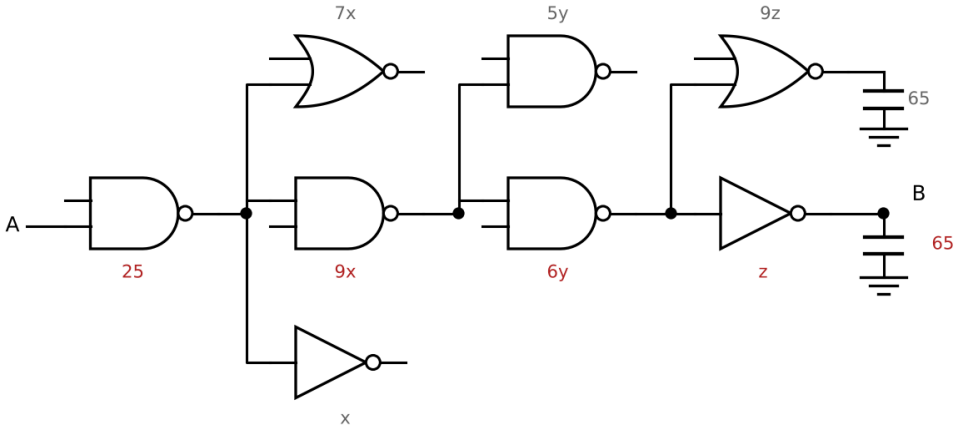


Figure 18: Repeated from the question: Figure 3c of the paper (redrawn)

Reading the circuit. The labels are input capacitances. The path A to B has 4 stages: NAND2 (input 25), NAND2 (9x), NAND2 (6y), inverter (z), load 65.

- The first gate also drives NOR2 7x and inverter x.
- The 9x gate also drives NAND2 5y.
- The 6y gate also drives NOR2 9z.

i) Logical effort:

$$G = \frac{4}{3} \times \frac{4}{3} \times \frac{4}{3} \times 1 = \frac{64}{27} = 2.37$$

ii) Branching effort:

$$b_1 = \frac{9x + 7x + x}{9x} = \frac{17}{9}, \quad b_2 = \frac{6y + 5y}{6y} = \frac{11}{6}, \quad b_3 = \frac{z + 9z}{z} = 10$$

$$B = \frac{17}{9} \times \frac{11}{6} \times 10 = 34.63$$

iii) Electrical effort:

$$E = H = \frac{C_{out}}{C_{in}} = \frac{65}{25} = 2.6$$

iv) Path effort:

$$F = GBH = 2.37 \times 34.63 \times 2.6 = 213.4$$

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v) Stage effort:

$$\hat{f} = F^{1/4} = 213.4^{1/4} = 3.82$$

vi) Parasitic delay:

$$P = 2 + 2 + 2 + 1 = 7$$

vii) Minimum path delay:

$$\hat{D} = N\hat{f} + P = 4(3.82) + 7 = 22.3\tau$$

$$\boxed{D = 22.3\ \tau\ \ (= 4.5 \text{ FO4 delays, with 1 FO4} = 5\tau)}$$

viii) Sizes, from the load back, using $C_{in} = g\,C_{out}/\hat{f}$:

$$z = \frac{1 \times 65}{3.82} = 17.0$$

The 6y gate drives $z + 9z = 10z = 170$:

$$6y = \frac{(4/3)(170)}{3.82} = 59.3 \ \Rightarrow\ y = 9.89$$

The 9x gate drives $6y + 5y = 11y = 108.8$:

$$9x = \frac{(4/3)(108.8)}{3.82} = 37.9 \ \Rightarrow\ x = 4.22$$

Check at the first gate: it drives $9x + 7x + x = 17x = 71.7$, so $C_{in} = (4/3)(71.7)/3.82 = 25.0$, which is the given input. ✔

$$\boxed{x = 4.22, \quad y = 9.89, \quad z = 17.0}$$

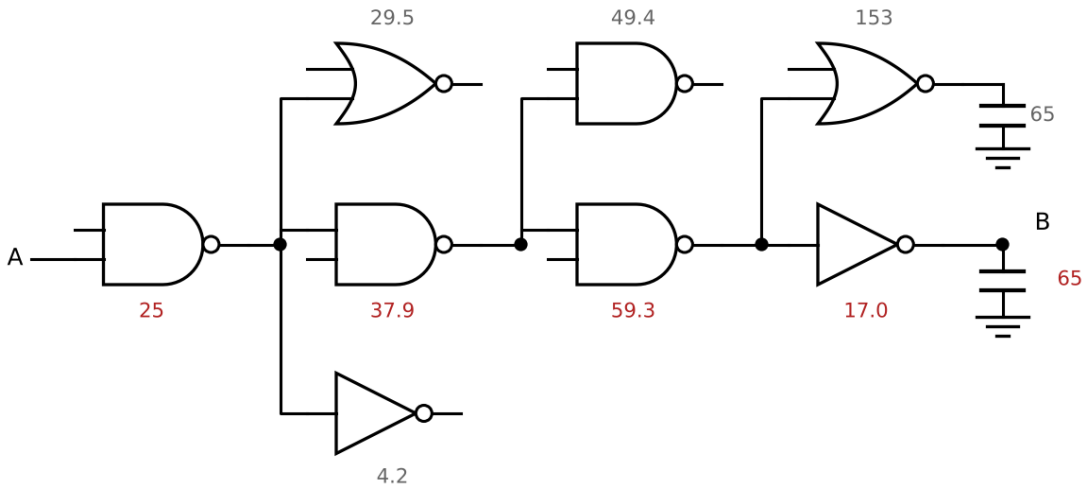


Figure 19: The circuit with the input capacitance of every gate

Reading the figure. The numbers are input capacitances: 37.9 for 9x, 59.3 for 6y, 17.0 for z, and off the path 29.5 (7x), 4.2 (x), 49.4 (5y), 153 (9z).

3. Keyword Explanation

1. Branching effort

How much a node's drive is split: total capacitance on the node divided by the capacitance on the path.

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2. Path effort

$F = GBH$, the total effort of the path. It does not depend on how the sizes are chosen.

4. Notes and Remarks

- The paper writes the electrical effort as E; Weste and Harris call it H.
- The stage electrical efforts are 2.87, 2.87, 2.87 and 3.82; with the g of each stage, all four f equal 3.82.
- The 7x, x and 5y gates have no output drawn; only their input capacitance matters here.
- The path's $B = 34.6$ is large because of the 9z gate: it takes 90 % of the drive at the last branch.

3.(d) Find the number of stages for minimum delay. Marks: 7

1. Theory

1. The question (Core Idea)

A logic path has a given path effort F. We can add inverters to it without changing its logic (an even number keeps the polarity). More stages share the effort, but each adds parasitic delay. There must be a best number. The paper does not name a circuit, so we find the rule, then use it on the F of 3(c).

2. Setting up the delay

Start with n_1 logic stages, with parasitic sum $\sum p_i$. Add $N - n_1$ inverters, each with $g = 1$ and parasitic delay p_{inv}. Inverters do not change F. With equal stage efforts:

$$D(N) = NF^{1/N} + \sum_{i=1}^{n_1} p_i + (N - n_1)\, p_{inv}$$

3. Differentiating, step by step

Treat N as a real number. To differentiate $NF^{1/N}$, use the product rule. For the second factor, $F^{1/N} = e^{(\ln F)/N}$, so

$$\frac{d}{dN} F^{1/N} = F^{1/N} \cdot \left(-\frac{\ln F}{N^2} \right)$$

Then

$$\frac{dD}{dN} = F^{1/N} + NF^{1/N} \left(-\frac{\ln F}{N^2} \right) + p_{inv} = F^{1/N} - \frac{\ln F}{N} F^{1/N} + p_{inv}$$

Let $\rho = F^{1/N}$, the best stage effort. Then $\ln \rho = (\ln F)/N$. Setting $dD/dN = 0$:

$$\rho - \rho \ln \rho + p_{inv} = 0 \;\Rightarrow\; p_{inv} + \rho(1 - \ln \rho) = 0$$

4. Solving for ρ

The equation has no closed form, so it is solved numerically. For $p_{inv} = 0$: $\ln \rho = 1$, so $\rho = e = 2.72$. For $p_{inv} = 1$: try $\rho = 3.59$: $\ln 3.59 = 1.278$, and $1 + 3.59(1 - 1.278) = 1 - 1.0 \approx 0$. ✔ So $\rho = 3.59$. For $p_{inv} = 2$: $\rho = 4.32$.
Designers round this to **4** ("about four per stage").

5. The best number of stages

Since $\rho = F^{1/N}$:

$$\hat{N} = \log_\rho F$$

2. Answer

Derivation. With N stages of which $N - n_1$ are added inverters, each stage bearing $F^{1/N}$:

$$D = NF^{1/N} + \sum_{i=1}^{n_1} p_i + (N - n_1)\, p_{inv}$$

Differentiating with respect to N and setting the result to zero, with $\rho = F^{1/N}$:

$$\frac{dD}{dN} = -F^{1/N} \ln F^{1/N} + F^{1/N} + p_{inv} = 0 \;\Rightarrow\; p_{inv} + \rho(1 - \ln \rho) = 0$$

For $p_{inv} = 1$, the numerical solution is $\rho = 3.59$. The number of stages for minimum delay is

$$\boxed{\hat{N} = \log_\rho F = \log_{3.59} F \approx \log_4 F}$$

For the path of Figure 3c ($F = 213.4$):

$$\hat{N} = \frac{\ln 213.4}{\ln 3.59} = \frac{5.363}{1.278} = 4.2 \;\Rightarrow\; N = 4$$

$\boxed{N = 4 \text{ stages, the number the circuit already has}}$

N	$F^{1/N}$	$D = NF^{1/N} + P$
3	5.98	23.9
4	**3.82**	**22.3**
5	2.92	22.6
6	2.45	23.7
7	2.15	25.1

Table 6: Delay against the number of stages, for F = 213.4 (P = 7 at N = 4, changing by 1 for each inverter)

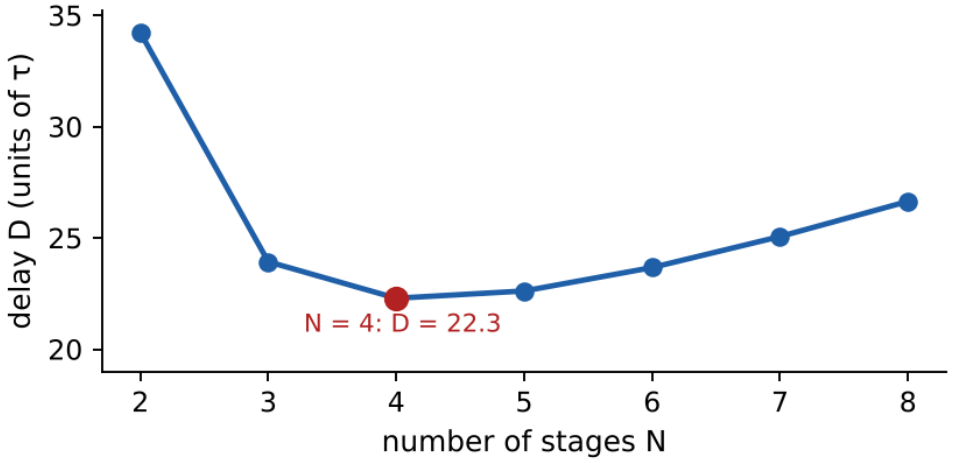


Figure 20: Path delay against the number of stages for the path of Figure 3c

Reading the figure. The delay falls steeply at first, reaches its minimum at $N = 4$ and then rises slowly. Being one stage off costs little, which is why the rule "about 4 per stage" is enough.

3. Keyword Explanation

1. Best stage effort ρ

The value of f that gives minimum delay when stages may be added freely: 3.59 for $p_{inv} = 1$.

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4. Notes and Remarks

- $N = 3$ here is only a comparison, since one fewer stage means removing an inverter, which also inverts the output. To keep polarity, inverters are added or removed in pairs.
- The paper's question is one line and worth 7 marks, so it was taken as the general derivation, then applied to Figure 3c.
- $\log_4 213.4 = 3.87$ also rounds to 4.

5. Final Concise Answer

(a) VHDL is the IEEE-standard language that describes digital hardware for simulation and synthesis. It is portable, supports several levels of description, allows verification before fabrication, shortens design through synthesis, and is reusable.

(b) pMOS series, width 6 each; nMOS parallel, width 1 each. $C_{in} = 7C$, $n_1 = n_2 = 12C$, output $(9 + 7h)C$. $t_{pdr} = (21 + 7h)RC$; $t_{pdf} = (9 + 7h)RC$.

(c) $G = 64/27 = 2.37$; $B = 34.63$; $E = H = 2.6$; $F = 213.4$; $f = 3.82$; $P = 7$; $D = 22.3\tau$; $x = 4.22$, $y = 9.89$, $z = 17.0$.

(d) $p_{inv} + \rho(1 - \ln \rho) = 0$ gives $\rho = 3.59$, so $\hat{N} = \log_{3.59} F$. For $F = 213.4$: $\hat{N} = 4.2$, so 4 stages.

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4. (a) Explain why NMOS network is not used as pull-up network. Marks: 5

(b) What are the sources of power dissipation in CMOS logic circuit? Marks: 5

(c) What is activity factor? Figure 4.1 shows a 4-input AND gate built using a tree and a chain of gates. Determine the activity factors at each node in the circuit assuming the input probabilities PA = PB = PC = PD = 0.5. Marks: 7

(d) A digital system-on-chip in a 1 V 65 nm process (with 50 nm drawn channel lengths and λ = 25 nm) has 1 billion transistors, of which 50 million are in logic gates and the remainder in memory arrays. The average logic transistor width is 12 λ and the average memory transistor width is 4 λ. The memory arrays are divided into banks and only the necessary bank is activated so the memory activity factor is 0.02. The static CMOS logic gates have an average activity factor of 0.1. Assume each transistor contributes 1 fF/Rm [*sic*: should be fF/µm] of gate capacitance and 0.8 fF/µm of diffusion capacitance. Neglect wire capacitance for now (though it could account for a large fraction of total power). **Estimate the switching power when operating at 1 GHz.** If Subthreshold leakage for OFF devices is 100 nA/µm for low-threshold devices and 10 nA/µm for high-threshold devices. Gate leakage is 5 nA/µm. Junction leakage is negligible. Memories use low-leakage devices everywhere. Logic uses low-leakage devices in all but 5% of the paths that are most critical for performance. **Estimate the static power consumption.** Marks: 8

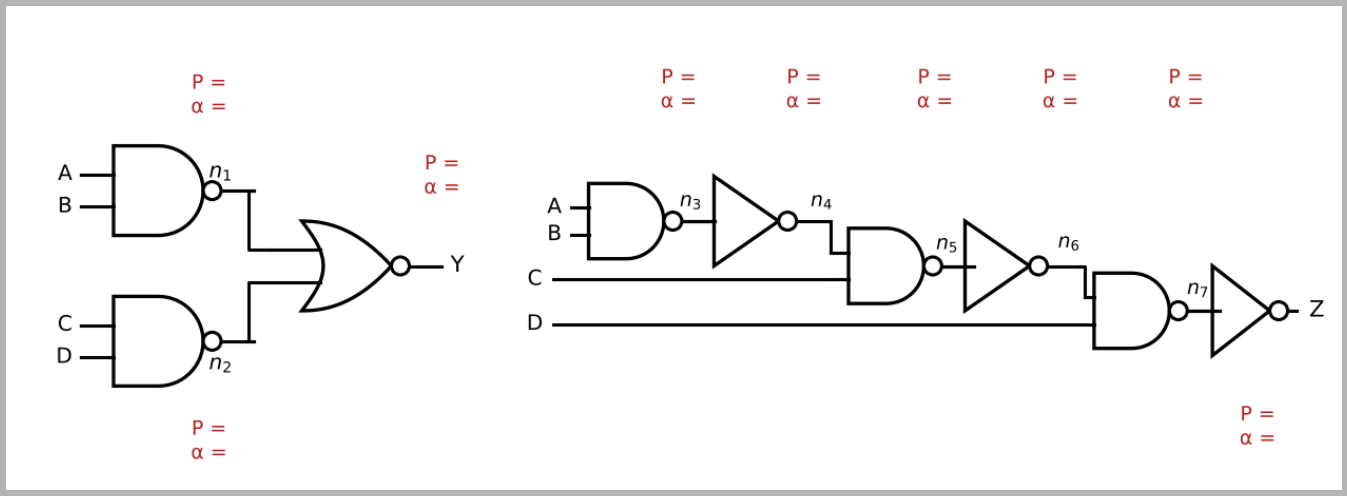


Figure 21: Figure: 4.1 of the paper (redrawn): the 4-input AND as a tree and as a chain

4.(a) Explain why NMOS network is not used as pull-up network. Marks: 5

1. Theory

1. Pull-up and pull-down (Core Idea)

A static CMOS gate has a **pull-down network** from the output to ground and a **pull-up network** from the output to V_{DD}. Exactly one conducts at a time. Static CMOS uses nMOS for pull-down and pMOS for pull-up. Why not the other way round?

2. When an nMOS turns off

An nMOS conducts while its gate is at least V_t above its **source**, and its source is whichever terminal is lower.

- Passing a **0**: the source stays at 0, so $V_{GS} = V_{DD}$ all the way down. The transistor stays strongly ON and the output reaches 0. An nMOS passes a **strong 0**.
- Passing a **1**: the drain is on V_{DD} and the output is the source. As the output rises, $V_{GS} = V_{DD} - V_{out}$ falls. When $V_{out} = V_{DD} - V_t$, $V_{GS} = V_t$ and the transistor turns off. An nMOS passes a **weak 1**.

The pMOS is the mirror: a strong 1 and a weak 0 (it stops at $|V_{tp}|$).

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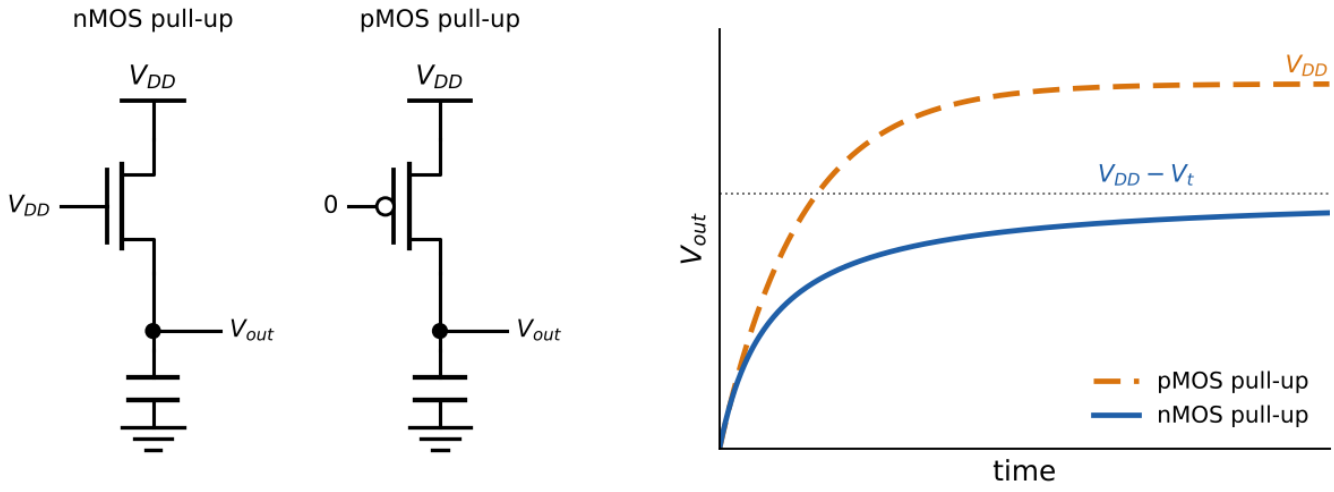


Figure 22: nMOS and pMOS used as a pull-up charging a load, and the output against time

Reading the figure. With the nMOS, the gate is at V_{DD} and the source is the output. The solid blue curve rises more and more slowly and flattens at $V_{DD} - V_t$. With the pMOS, the gate is at 0, the source stays at V_{DD}, so V_{SG} stays large, and the dashed orange curve reaches V_{DD} quickly.

3. The body effect makes it worse

The output, which is the source, is above the body (at ground). So $V_{SB} > 0$, which raises the threshold (see 8(a)). The output stops lower than a simple $V_{DD} - V_{t0}$.

2. Answer

An nMOS turns on only while $V_{GS} > V_t$. Used as a pull-up, its drain is at V_{DD} and its source is the output, so with its gate at V_{DD}:

$$V_{GS} = V_{DD} - V_{out}$$

As the output rises, V_{GS} falls, and the transistor turns off when $V_{GS} = V_t$. So the output stops at

$$V_{OH} = V_{DD} - V_t$$

This is a **degraded 1** (threshold drop). For example, with $V_{DD} = 1$ V and $V_t = 0.3$ V (illustrative) the output stops at 0.7 V.

The results are:

1. **Reduced noise margin:** the output is not a full logic 1, and chained gates may misread it.
2. **Worse with the body effect:** the source is above the body, so V_t rises and the output stops lower.
3. **Slow:** as V_{out} rises, V_{GS} falls, so the current drops, and the last part of the transition is very slow.
4. **Static current in the next gate:** a weak 1 of $V_{DD} - V_t$ at an input leaves the next gate's pMOS slightly ON ($V_{SG} = V_t$), so both of its networks conduct and it draws static current.
5. **Wrong logic form:** a pull-up of nMOS would turn on for high inputs, so the gate could not invert, while a single-stage static CMOS gate is inverting by nature. Both networks would conduct for the same input.

A pMOS has none of these: its source stays at V_{DD}, V_{SG} stays large, and it passes a **strong 1**. So pull-up networks use pMOS, and pull-down networks use nMOS, which passes a strong 0.

3. Keyword Explanation

1. Threshold drop

The loss of V_t when a transistor passes the value it is weak at.

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2. Strong and weak level

A level is strong when the passing transistor stays fully ON while it reaches that level, so the node reaches the rail.

4. Notes and Remarks

- Weste and Harris list threshold drops among the DC effects: nMOS pull up only to $V_{DD} - V_{tn}$, and pMOS pull down only to $|V_{tp}|$.
- The saturated-load inverter of 2(c) is a real example.

4.(b) What are the sources of power dissipation in CMOS logic circuit? Marks: 5

1. Theory

1. Why does a chip use power at all? (Core Idea)

Power is the rate at which energy is taken from the supply: $P(t) = I_{DD}(t)V_{DD}$. A CMOS gate ideally draws no current when idle. Real chips draw power in two ways: when nodes **switch** (dynamic power) and all the time through **leakage** (static power).

2. Energy of one switching event

Charging a load C from 0 to V_{DD}, the supply delivers charge $Q = CV_{DD}$ at voltage V_{DD}:

$$E_{supply} = \int_0^\infty I(t)V_{DD}\,dt = V_{DD}\int_0^\infty C\frac{dV}{dt}\,dt = CV_{DD}^2$$

The capacitor stores only

$$E_C = \int_0^{V_{DD}} CV\,dV = \frac{1}{2}CV_{DD}^2$$

The other half is heat in the pMOS. On the falling edge, the stored half turns to heat in the nMOS.

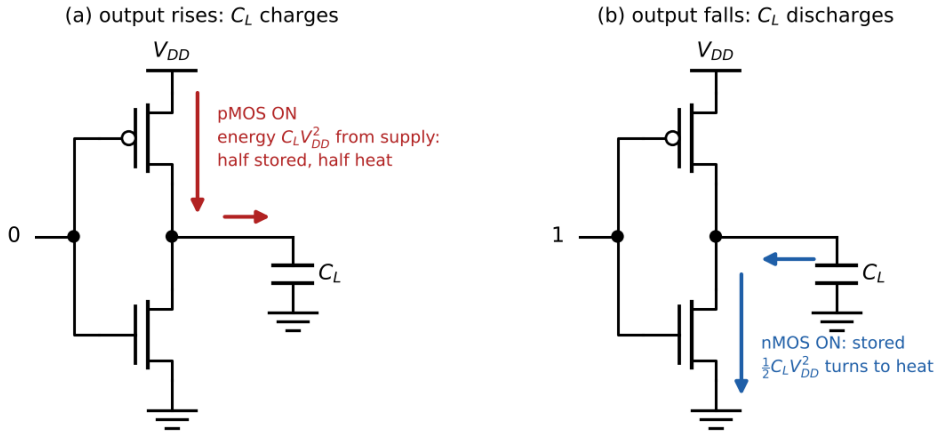


Figure 23: Output rising and falling: energy from the supply is stored and then turned to heat (reused from the 2020-21 paper's solution)

Reading the figure. In (a) the pMOS is ON and the load charges: half the supply's energy is stored and half is heat. In (b) the nMOS is ON and the stored energy goes to heat.

3. From energy to power

If a node makes one rising transition in $1/\alpha$ cycles, it takes αf full cycles per second, each costing CV_{DD}^2:

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$$P_{switching} = \alpha\, C\, V_{DD}^2\, f$$

2. Answer

Total power is dynamic plus static, $P_{total} = P_{dynamic} + P_{static}$.

Dynamic power (only while nodes switch):

1. **Switching power:** charging and discharging the load, gate, diffusion and wire capacitances. It is usually the largest part:

$$P_{switching} = \alpha C V_{DD}^2 f$$

where α is the activity factor, C the switched capacitance, f the clock frequency.

2. **Short-circuit power:** while an input changes, the pMOS and nMOS are both partly ON for a moment, and current flows straight from V_{DD} to ground. It is normally under 10 % of the dynamic power.

Static power (even when nothing switches):

$$P_{static} = (I_{sub} + I_{gate} + I_{junct} + I_{contention})V_{DD}$$

3. **Subthreshold leakage I_{sub}:** current through OFF transistors; exponential in $-V_t$ and rising with temperature.
4. **Gate leakage I_{gate}:** electrons tunnel through a thin gate oxide.
5. **Junction leakage I_{junct}:** reverse-biased source and drain junctions leak to the body, by band-to-band tunnelling too. It is usually small.
6. **Contention current $I_{contention}$:** only in ratioed circuits such as pseudo-nMOS, where pull-up and pull-down fight. Static CMOS has none.

3. Keyword Explanation

1. Activity factor

The probability that a node makes a rising transition in a cycle; see 4(c).

2. Short-circuit current

The brief current through both networks during an input transition.

4. Notes and Remarks

- Weste and Harris estimate short-circuit power by adding 10 % to the switching power.
- Power falls fastest with V_{DD}, because of the square term.
- Static power is estimated in 4(d).

4.(c) What is activity factor? Figure 4.1 shows a 4-input AND gate built using a tree and a chain of gates. Determine the activity factors at each node in the circuit assuming the input probabilities PA = PB = PC = PD = 0.5. Marks: 7

1. Theory

1. What is the activity factor? (Core Idea)

A node burns energy only when it switches. The **activity factor** α of a node is the probability that it goes from 0 to 1 in a cycle. A clock has $\alpha = 1$; random data has $\alpha = 0.25$.

2. How to find it from probabilities

Let P be the probability that the node is 1, so $\overline{P} = 1 - P$ is the probability it is 0. To rise, it must be 0 in one cycle and 1 in the next. If cycles are independent:

$$\alpha = \overline{P}\,P = P(1-P)$$

With $P = 0.5$: $\alpha = 0.25$, the highest possible.

3. Output probability of a gate

For independent inputs:

Gate	P_Y
NAND2	$1 - P_A P_B$
NOR2	$\overline{P_A}\,\overline{P_B}$
AND2	$P_A P_B$
Inverter	$1 - P_A$

Table 7: Output probabilities (Weste and Harris Table 5.1)

Reason for NAND2: the output is 0 only when both inputs are 1. Reason for NOR2: the output is 1 only when both inputs are 0.

2. Answer

Activity factor: the probability that a node makes a 0-to-1 transition in a clock cycle:

$$\alpha = P(1-P)$$

where P is the probability that the node is 1. It sets how often the node's capacitance is charged in $P = \alpha C V_{DD}^2 f$.

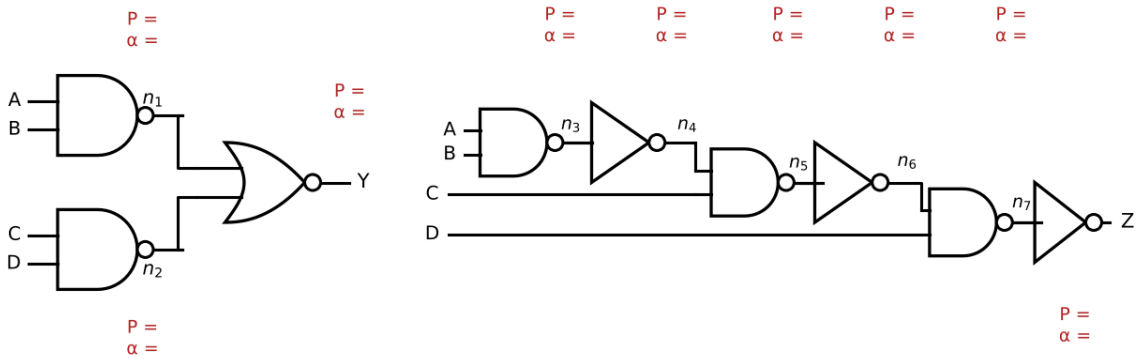


Figure 24: Repeated from the question: Figure 4.1 (redrawn)

Tree (a). n_1 and n_2 are NAND2 outputs of independent inputs:

$$P_{n1} = P_{n2} = 1 - (0.5)(0.5) = \frac{3}{4}, \qquad \alpha = \frac{3}{4}\cdot\frac{1}{4} = \frac{3}{16}$$

The last gate is a NOR (an AND with bubbles on its inputs), so Y = 1 only when $n_1 = n_2 = 0$:

$$P_Y = \frac{1}{4}\cdot\frac{1}{4} = \frac{1}{16}, \qquad \alpha_Y = \frac{1}{16}\cdot\frac{15}{16} = \frac{15}{256}$$

Chain (b).

n_3 is a NAND2 of A and B: $P_{n3} = 1 - (0.5)(0.5) = 3/4$, $\alpha = 3/16$.
n_4 is its inverse: $P_{n4} = 1/4$, $\alpha = 3/16$.
n_5 is a NAND2 of n_4 and C: $P_{n5} = 1 - (1/4)(1/2) = 7/8$, $\alpha = (7/8)(1/8) = 7/64$.
n_6 is its inverse: $P_{n6} = 1/8$, $\alpha = 7/64$.

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n_7 is a NAND2 of n_6 and D: $P_{n7} = 1 - (1/8)(1/2) = 15/16$, $\alpha = (15/16)(1/16) = 15/256$.
Z is its inverse: $P_Z = 1/16$, $\alpha = 15/256$.

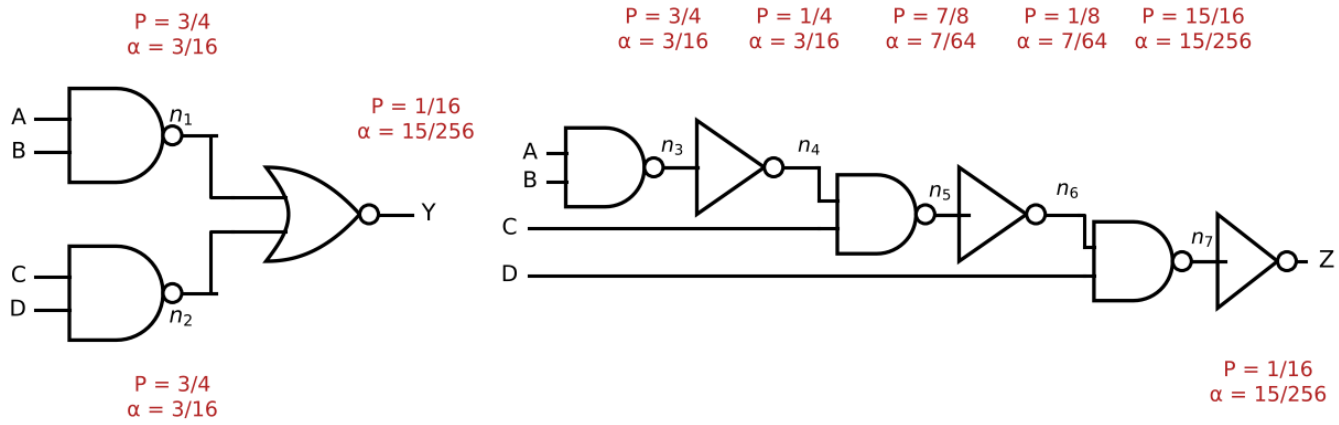


Figure 25: Figure 4.1 with the probability and activity factor of every node

Node	n_1, n_2	Y	n_3	n_4	n_5	n_6	n_7	Z
P	3/4	1/16	3/4	1/4	7/8	1/8	15/16	1/16
α	3/16	15/256	3/16	3/16	7/64	7/64	15/256	15/256

Table 8: Probabilities and activity factors

Interpretation: both outputs have the same α, $15/256 = 0.059$, since both are the same function. The chain's activity falls along the chain, because the later nodes rarely change.

3. Keyword Explanation

1. Switching probability

The probability P that a node is 1 in a given cycle.

4. Notes and Remarks

- These values assume zero gate delay. In a real chain, unequal arrival times cause **glitches** that raise the true activity. Weste and Harris note chains are especially prone to this, as in the change ABCD = 1101 to 0111.
- The paper's figure draws the tree's last gate as an AND with input bubbles, which is a NOR by De Morgan's law; the redrawn figure uses the NOR symbol. In the chain, the paper puts the inverter's bubble at its input; the redrawn figure puts it at the output, and n_3 to n_7 keep their meaning.
- Summing the node activities gives 0.43 for the tree and 0.71 for the chain, counting the nodes in the figure.

4.(d) ⚠ Note on the paper: The unit is printed as "fF/Rm" in the first place, and the other places are overstruck. Every one is the unit fF/µm or nA/µm: the "R" is the µ of the book's example, wrongly copied. The solution uses µm.

1. Theory

1. Switching power from transistor counts (Core Idea)

To estimate power before a design exists, we count transistor **width**. Total switched capacitance is width times capacitance per width. Each transistor's terminals give gate and diffusion capacitance, so $(1 + 0.8)$ fF/µm counts for each transistor.

Memory and logic have different activity, so they are kept apart:

$$P_{switching} = (\alpha_{logic} C_{logic} + \alpha_{mem} C_{mem}) V_{DD}^2 f$$

2. Static power

Leakage scales with width too. In static CMOS each stack has some transistors ON and some OFF. "Neglecting the benefits of series stacks", half the transistors are OFF and half are ON:

- **Subthreshold leakage** flows in **OFF** devices.
- **Gate leakage** flows in **ON** devices, whose channel faces the gate.

So each is multiplied by one half.

3. Which threshold is "low-leakage"?

A high threshold means less subthreshold current. So **low-leakage devices are the high-threshold devices** (10 nA/µm). The fast leaky ones are the low-threshold devices, and they are kept for the 5 % critical paths.

2. Answer

Given:

- $V_{DD} = 1$ V, $f = 1$ GHz $= 10^9$ Hz, $\lambda = 25$ nm $= 0.025$ µm
- Logic: 50×10^6 transistors, width 12λ, $\alpha = 0.1$
- Memory: $(1000 - 50) \times 10^6 = 950 \times 10^6$ transistors, width 4λ, $\alpha = 0.02$
- Capacitance: $1 + 0.8 = 1.8$ fF/µm
- I_{sub}: 100 nA/µm (low V_t), 10 nA/µm (high V_t); I_{gate}: 5 nA/µm

Step 1: Total widths

$$W_{logic} = (50 \times 10^6)(12)(0.025\ \mu\text{m}) = 15 \times 10^6\ \mu\text{m}$$

$$W_{mem} = (950 \times 10^6)(4)(0.025\ \mu\text{m}) = 95 \times 10^6\ \mu\text{m}$$

Step 2: Capacitances

$$C_{logic} = (15 \times 10^6\ \mu\text{m})(1.8\ \text{fF}/\mu\text{m}) = 27\ \text{nF}$$

$$C_{mem} = (95 \times 10^6\ \mu\text{m})(1.8\ \text{fF}/\mu\text{m}) = 171\ \text{nF}$$

Step 3: Switching power

$$P_{switching} = \left[(0.1)(27 \times 10^{-9}) + (0.02)(171 \times 10^{-9})\right](1.0)^2(10^9)$$

$$= (2.7 + 3.42)\ \text{W}$$

$$\boxed{P_{switching} = 6.12\ \text{W} \approx 6.1\ \text{W}}$$

Memory uses more than logic in spite of its low activity, because it holds six times the capacitance.

Step 4: Widths by threshold

Logic uses low-leakage (high-threshold) devices except on 5 % of paths, and memory uses high-threshold devices everywhere.

$$W_{low\,V_t} = 0.05 \times 15 \times 10^6 = 0.75 \times 10^6\ \mu\text{m}$$

$$W_{high\,V_t} = 0.95 \times 15 \times 10^6 + 95 \times 10^6 = 109.25 \times 10^6\ \mu\text{m}$$

Step 5: Subthreshold current

Half the transistors are OFF:

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$$I_{sub} = \frac{1}{2}\left[(0.75\times10^6)(100\text{ nA}) + (109.25\times10^6)(10\text{ nA})\right] = \frac{0.075+1.0925}{2}\text{ A} = 583.75\text{ mA}$$

Step 6: Gate leakage

Half the transistors are ON:

$$I_{gate} = \frac{1}{2}(0.75+109.25)\times10^6\times5\text{ nA} = 275\text{ mA}$$

Step 7: Static power

$$P_{static} = (I_{sub}+I_{gate})V_{DD} = (583.75+275)\text{ mA}\times1\text{ V}$$

$$\boxed{P_{static} = 858.75\text{ mW} \approx 859\text{ mW}}$$

Final Answer: switching power 6.1 W; static power 859 mW, which is 14 % of the switching power.

3. Keyword Explanation

1. Drawn channel length

The channel length on the layout, 50 nm here. The "65 nm process" is the name of the node.

2. λ-based width

Layout sizes given as multiples of λ, which is half the drawn channel length; here $\lambda = 25$ nm.

4. Notes and Remarks

- This is Examples 5.1 and 5.4 of Weste and Harris. The book says "15 %" for the ratio; exact values give 14.0 %.
- The figure ignores wire capacitance, which could add a large share, and short-circuit power, which adds about 10 %.
- The paper's "1 billion" is taken as 10^9, so the memory holds 950 million transistors.

5. Final Concise Answer

(a) An nMOS pull-up has its source at the output, so $V_{GS} = V_{DD} - V_{out}$ and it turns off at $V_{out} = V_{DD} - V_t$: a degraded 1, made worse by the body effect, slow, and it leaves static current in the next gate. A pMOS passes a strong 1.

(b) Dynamic: switching power $\alpha C V_{DD}^2 f$ and short-circuit power. Static: subthreshold, gate and junction leakage, and contention current in ratioed circuits.

(c) $\alpha = P(1-P)$. Tree: n_1, n_2: 3/16; Y: 15/256. Chain: n_3, n_4: 3/16; n_5, n_6: 7/64; n_7, Z: 15/256.

(d) $C_{logic} = 27$ nF, $C_{mem} = 171$ nF, $P_{switching} = 6.1$ W. $I_{sub} = 583.75$ mA, $I_{gate} = 275$ mA, $P_{static} = 859$ mW.

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5. (a) Draw the stick diagram of the following Boolean function: $f = \overline{A.B.C}$ with their symbolic diagram and truth table. Marks: 8
   (b) Write down the advantages of CMOS. Marks: 5
   (c) Explain max-delay constraint and min-delay constraint in sequential circuits. Marks: 6
   (d) Sketch logic circuit of flip-flop with synchronous reset and explain its operation. Marks: 6

5.(a) Draw the stick diagram of the following Boolean function: $f = \overline{A.B.C}$ with their symbolic diagram and truth table. Marks: 8

1. Theory

1. Building the gate (Core Idea)

A static CMOS gate has an inverted output, and $f = \overline{ABC}$ is already inverted, so a single gate does it: a 3-input NAND.

- **Pull-down (nMOS):** $f = 0$ only when $A = B = C = 1$. AND means **series**: three nMOS in series from the output to ground.
- **Pull-up (pMOS):** the dual network. Series becomes **parallel**: three pMOS in parallel from V_{DD} to the output. Any one input at 0 pulls the output high.

For unit drive, the nMOS each have width 3 (three in series give $3 \times R/3 = R$), and the pMOS each have width 2 (one alone gives R).

2. What is a stick diagram?

A **stick diagram** is a simplified layout: each layer is a coloured line, with no exact widths.

- Yellow (solid): p-diffusion. Green (dashed): n-diffusion.
- Red (dash-dot): polysilicon. A transistor exists wherever polysilicon crosses diffusion.
- Blue (solid, thin): metal 1. An X is a contact.

The neatest layout has one unbroken strip of p-diffusion and one of n-diffusion, with each input's polysilicon running vertically through both. That needs an order of the inputs in which the transistors form a chain (an **Euler path**) in both networks.

3. The Euler path

For the order A, B, C:

- **n-strip:** GND –A– n_1 –B– n_2 –C– f. This is the series chain itself.
- **p-strip:** V_{DD} –A– f –B– V_{DD} –C– f. A lies between V_{DD} and f; B between f and V_{DD}; C between V_{DD} and f. So all three are in parallel, with shared diffusion.

2. Answer

Symbolic diagram: three parallel pMOS and three series nMOS.

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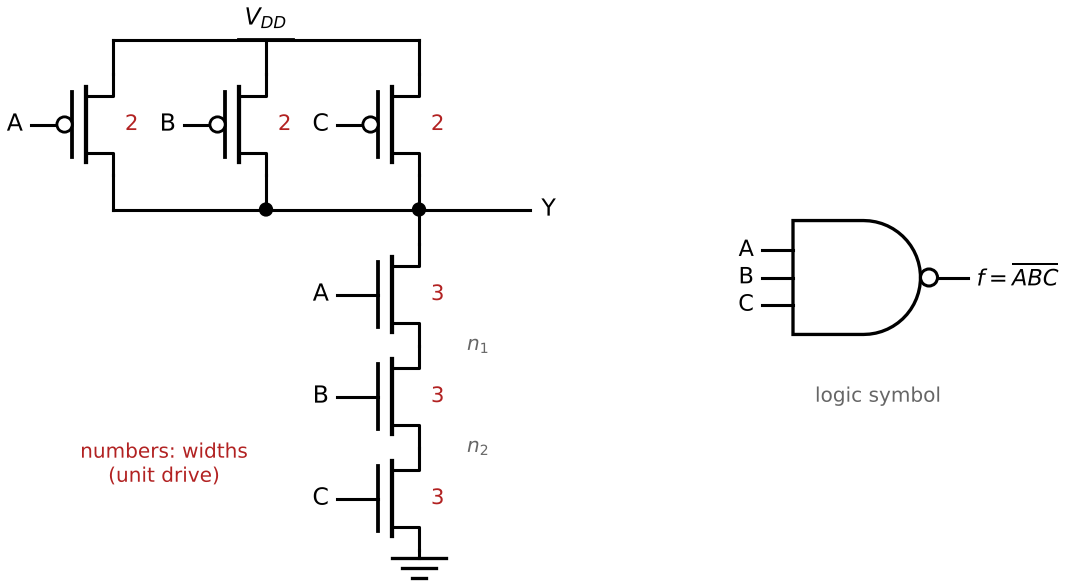


Figure 26: Symbolic (transistor) diagram of the 3-input NAND, with its logic symbol

Truth table:

A	**B**	**C**	$A.B.C$	$f = \overline{A.B.C}$
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0

Table 9: Truth table of the 3-input NAND

Stick diagram (gate order A, B, C on both strips):

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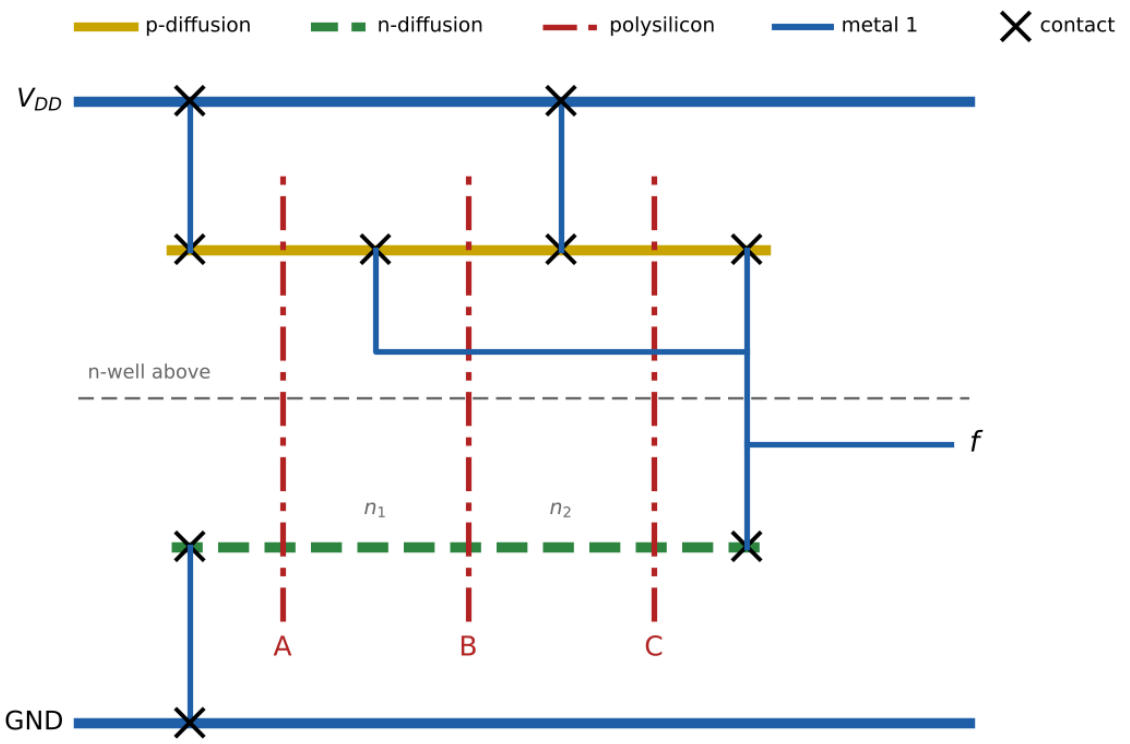


Figure 27: Stick diagram of the 3-input NAND

Reading the stick diagram. Three vertical polysilicon lines A, B, C cross both strips, making six transistors. On the p-strip the contacts from left to right are V_{DD}, f, V_{DD}, f. On the n-strip, the left end goes to GND and the right end is f. The nodes n_1 and n_2 need no contact. Metal joins the two f contacts of the p-strip and the f contact of the n-strip, and carries the output to the right.

3. Keyword Explanation

1. Dual network

The pMOS network made from the nMOS network by swapping series and parallel connections; it conducts exactly when the nMOS network does not.

2. Euler path

A path that uses every edge (transistor) of a graph exactly once; a common Euler path in the nMOS and pMOS graphs gives unbroken diffusion strips.

4. Notes and Remarks

- The input order changes delay: the input nearest the output is usually the fastest.

5.(b) Write down the advantages of CMOS. Marks: 5

1. Theory

1. What makes CMOS special? (Core Idea)

CMOS uses **complementary** pairs: for each logic input one pMOS and one nMOS. In steady state exactly one of the two networks conducts, so there is **no path from V_{DD} to ground**. Compare the saturated-load inverter of 2(c), which draws current whenever its output is low and never reaches a full 1.

2. Answer

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1. **Very low static power:** with one network always off, ideally no DC current flows; power is used mainly when switching.
2. **Full rail-to-rail swing:** $V_{OH} = V_{DD}$ and $V_{OL} = 0$, because pMOS pass strong 1s and nMOS strong 0s.
3. **Ratioless logic:** the output levels do not depend on the transistor sizes, so minimum sizes can be used.
4. **High noise margins**, close to $V_{DD}/2$ for a symmetric inverter, and a sharp transfer curve.
5. **High input impedance:** the gate draws no DC current, so fan-out is large (limited only by delay).
6. **High integration density:** low power per gate allows billions of transistors on a chip, with scaling.
7. **Good supply range and robustness:** works at low supply voltages, and is stable with temperature and process changes.

3. Keyword Explanation

1. Ratioless

A circuit whose logic levels do not depend on the relative sizes of its transistors.

4. Notes and Remarks

- Costs in return: about twice the transistors of an nMOS gate, more process steps (n-well), and a risk of latch-up.

5.(c) Explain max-delay constraint and min-delay constraint in sequential circuits. Marks: 6

1. Theory

1. Sequential circuits and the terms (Core Idea)

A synchronous circuit passes data from flip-flop to flip-flop through combinational logic, once per clock period T_c. Data has to arrive **in time** for the next edge, but **not so early** that it overruns the edge that launched it.
The times involved:

- t_{pcq}: clock-to-Q **propagation** delay, the longest time from the clock edge to a settled Q.
- t_{ccq}: clock-to-Q **contamination** delay, the shortest time before Q may start changing.
- t_{pd}: longest delay through the logic. t_{cd}: shortest delay through the logic.
- t_{setup}: D must be stable this long *before* the edge.
- t_{hold}: D must stay stable this long *after* the edge.

2. Why both are needed

Take two flip-flops F1 and F2 on the same clock.

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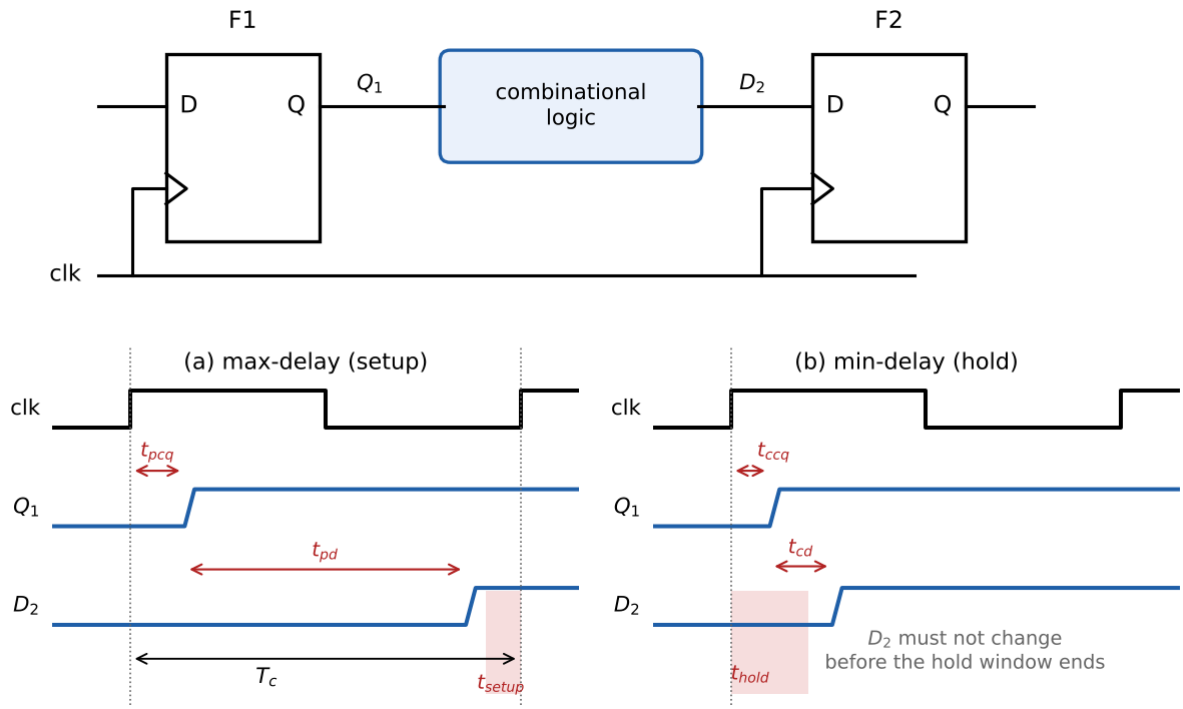


Figure 28: Two flip-flops with logic between them, and (a) the max-delay and (b) the min-delay timing

Reading the figure. In (a), data leaves F1 t_{pcq} after the edge and takes t_{pd} more to reach D_2. It must arrive before the shaded setup window of the **next** edge. In (b), look at the *same* edge at both flip-flops. The new data may start to change Q_1 after t_{ccq} and D_2 after t_{cd} more. That must come after F2's shaded hold window, or F2 captures the new data one cycle too early.

2. Answer

Max-delay (setup) constraint. Data launched by one clock edge must reach the next flip-flop and settle t_{setup} before the **next** edge. Adding up along the slowest path:

$$T_c \geq t_{pcq} + t_{pd} + t_{setup}$$

Or, for the logic:

$$t_{pd} \leq T_c - \underbrace{(t_{pcq} + t_{setup})}_{\text{sequencing overhead}}$$

If it fails (a **setup-time** or **max-delay failure**), the flip-flop captures a wrong value. It can be fixed by making the logic faster or by **slowing the clock**.

Min-delay (hold) constraint. New data launched by a clock edge must not reach the next flip-flop until t_{hold} after the *same* edge. Adding up along the fastest path:

$$t_{ccq} + t_{cd} \geq t_{hold}$$

Or, for the logic:

$$t_{cd} \geq t_{hold} - t_{ccq}$$

If it fails (a **hold-time** or **min-delay failure**, also called a race), data runs through two flip-flops on one edge. The clock period is not in this inequality, so it **cannot** be fixed by slowing the clock: the logic must be redesigned, for example by adding buffers. So designers are very careful with it, since it means refabricating the chip.

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Example

(Weste and Harris's flip-flop, Examples 10.1 and 10.3): $t_{pcq} = 90$ ps, $t_{setup} = 62$ ps, $t_{pd} = 1000$ ps give $T_c \geq 90 + 1000 + 62 = 1152$ ps. With $t_{hold} = -10$ ps and $t_{ccq} = 75$ ps, $t_{cd} \geq -85$ ps, so hold is safe even with no logic between the flip-flops.

3. Keyword Explanation

1. Contamination delay

The shortest time after which an output may start to change.

2. Sequencing overhead

The part of the cycle used by the flip-flops, not by logic: $t_{pcq} + t_{setup}$.

4. Notes and Remarks

- With clock skew t_{skew}, the constraints become $T_c \geq t_{pcq} + t_{pd} + t_{setup} + t_{skew}$ and $t_{cd} \geq t_{hold} - t_{ccq} + t_{skew}$. Skew tightens both.
- Back-to-back flip-flops are safe only if $t_{ccq} \geq t_{hold}$.

5.(d) Sketch logic circuit of flip-flop with synchronous reset and explain its operation. Marks: 6

1. Theory

1. Latches and flip-flops (Core Idea)

A **latch** is transparent (its output follows its input) while its clock is at one level, and holds its value at the other level. A **flip-flop** is two latches in a row, a **master** and a **slave**, on opposite clock phases. For a positive-edge flip-flop, the master is transparent while clk = 0 and the slave while clk = 1. At the rising edge the master closes on D and the slave opens, so Q changes only at the rising edge.

2. What a reset does

Reset forces the stored value to 0, for example at power-up, so that the chip starts from a known state.

- **Synchronous reset:** reset takes effect only at a clock edge. It is just a change to the D input.
- **Asynchronous reset:** reset forces Q to 0 at once, whatever the clock does. It must go inside the latch loops.

For synchronous reset, we want the input to the flip-flop to be

$$D_{in} = D \cdot \overline{reset}$$

When reset = 1 the flip-flop sees 0; when reset = 0 it sees D. One AND gate does it.

2. Answer

Circuit: an AND gate ahead of a positive-edge master-slave flip-flop built of transmission gates.

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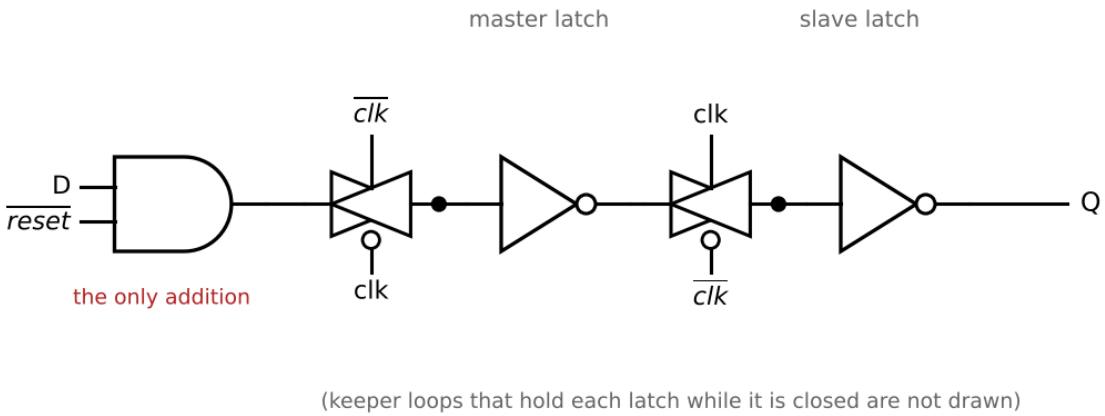


Figure 29: Flip-flop with synchronous reset: the AND gate on D, then the master and slave latches

Operation:

- **reset = 0:** $\overline{reset} = 1$, so the AND gate passes D. The flip-flop behaves normally.
- **clk = 0:** the first transmission gate is ON, so the master follows $D \cdot \overline{reset}$ and the slave gate is OFF, so Q holds.
- **clk rises:** the master gate turns OFF and holds the value; the slave gate turns ON, and the value goes to Q. So on each rising edge $Q \leftarrow D \cdot \overline{reset}$.
- **reset = 1:** the AND output is 0 whatever D is. Nothing happens at once; at the next rising edge the 0 reaches Q. So reset acts only **with the clock**.

$$Q_{next} = D \cdot \overline{reset} \quad \text{(at the rising edge)}$$

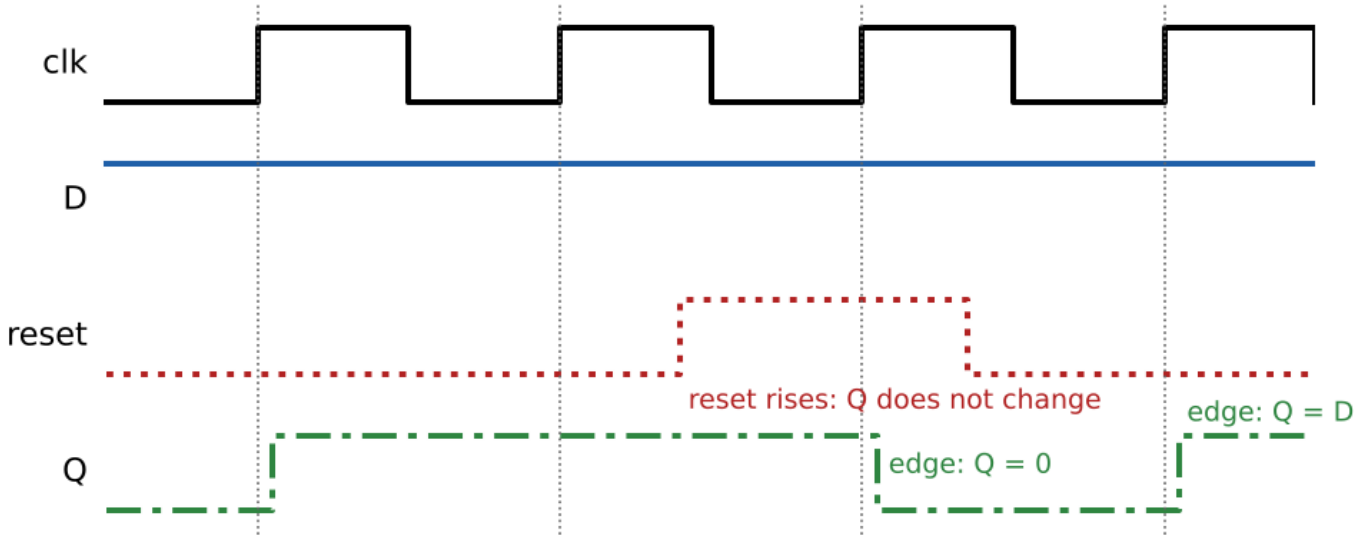


Figure 30: Timing with D = 1 throughout: Q changes only at rising clock edges; reset takes effect at the next edge

Reading the timing diagram. D stays 1. At the first edge Q becomes 1. Reset rises between edges and Q does not change. At the next edge Q falls to 0, because reset is still high. Reset falls before the last edge, and at that edge Q takes D = 1 again.

Features: the reset must be stable around the edge like any data input (it has setup and hold times). A clock must be running for reset to work. The cost is one small gate at the input, and no change inside the latches.

3. Keyword Explanation

1. Transmission gate

An nMOS and a pMOS in parallel, driven by complementary signals; a switch that passes both 0 and 1 well.

2. Master-slave flip-flop

Two latches in series on opposite clock phases, forming an edge-triggered storage element.

4. Notes and Remarks

- Weste and Harris (Figure 10.24) draw resettable flip-flops at transistor level, with the reset gate inside the input stage; the logic is the same.
- The asynchronous form was solved in 4(b) of the 2020-21 paper, as extra reading.
- In VHDL, synchronous reset means the test `if reset = '1'` sits **inside** the `clk'event` branch.

5. Final Concise Answer

(a) $f = \overline{ABC}$ is a NAND3: three parallel pMOS (width 2) and three series nMOS (width 3). $f = 0$ only for A = B = C = 1. Stick diagram with gate order A, B, C: p-strip V_{DD}, f, V_{DD}, f; n-strip GND, n_1, n_2, f.
(b) Low static power, full swing, ratioless, high noise margins, high input impedance and fan-out, high density, wide supply range.
(c) Max-delay: $T_c \geq t_{pcq} + t_{pd} + t_{setup}$; a failure is fixed by a slower clock. Min-delay: $t_{cd} \geq t_{hold} - t_{ccq}$; a failure needs a redesign.
(d) AND gate on D with $\overline{reset}$, then a master-slave flip-flop: $Q_{next} = D \cdot \overline{reset}$ at the rising edge, so reset acts only at a clock edge.

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6. (a) Explain the principle of dynamic inverter. Marks: 5
   (b) Design and implement full adder logic circuit. Marks: 6
   (c) Define PLA and PAL. Sketch a 3-input and 2-output PLA implementing the following logic expressions- $s = a\bar{b}\,\bar{c} + \bar{a}b\bar{c} + \bar{a}\,\bar{b}c + abc$, $c = ab + bc + ac$ Marks: 7
   (d) Explain how data is written and read in a DRAM sub-array. Marks: 7

6.(a) Explain the principle of dynamic inverter. Marks: 5

1. Theory

1. Why dynamic? (Core Idea)

A static CMOS gate has a pull-down network of nMOS and a matching pull-up network of pMOS. The pMOS is slow and adds input capacitance. A **dynamic** gate keeps only the nMOS network and replaces the whole pull-up with **one clocked pMOS**. It stores the output as charge on the output capacitance. The value lasts only a short time, so the gate works in two clock phases. A **dynamic inverter** is the simplest case, one input A.

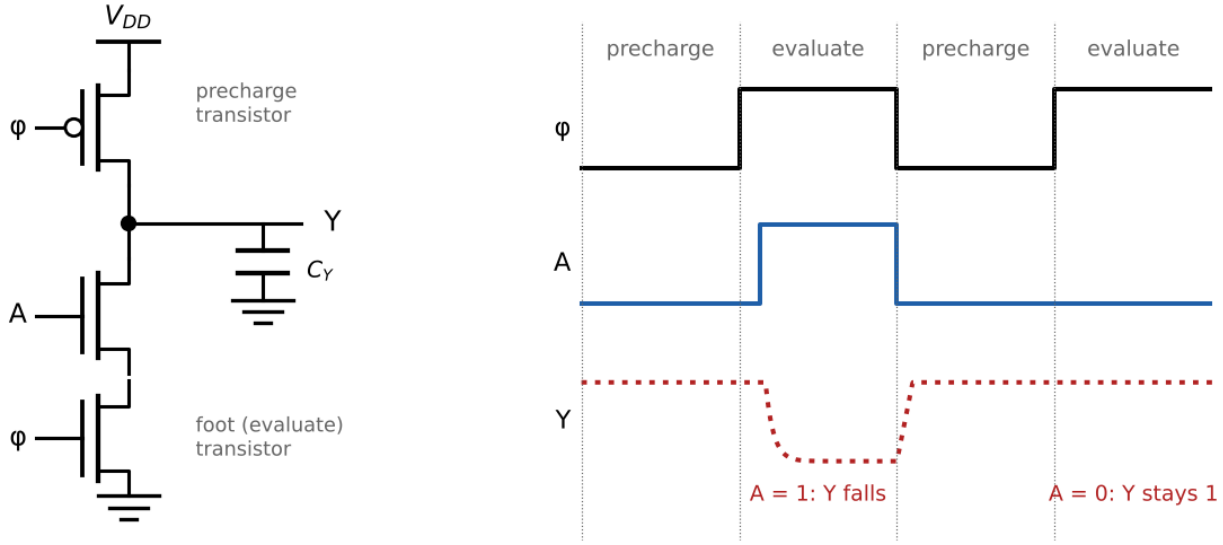


Figure 31: (a) A footed dynamic inverter; (b) its waveforms over two cycles

Reading the figure. While φ is low (precharge), the pMOS charges Y to V_{DD} and the foot transistor keeps the path to ground open. When φ goes high (evaluate), the pMOS turns off and the foot turns on. In the first cycle A = 1, so Y falls to 0. In the second cycle A = 0, there is no path to ground, and Y keeps its precharged 1 on C_Y.

2. Answer

A dynamic inverter has a clocked pMOS (precharge transistor) from V_{DD} to the output Y, and an nMOS with input A from Y to ground, usually in series with a clocked nMOS (foot). It works in two phases of the clock φ:

1. **Precharge ($\varphi = 0$):** the pMOS is ON and charges Y to V_{DD}. The foot is OFF, so no current flows, whatever A is.
2. **Evaluate ($\varphi = 1$):** the pMOS is OFF and the foot is ON. If A = 1, the nMOS discharges Y to 0. If A = 0, there is no path, and Y **stays high**, held only by the charge on C_Y.

So during evaluation $Y = \overline{A}$.

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Without the foot, A = 1 during precharge would turn both pMOS and nMOS on (contention), so the foot is added when A cannot be guaranteed low then.

Advantages: lower input capacitance (A drives only an nMOS), so it is fast; no contention during switching; zero static power; full swing.

Limitations:

- **Monotonicity:** in evaluate, inputs may only rise. If A starts at 1 and then falls, Y has already discharged and cannot recover until the next precharge.
- **Leakage:** the high output floats, so leakage slowly discharges it, and the clock cannot be too slow. A weak **keeper** pMOS holds it.
- **Charge sharing** with internal nodes, in larger gates.
- **Clock power and noise:** the node may switch every cycle (α near 1), and the floating output is sensitive to noise.

3. Keyword Explanation

1. Foot transistor

The clocked nMOS at the bottom of the network that blocks the path to ground during precharge.

2. Keeper

A small pMOS, controlled through an inverter on the output, that replaces charge lost by leakage and keeps a high output high.

4. Notes and Remarks

- Domino logic adds a static inverter after each dynamic gate so that the outputs rise monotonically; see Weste and Harris Section 9.2.4.
- Weste and Harris make the precharge transistor weak (twice unit resistance), which lowers the clock load.

6.(b) Design and implement full adder logic circuit. Marks: 6

1. Theory

1. What does a full adder do? (Core Idea)

A **full adder** adds three bits: A, B and the carry in C_{in}. The result is a value 0 to 3, written with two bits: the **sum** S and the **carry out** C_{out}.

2. Getting the equations

From the truth table in the Answer:

Sum. S = 1 when an odd number of inputs are 1:

$$S = \overline{A}\,\overline{B}C_{in} + \overline{A}B\overline{C_{in}} + A\overline{B}\,\overline{C_{in}} + ABC_{in}$$

Group the first two terms on $\overline{A}$ and the others on A:

$$S = \overline{A}(\overline{B}C_{in} + B\overline{C_{in}}) + A(\overline{B}\,\overline{C_{in}} + BC_{in}) = \overline{A}(B \oplus C_{in}) + A\overline{(B \oplus C_{in})} = A \oplus B \oplus C_{in}$$

Carry. $C_{out} = 1$ when at least two inputs are 1 (the **majority** function):

$$C_{out} = AB + BC_{in} + AC_{in}$$

Rewritten for the two half adders, since either AB gives a carry or exactly one of A, B does and $C_{in} = 1$:

$$C_{out} = AB + C_{in}(A \oplus B)$$

3. The mirror adder

Direct CMOS needs one gate per function. The trick is that S can reuse $\overline{C_{out}}$. If there is a carry, S = 1 only with all three inputs 1. If no carry, S = 1 with any input 1:

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$$S = ABC_{in} + (A + B + C_{in})\overline{C_{out}}$$

Both networks of each stage have the same shape, so it is called the **mirror adder**.

2. Answer

Truth table:

A	B	C_{in}	C_{out}	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Table 10: Truth table of the full adder

Design equations:

$$S = A \oplus B \oplus C_{in}, \qquad C_{out} = AB + C_{in}(A \oplus B) = AB + BC_{in} + AC_{in}$$

Logic implementation: two half adders and an OR gate.

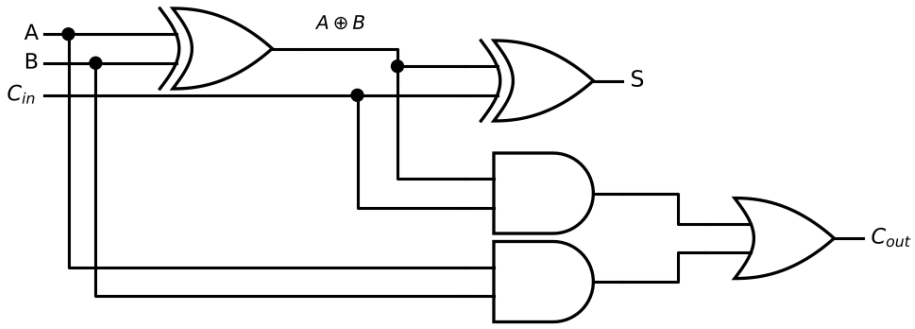


Figure 32: Full adder from two XOR gates, two AND gates and an OR gate

CMOS implementation (mirror adder): the carry stage makes $\overline{C_{out}}$; the sum stage uses it, from $S = ABC_{in} + (A + B + C_{in})\overline{C_{out}}$. C in the figure is C_{in}.

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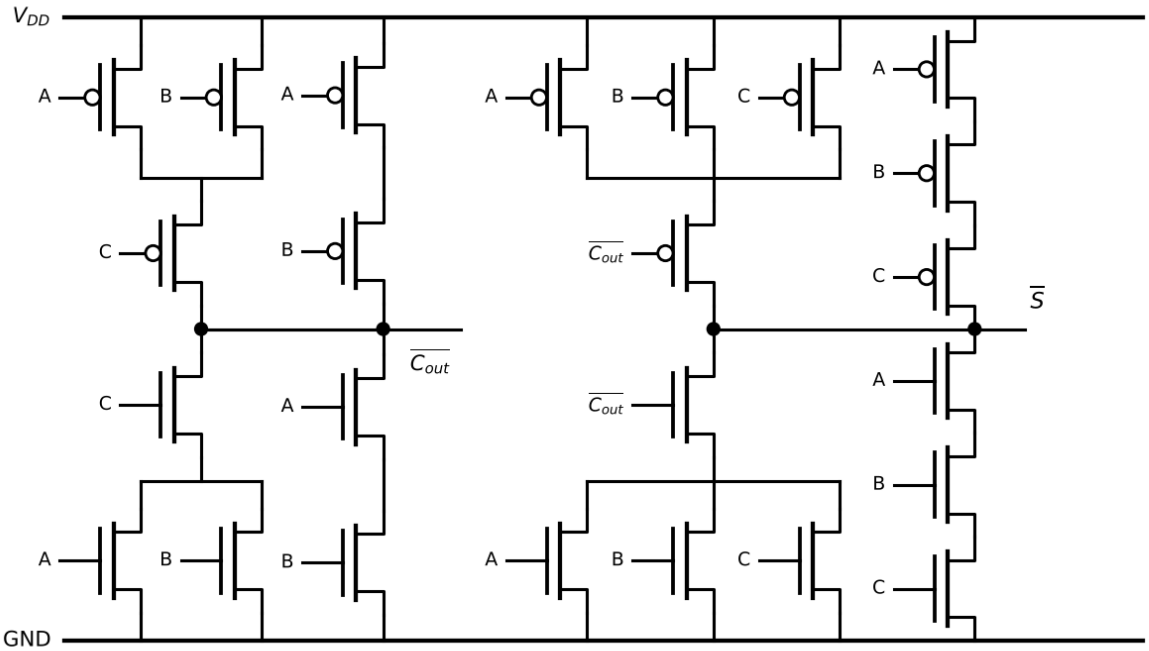


Figure 33: Mirror adder: carry stage (left) and sum stage (right)

Reading the figure. In the left stage, the lower half pulls $\overline{C_{out}}$ low when $C(A+B)+AB=1$, which is the carry. The upper half has the same shape in pMOS. In the right stage, the lower half pulls $\overline{S}$ low when $(A+B+C)\overline{C_{out}}+ABC=1$. The stages have 10 and 14 transistors. Two output inverters (not drawn) turn $\overline{C_{out}}$ and $\overline{S}$ into C_{out} and S, which brings the total to 28.

3. Keyword Explanation

1. Half adder

Adds two bits: sum $A \oplus B$ and carry AB.

2. Majority function

A function that is 1 when most of its inputs are 1; the carry is the 3-input majority.

4. Notes and Remarks

- Weste and Harris (Section 11.2.1) give the same mirror adder: its pMOS network is identical to the nMOS network instead of being the conduction complement, which is possible because addition is symmetric (complementing all inputs complements the output). A plain gate and XOR version takes 32 transistors.
- The sum needs $\overline{C_{out}}$, so the sum is one gate later than the carry.
- 8(c) builds the same carry function from a term test question, and 6(c) makes the adder in a PLA.

6.(c) Define PLA and PAL. Sketch a 3-input and 2-output PLA implementing the following logic expressions- $s = a\bar{b}\,\bar{c} + \bar{a}b\bar{c} + \bar{a}\,\bar{b}c + abc$, $c = ab + bc + ac$ **Marks: 7**

1. Theory

1. Two-level logic (Core Idea)

Any logic function can be written as a **sum of products**: an OR of AND terms. So a general block needs a set of AND gates feeding a set of OR gates. Programmable logic builds both as regular grids.

- The **AND plane** has one vertical line for each input and its complement, and one horizontal **product line** per product term.
- The **OR plane** has one vertical line per output, each joining the product lines it needs.

A dot at a crossing is a connection (in silicon, a transistor).

2. Why the two names?

The difference is which plane can be chosen.

Device	AND plane	OR plane
ROM	fixed (all minterms)	programmable
PAL	programmable	fixed
PLA	programmable	programmable

Table 11: ROM, PAL and PLA

A PAL is simpler and faster, but each output has a fixed set of product terms. A PLA can share product terms between outputs.

3. Counting the product lines

The expressions have 4 product terms for s and 3 for c, with none shared: 7 product lines. Three inputs give 6 vertical lines. Note that the c here is the output, while $\overline{c}$ and c on the input lines are the third input.

2. Answer

PLA (programmable logic array): a programmable logic device with both a programmable AND plane and a programmable OR plane, which implements sum-of-products functions and can share product terms among outputs.

PAL (programmable array logic): a programmable logic device with a programmable AND plane and a **fixed** OR plane; each output has a fixed group of product terms.

PLA with 3 inputs, 2 outputs (s and c; this is a full adder, the example in Weste and Harris Figure 12.75):

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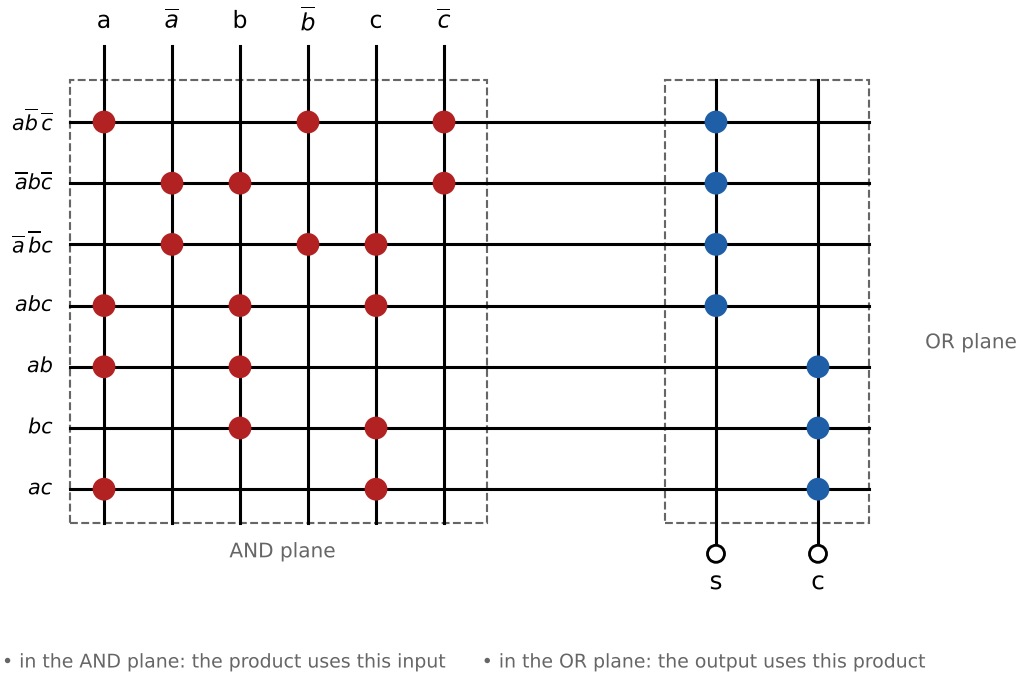


Figure 34: PLA for s and c: the dots in the left box program the AND plane, those in the right box the OR plane

Product line	Inputs joined	Output
$a\overline{b}\,\overline{c}$	a, $\overline{b}$, $\overline{c}$	s
$\overline{a}b\overline{c}$	$\overline{a}$, b, $\overline{c}$	s
$\overline{a}\,\overline{b}c$	$\overline{a}$, $\overline{b}$, c	s
abc	a, b, c	s
ab	a, b	c
bc	b, c	c

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ac	a, c	c

Table 12: Programming of the PLA

Reading the figure. Each horizontal product line is high only when all inputs with a dot in the AND plane are high: that is the AND. Each output column is high when any product line with a dot in the OR plane is high: that is the OR.

3. Keyword Explanation

1. Product line

A horizontal line of the AND plane, carrying one product term.

2. Pseudo-nMOS PLA

A PLA built from wide NOR gates with always-on pMOS pull-ups: small but with static power.

4. Notes and Remarks

- In silicon, both planes are usually NOR gates, since NOR of inverted inputs is an AND (Weste and Harris Figure 12.76).
- Also see the XOR PLA in 6(e) of the 2020-21 paper, as extra reading.

6.(d) Explain how data is written and read in a DRAM sub-array. Marks: 7

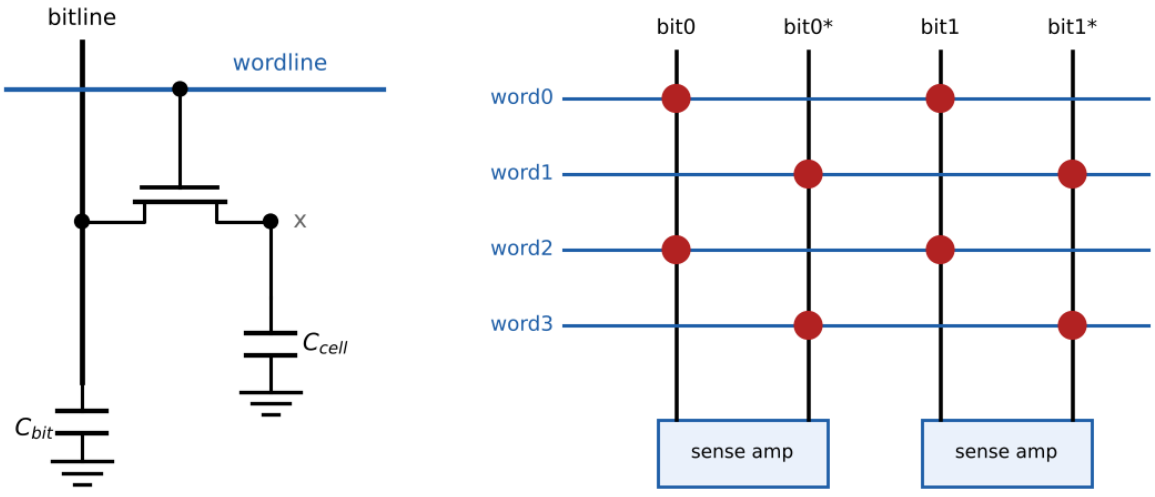
1. Theory

1. The 1T cell (Core Idea)

A **DRAM** stores a bit as charge on a tiny capacitor. Each cell is one nMOS **access transistor** and one capacitor C_{cell}: a charged capacitor is a 1, an empty one a 0. The **wordline** turns the transistor on, and the **bitline** carries the data. The cell is far smaller than a 6T SRAM cell, but charge leaks, so each cell must be read and rewritten every few milliseconds: this is **refresh**.

2. The sub-array

A DRAM is divided into **sub-arrays**, typically 256 words by 512 bits. A wordline drives one row, and all cells on that row connect to their bitlines. Many cells share a bitline, so the bitline capacitance C_{bit} is about ten times C_{cell}.



(a) 1T cell

(b) folded subarray: each dot is one cell

Figure 35: (a) A 1T cell; (b) a folded-bitline sub-array, in which each sense amplifier compares two neighbouring bitlines

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Reading the figure. In (a), raising the wordline joins C_{cell} to the bitline. In (b), each sense amplifier has a pair of neighbouring bitlines. A wordline connects a cell to only one of the pair, so the other serves as a quiet reference.

3. Reading is charge sharing

The bitline is first precharged to $V_{DD}/2$ and left floating. When the wordline rises, charge spreads between the cell at V_{cell} and the bitline at $V_{DD}/2$. Charge is conserved:

$$C_{cell}V_{cell} + C_{bit}\frac{V_{DD}}{2} = (C_{cell} + C_{bit})V_{final}$$

Solve for the change $\Delta V = V_{final} - V_{DD}/2$:

$$\Delta V = \frac{C_{cell}}{C_{cell} + C_{bit}}\left(V_{cell} - \frac{V_{DD}}{2}\right) = \pm\frac{V_{DD}}{2}\frac{C_{cell}}{C_{cell} + C_{bit}}$$

The sign is + for a stored 1 ($V_{cell} = V_{DD}$) and − for a stored 0. With $C_{cell} = 30$ fF, $C_{bit} = 300$ fF and $V_{DD} = 1$ V (illustrative): $\Delta V = \pm 45$ mV. This is small, so a **sense amplifier** is needed.

4. The sense amplifier

It is two cross-coupled inverters between bit and bit*, with its supply terminals pulled apart. Starting from a small difference, positive feedback drives the higher line to V_{DD} and the lower to 0.

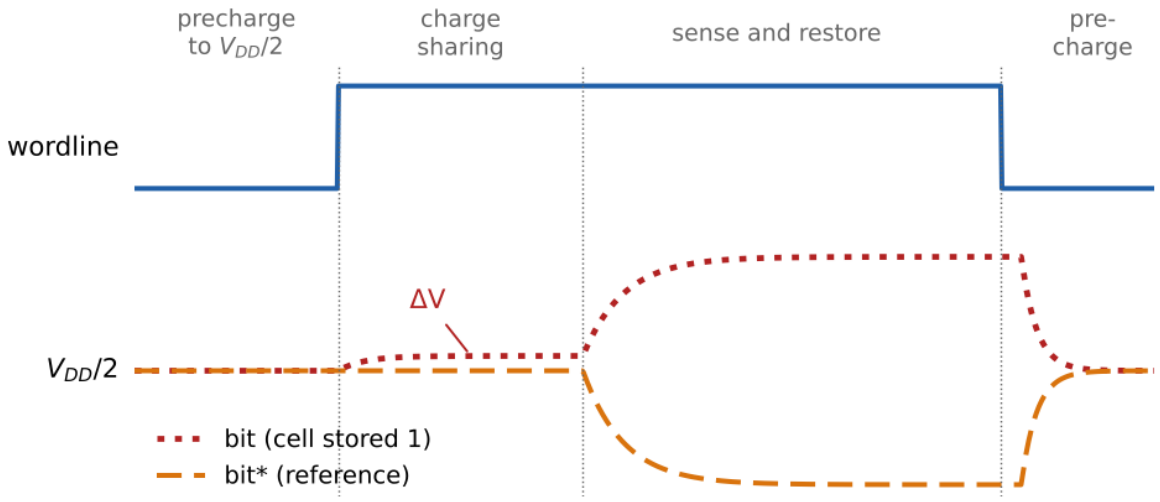


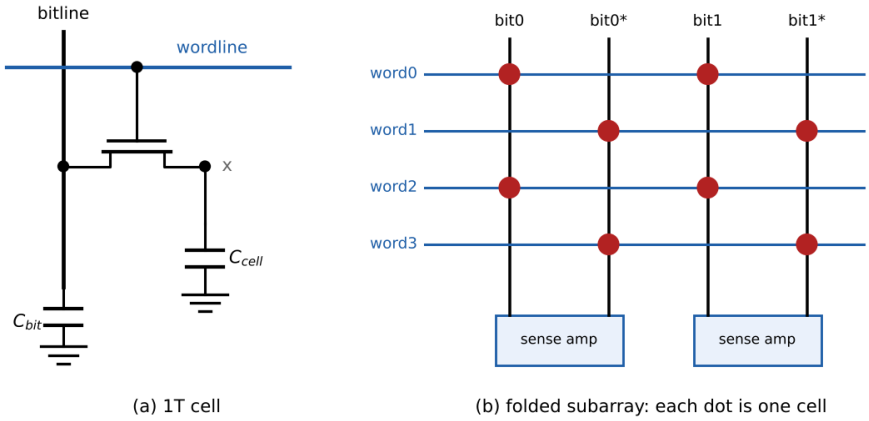
Figure 36: Read of a stored 1: the wordline rises, charge sharing gives ΔV on bit, then the sense amplifier drives bit to VDD and bit* to 0, which also restores the cell

Reading the figure. Both bitlines start at $V_{DD}/2$. When the wordline rises, bit (dotted red) moves up by the small ΔV, and bit* (dashed orange) does not move, as it has no selected cell. The sense amplifier then amplifies the difference. Because the wordline is still high, the full level returns to the cell. Finally the wordline falls and the lines are precharged.

2. Answer

A sub-array has rows of wordlines and columns of bitlines, with 1T cells at crossings, and sense amplifiers and write drivers at the column ends.

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(a) 1T cell (b) folded subarray: each dot is one cell

Figure 37: 1T cell and folded sub-array

Write:

1. The address selects a column, and the write driver forces the bitline to V_{DD} (for a 1) or to 0 (for a 0).
2. The row decoder raises the wordline, so the access transistor connects the bitline to C_{cell}.
3. The capacitor charges or discharges to the bitline's level.
4. The wordline falls, and the charge is trapped.

The nMOS passes a weak 1, so the wordline is often raised to $V_{DDP} = V_{DD} + V_t$ to store a full V_{DD}.

Read:

1. **Precharge:** bit and bit* are precharged to $V_{DD}/2$ and left floating.
2. **Charge sharing:** the wordline rises and the cell shares charge with its bitline, so the bitline changes by

$$\Delta V = \pm \frac{V_{DD}}{2} \frac{C_{cell}}{C_{cell} + C_{bit}}$$

(+ for a stored 1, − for a stored 0). bit* stays at $V_{DD}/2$ as the reference.

3. **Sensing:** $C_{cell} \ll C_{bit}$, so ΔV is tiny. The cross-coupled sense amplifier is enabled: V_n falls and then V_p rises, and it drives the higher line to V_{DD} and the lower to 0.
4. **Restore:** the read destroyed the charge, but the wordline is still high, so the full bitline level is written back into the cell.
5. The data goes out through the column multiplexer, the wordline falls, and the bitlines are precharged again.

Refresh: charge leaks away, so every row is read and restored periodically.

3. Keyword Explanation

1. Charge sharing

Two capacitors at different voltages, once joined, settle at a common voltage with their total charge unchanged.

2. Folded bitlines

Bitline pairs placed side by side, so noise couples to both equally and the sense amplifier rejects it.

4. Notes and Remarks

- A read is **destructive**, unlike an SRAM read, which is why the restore step is part of every read.
- Open bitlines are denser but noisier; twisted bitlines reduce coupling from neighbours (Weste and Harris Section 12.3.1).

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- A typical target is $C_{cell} = 30$ fF, reached by trench or stacked capacitors. $C_{bit} = 300$ fF is an illustrative value from "an order of magnitude higher".

5. Final Concise Answer

(a) Precharge ($\varphi = 0$): the clocked pMOS charges Y to V_{DD}, foot OFF. Evaluate ($\varphi = 1$): if A = 1, Y discharges, and if A = 0, Y stays high on C_Y. So $Y = \overline{A}$; input must be monotonic; leakage needs a keeper.

(b) $S = A \oplus B \oplus C_{in}$; $C_{out} = AB + C_{in}(A \oplus B) = AB + BC_{in} + AC_{in}$. Built from two half adders and an OR, or the 28-transistor mirror adder with $S = ABC_{in} + (A + B + C_{in})\overline{C_{out}}$.

(c) PLA: programmable AND and OR planes. PAL: programmable AND, fixed OR. The PLA has 3 inputs (6 vertical lines), 7 product lines and 2 output columns: four products to s, three to c.

(d) Write: driver forces the bitline, wordline high, cell takes the level. Read: precharge to $V_{DD}/2$, wordline high, charge sharing gives $\Delta V = \pm(V_{DD}/2)C_{cell}/(C_{cell}+C_{bit})$, sense amplifier amplifies against bit*, and the full level restores the cell.

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7. (a) Briefly describe steps of photo-lithography. Marks: 8
   (b) Explain the silicon oxidation model mentioning its limiting cases. Marks: 6
   (c) Briefly explain vapor phase epitaxy. Marks: 6

7.(a) Briefly describe steps of photo-lithography. Marks: 8

1. Theory

1. What is photolithography? (Core Idea)

Every layer of a chip has a pattern: where the gates go, where the contacts go, where the wires run. The pattern is drawn on a **mask** (a glass plate with an opaque chromium pattern). **Photolithography** copies it onto the wafer using light, much like printing a photograph from a negative:

- The wafer is coated with **photoresist**, a light-sensitive polymer.
- Light shines through the mask. Where the mask is clear, the resist is exposed.
- A developer dissolves either the exposed resist (**positive resist**) or the unexposed resist (**negative resist**).

The resist left on the wafer then protects the layer below during etching or ion implantation.

2. What limits how small a line can be?

The smallest line a projection system can print is about

$$W_{min} = k_1 \frac{\lambda}{NA}$$

where λ is the wavelength of the light (365 nm i-line, 248 nm KrF, 193 nm ArF, 13.5 nm EUV), NA is the numerical aperture of the lens, and k_1 is a process factor (about 0.25 to 0.8). This is why the industry moved to shorter wavelengths.

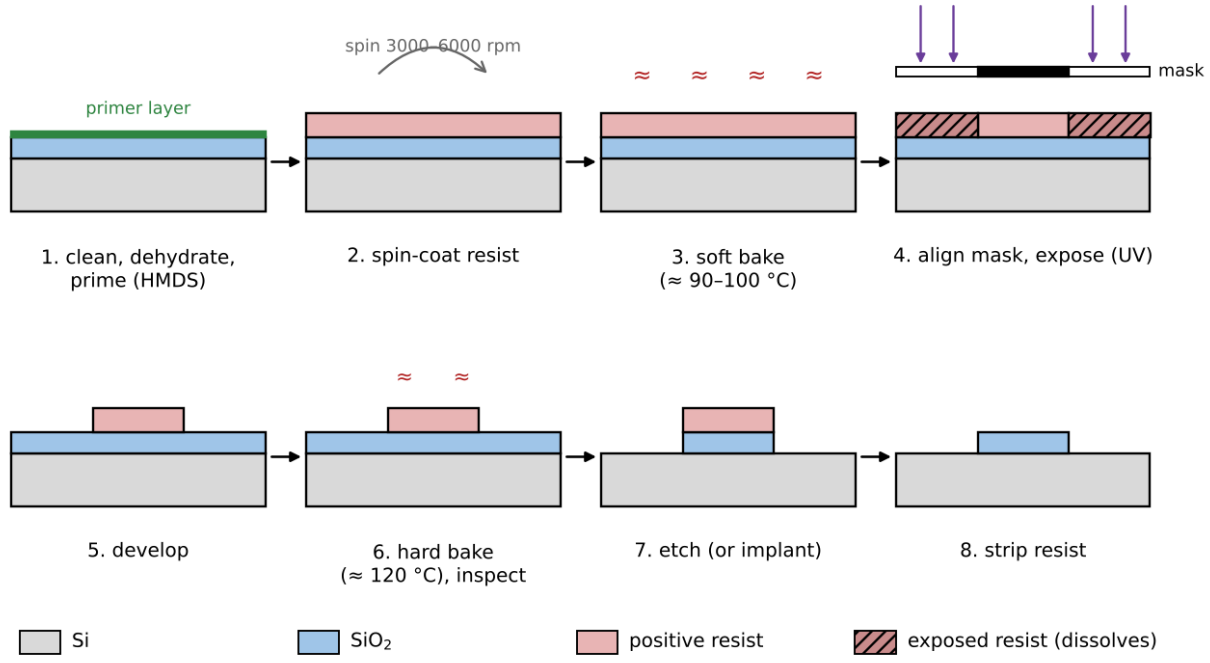


Figure 38: The steps of photolithography with a positive resist, followed by etching and resist strip (reused from the 2019-20 paper's solution)

Reading the figure. Follow the panels from 1 to 8. After cleaning and priming (1), the resist is spun on (2) and dried (3). The mask lets UV light through only at its clear ends (4), so the hatched resist

there is exposed. The developer removes the exposed resist (5), the rest is hardened (6), the oxide not covered by resist is etched (7), and the resist is removed (8).

2. Answer

Photolithography transfers a pattern from a mask to a photoresist layer on the wafer by exposing the resist to light and developing it. The patterned resist then masks the wafer for etching or ion implantation.

Steps:

1. **Wafer cleaning and priming:** the wafer is cleaned, baked at about 150–200 °C to drive off moisture, and primed with HMDS so the resist sticks.
2. **Resist coating (spin coating):** liquid resist is put on the wafer, which is spun at about 3000–6000 rpm for 30–60 s, giving a uniform film about 0.5–2 µm thick.
3. **Soft bake:** about 90–100 °C for 1 min, to drive the solvent out and improve adhesion.
4. **Alignment and exposure:** the mask is aligned to the patterns already on the wafer, then the resist is exposed to UV light through it. Exposure tools are **contact**, **proximity** and **projection** (step-and-repeat, usually reduced 4×, the one used today).
5. **Post-exposure bake (when needed):** about 110 °C; it smooths standing-wave ripples and completes the reaction in chemically amplified resists.
6. **Development:** developer (TMAH for positive resist) dissolves the exposed regions of a positive resist (the unexposed regions of a negative resist).
7. **Hard bake:** about 120–150 °C, to harden the resist so that it survives etching or implantation.
8. **Inspection:** line width, alignment and defects are checked; a bad layer is stripped and redone.

After lithography the wafer is **etched** or implanted through the openings, and the resist is **stripped** in an O_2 plasma or a solvent.

3. Keyword Explanation

1. HMDS

Hexamethyldisilazane, an adhesion promoter that replaces the water-loving OH groups on the oxide surface so that the resist does not lift off in the developer.

2. Critical dimension

The width of the smallest line in a layer, measured after development to check the process.

4. Notes and Remarks

- Also solved in 1(c) of the 2019-20 paper, as extra reading.

7.(b) Explain the silicon oxidation model mentioning its limiting cases. Marks: 6

1. Theory

1. Why silicon oxidation matters (Core Idea)

Silicon heated in oxygen or steam grows silicon dioxide:

$$Si + O_2 \rightarrow SiO_2 \qquad Si + 2H_2O \rightarrow SiO_2 + 2H_2$$

The oxide grows at the **Si/SiO$_2$ interface**, not on top. So oxygen must first get through the oxide already there. As the oxide thickens, the journey lengthens and the growth slows. A good model must explain this slowing down. That is the **Deal-Grove (linear-parabolic) model**.

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2. Three steps in series

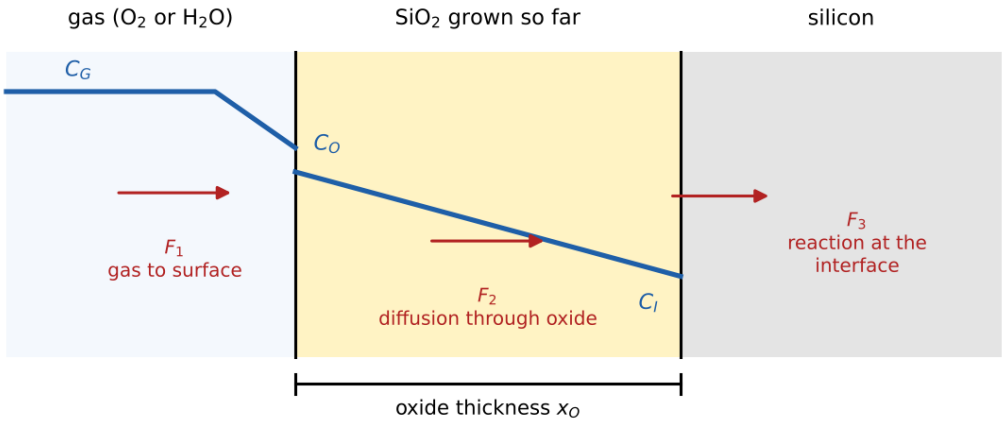


Figure 39: Oxidant concentration from the gas through the oxide to the silicon, and the three fluxes

Reading the figure. The concentration of oxidant falls from the gas (C_G), through the surface value C_O, across the oxide, to C_I at the silicon. The three red arrows are the fluxes in series.
The fluxes (molecules per unit area per unit time) are:

$$F_1 = h(C^* - C_O), \qquad F_2 = D\,\frac{C_O - C_I}{x_O}, \qquad F_3 = k_S C_I$$

where h is the gas-phase transfer coefficient, C^* the solubility of the oxidant in SiO_2, D its diffusivity in the oxide, x_O the oxide thickness and k_S the interface reaction rate constant.

3. Deriving the law, step by step

In steady state $F_1 = F_2 = F_3 = F$.
From $F_2 = F_3$:

$$C_O = C_I\left(1 + \frac{k_S x_O}{D}\right)$$

Put this in $F_1 = F_3$:

$$h\left[C^* - C_I\left(1 + \frac{k_S x_O}{D}\right)\right] = k_S C_I \;\Rightarrow\; C_I = \frac{C^*}{1 + k_S/h + k_S x_O/D}$$

So

$$F = k_S C_I = \frac{k_S C^*}{1 + k_S/h + k_S x_O/D}$$

Each unit volume of oxide takes up N_1 oxidant molecules, so the growth speed is

$$\frac{dx_O}{dt} = \frac{F}{N_1}$$

Separate the variables, then multiply both sides by $2D/k_S$:

$$\left[2D\left(\frac{1}{k_S} + \frac{1}{h}\right) + 2x_O\right] dx_O = \frac{2DC^*}{N_1}\,dt$$

With $A = 2D(1/k_S + 1/h)$ and $B = 2DC^*/N_1$, integrate from x_i at $t = 0$:

$$x_O^2 + Ax_O - (x_i^2 + Ax_i) = Bt$$

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Calling $\tau = (x_i^2 + Ax_i)/B$:

$$x_O^2 + Ax_O = B(t + \tau)$$

4. Which term wins?

A quadratic in x_O has a small-x_O end, where the Ax_O term rules, and a large-x_O end, where the x_O^2 term rules. These are the two limiting cases.

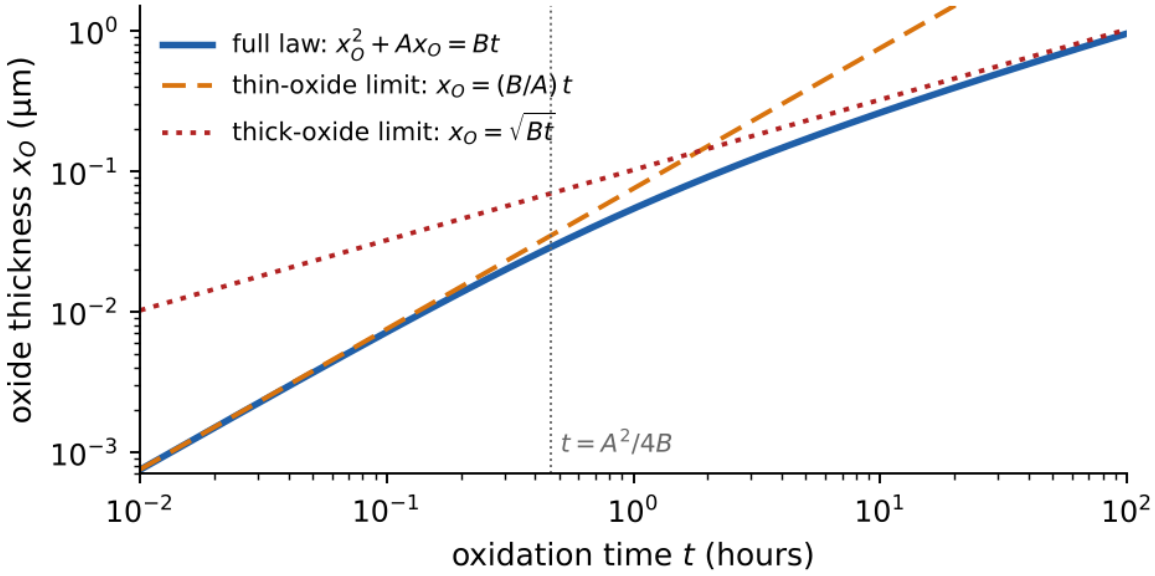


Figure 40: Oxide thickness against time for dry O2 at 1000 °C on (111) silicon, with the linear and parabolic limits

Reading the figure. On log axes the thin-oxide limit and the thick-oxide limit are straight lines. The solid blue curve moves from the first to the second near $t = A^2/4B = 0.46$ h, for the values of Plummer's Table 6-2 at 1000 °C ($B = 0.0104$ μm^2/h, $B/A = 0.0752$ μm/h).

2. Answer

Model: oxidant reaches the silicon in three series steps: gas to oxide surface ($F_1 = h(C^* - C_O)$), diffusion through the oxide ($F_2 = D(C_O - C_I)/x_O$), reaction at the interface ($F_3 = k_S C_I$). In steady state they are equal, which gives

$$\frac{dx_O}{dt} = \frac{F}{N_1} = \frac{k_S C^*/N_1}{1 + k_S/h + k_S x_O/D}$$

Integrating:

$$x_O^2 + Ax_O = B(t + \tau)$$

with $A = 2D(1/k_S + 1/h)$, $B = 2DC^*/N_1$, $\tau = (x_i^2 + Ax_i)/B$.

Limiting cases:

1. **Short times, thin oxide:** $x_O \approx (B/A)(t + \tau)$. Growth is linear and **reaction limited**, since diffusion is easy through a thin oxide. B/A is the linear rate constant.
2. **Long times, thick oxide:** $x_O \approx \sqrt{B(t + \tau)}$. Growth is parabolic and **diffusion limited**, since oxygen arrives slowly through a thick oxide. B is the parabolic rate constant.

3. Keyword Explanation

1. Linear and parabolic rate constants

B/A (μm/h) sets the speed of thin-oxide growth; B (μm^2/h) sets the slowing of thick-oxide growth.

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4. Notes and Remarks

- The final's 1(b) asks the same thing, with the same model; this solution stands on its own.

7.(c) Briefly explain vapor phase epitaxy. Marks: 6

1. Theory

1. What is epitaxy? (Core Idea)

The word comes from Greek for "arranged upon". **Epitaxy** grows a new layer of crystal on a crystal, so that the new atoms continue the lattice of the wafer below. The result is one single crystal with a new doping. If the conditions are not right, amorphous or polycrystalline silicon grows instead.

Vapor phase epitaxy (VPE) does this from gases: it is a form of chemical vapor deposition (CVD) in which the deposited film is a single crystal.

2. Why grow a layer at all?

The wafer is cut from one crystal with one doping. An epitaxial layer allows a lightly doped layer on a heavily doped substrate, which CMOS uses against latch-up and bipolar uses for collectors, with sharp doping steps and a clean crystal free of oxygen.

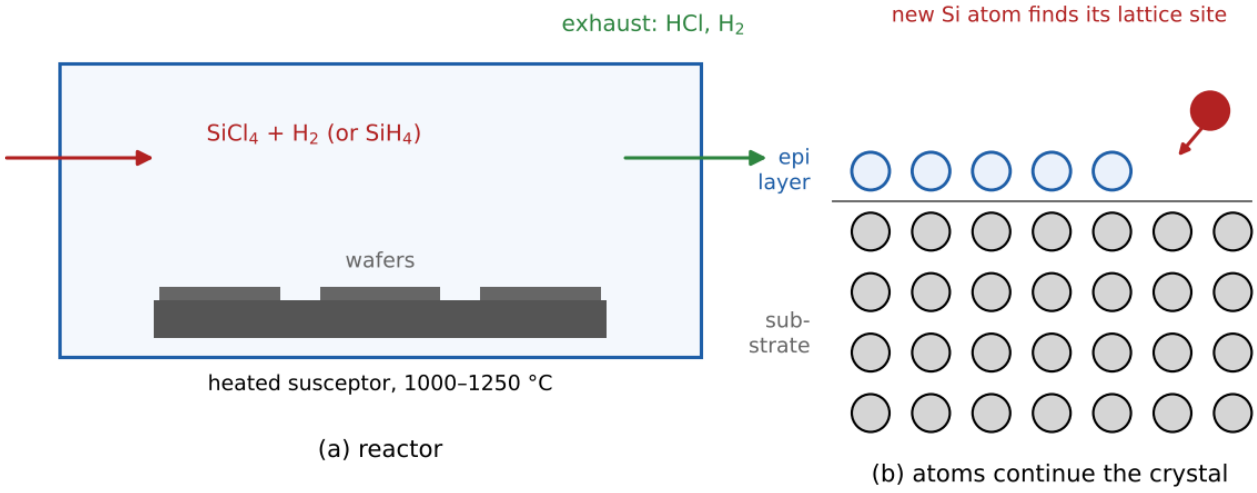


Figure 41: (a) A CVD epitaxy reactor; (b) new atoms continue the crystal of the substrate

Reading the figure. In (a), source gas flows over wafers on a hot susceptor, and the products leave at the exhaust. In (b), the blue atoms have joined the grey crystal in line with its lattice, and a red atom is still moving along the surface to its site.

3. The steps at the surface

Gas molecules reach the wafer through a thin stagnant boundary layer; they adsorb and react, and silicon atoms are left on the surface; these atoms **migrate** along the surface until they find a lattice site; the products (HCl, H_2) leave. The atoms need time and heat to reach their sites. So the temperature must be high and the growth rate not too fast. Too fast or too cold, and the atoms are buried before finding a site, and the film is polycrystalline.

4. Two rate regimes

At high temperature the surface reaction is fast, so the rate is set by how fast gas can reach the wafer (**mass-transfer limited**), and uniform gas flow matters. At low temperature the surface reaction sets the rate (**reaction limited**), and uniform temperature matters. High-temperature epitaxy lies in the mass-transfer regime, so wafers lie flat, not stacked.

2. Answer

Vapor phase epitaxy grows a single-crystal silicon layer on a silicon wafer from a gas, by CVD, so that the layer continues the crystal of the substrate.

Reactions:

$$SiH_4 \rightarrow Si + 2H_2 \qquad \text{(decomposition of silane)}$$

$$SiCl_4 + 2H_2 \rightarrow Si + 4HCl \qquad \text{(reduction of a chloride)}$$

$SiHCl_3$ and SiH_2Cl_2 are also used. Hydrogen is the carrier gas.

Process:

1. **Clean:** the wafer is cleaned, and an in situ HCl gas etch or H_2 bake removes native oxide. Any oxide on the surface prevents single-crystal growth.
2. **Deposition:** in a cold-wall reactor, wafers on a heated susceptor are held at about 1000–1250 °C (lowest for silane, highest for $SiCl_4$), at atmospheric pressure or about 80–200 torr.
3. **Surface action:** gas reaches the surface through the boundary layer, reacts and leaves Si atoms, which migrate to lattice sites. The products are pumped away.
4. **Doping:** dopant gases are added: arsine (AsH_3), phosphine (PH_3) or diborane (B_2H_6).

Key points:

- High temperature gives the atoms the energy to reach lattice sites, and desorbs native oxide; growth is mass-transfer limited.
- **Autodoping:** dopant leaving a heavily doped substrate enters the gas and is put back into the film. Lower pressure reduces it.
- **Selective epitaxy:** chlorine in the gas lets silicon grow on exposed silicon and not on oxide.
- **Low-temperature epitaxy** (500–900 °C) in an ultra-clean system limits dopant diffusion.

3. Keyword Explanation

1. Susceptor

The heated holder (often graphite) on which the wafers sit.

2. Autodoping

Unintended doping of the growing layer by dopant that escapes from the substrate.

4. Notes and Remarks

- Plummer, Deal and Griffin treat this in Section 9.3.1 (epitaxial silicon deposition) and Figure 9-8.
- In growth-rate order: SiH_4 is the fastest source and $SiCl_4$ the slowest.
- In molecular beam epitaxy, by contrast, the source is an evaporated beam in ultra-high vacuum, not a gas reaction.

5. Final Concise Answer

(a) Clean, dehydrate and prime with HMDS; spin coat; soft bake; align and expose; post-exposure bake; develop; hard bake; inspect; then etch or implant and strip.

(b) Deal-Grove: three equal fluxes give $x_O^2 + Ax_O = B(t+\tau)$. Thin oxide: linear, $x_O \approx (B/A)(t+\tau)$, reaction limited. Thick oxide: parabolic, $x_O \approx \sqrt{B(t+\tau)}$, diffusion limited.

(c) CVD growth of a single-crystal layer that continues the substrate: $SiH_4 \rightarrow Si + 2H_2$ or $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$ at 1000–1250 °C, on a clean surface, with AsH_3, PH_3 or B_2H_6 for doping.

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8. (a) Find the threshold voltage of MOSFET considering the body effect. Marks: 5

(b) Briefly explain the following terms: i) Drain induced barrier lowering, ii) Gate leakage. Marks: 5

(c) Implement the following expression using CMOS logic: $Y = AB + C(A + B)$. Marks: 5

(d) Find minimum number delay [*sic*: should be "minimum delay"] for the logic circuit below. Also find the gate size to achieve this delay. Marks: 5

(e) Find the stage effort for the ideal number of stages. Marks: 5

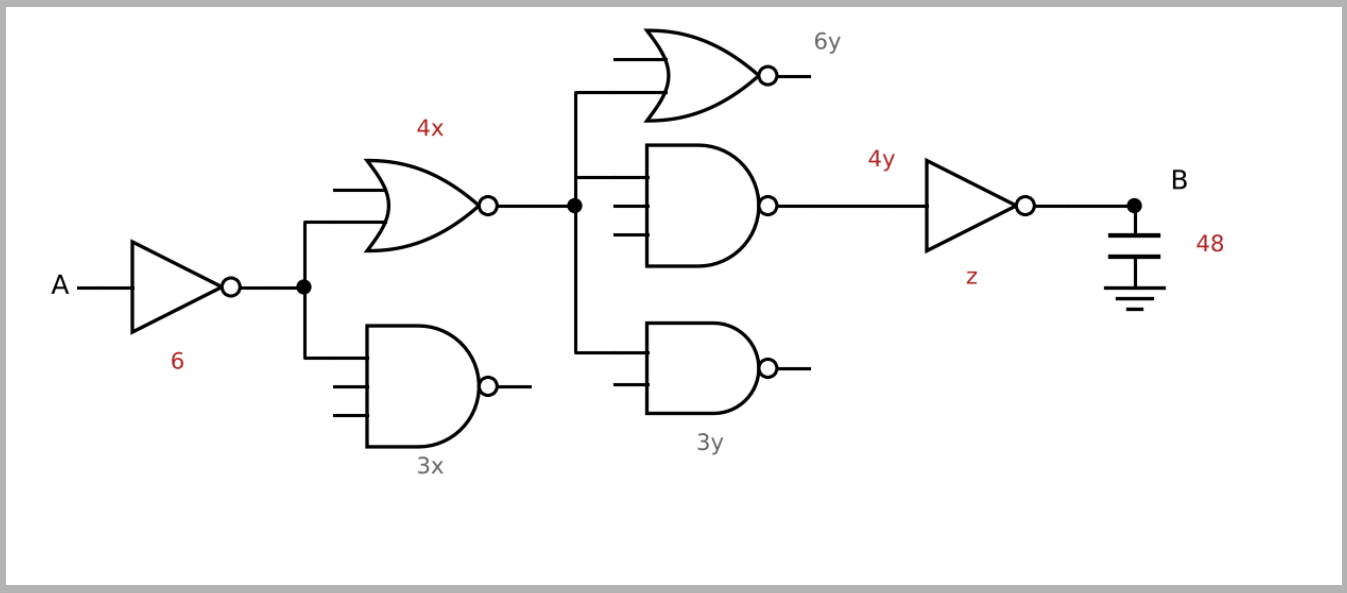


Figure 42: The figure of Term Test 2, question 4 (redrawn): inverter 6, then NOR2 4x and NAND3 3x, then NOR2 6y, NAND3 4y and NAND2 3y, then inverter z and the load 48; A and B are faint marks on the paper

8.(a) Find the threshold voltage of MOSFET considering the body effect. Marks: 5

1. Theory

1. Threshold voltage at zero bias (Core Idea)

An nMOS transistor conducts when the gate attracts enough electrons to invert the p-type surface. With the source and body both at 0, the threshold is

$$V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

where the four terms are the work-function difference between gate and channel, the voltage to bend the surface by $-2\phi_F$, the voltage to hold the depletion charge, and the voltage to cancel the fixed oxide charge.

2. What is the body effect?

The body is the fourth terminal. In a stack of transistors, a source can sit above the body, so $V_{SB} > 0$. Then the p-n junction under the channel is more reverse biased, and the depletion region is **wider**. More fixed charge has to be held by the gate, so the threshold rises. This is the **body effect**.

3. Deriving the term, step by step

The depletion region now holds the voltage $|-2\phi_F| + V_{SB}$ instead of $|2\phi_F|$. The depletion charge formula from Poisson's equation becomes

$$Q_B = -\sqrt{2qN_A\varepsilon_{Si}\left(|2\phi_F| + V_{SB}\right)}$$

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Only the depletion term of V_T changes:

$$V_T = \phi_{GC} - 2\phi_F - \frac{Q_B}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

Add and subtract Q_{B0}/C_{ox}:

$$V_T = \underbrace{\phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}}_{V_{T0}} - \frac{Q_B - Q_{B0}}{C_{ox}}$$

Now put in the two charges:

$$-\frac{Q_B - Q_{B0}}{C_{ox}} = \frac{\sqrt{2qN_A\varepsilon_{Si}}}{C_{ox}}\left(\sqrt{|2\phi_F| + V_{SB}} - \sqrt{|2\phi_F|}\right)$$

Name the coefficient

$$\gamma = \frac{\sqrt{2qN_A\varepsilon_{Si}}}{C_{ox}}$$

This gives the result in the Answer.

2. Answer

For $V_{SB} \neq 0$, the depletion charge grows, $Q_B = -\sqrt{2qN_A\varepsilon_{Si}\,|-2\phi_F + V_{SB}|}$, and only that term of the threshold changes:

$$V_T = \phi_{GC} - 2\phi_F - \frac{Q_B}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} = V_{T0} - \frac{Q_B - Q_{B0}}{C_{ox}}$$

where

$$V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

Hence

$$\boxed{V_T = V_{T0} + \gamma\left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}\right)}$$

$$\gamma = \frac{\sqrt{2qN_A\varepsilon_{Si}}}{C_{ox}}$$

where γ is the substrate-bias (body-effect) coefficient.

Illustration (Kang's device; the paper gives no values): $N_A = 10^{16}$ cm^{-3}, $t_{ox} = 500$ Å, $C_{ox} = 7.03\times10^{-8}$ F/cm^2, $\phi_F = -0.35$ V, $V_{T0} = 0.40$ V:

$$\gamma = \frac{\sqrt{2(1.6\times10^{-19})(11.7)(8.85\times10^{-14})(10^{16})}}{7.03\times10^{-8}} = 0.82\ \text{V}^{1/2}$$

For $V_{SB} = 2.5$ V:

$$V_T = 0.40 + 0.82\left(\sqrt{0.70 + 2.5} - \sqrt{0.70}\right) = 0.40 + 0.82(1.789 - 0.837) = 1.18\ \text{V}$$

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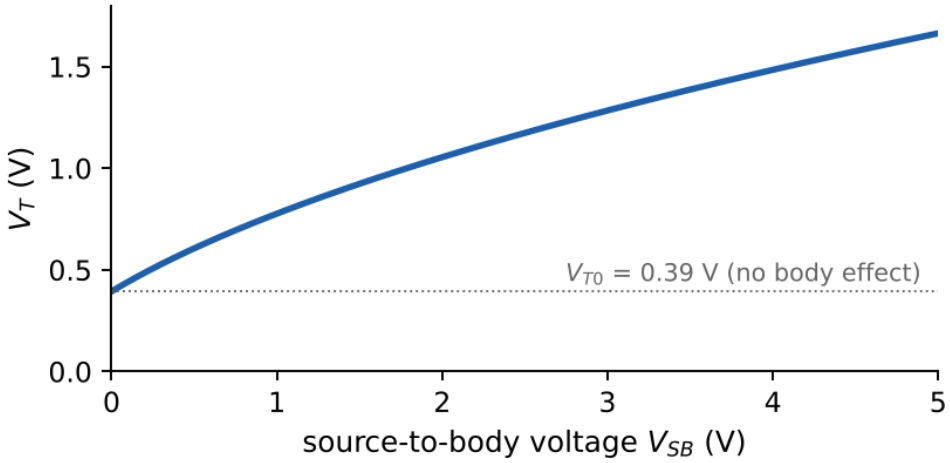


Figure 43: Threshold voltage against the source-to-body voltage for this device

Reading the figure. V_T starts at about 0.39 V and rises with a square-root shape. A source 2.5 V above the body has about three times the threshold.

3. Keyword Explanation

1. Body effect coefficient γ

The factor, in $V^{1/2}$, that sets how much the threshold rises with V_{SB}. It grows with the doping and with a thicker oxide.

4. Notes and Remarks

- For a p-channel device the signs change: ϕ_F, Q_B are positive; γ and V_{SB} are negative in Kang's convention, and V_T grows more negative.
- The curve uses exact arithmetic (0.394 V), and the text uses Kang's rounded 0.40 V.
- The final's 2(a) asked the $V_{SB} = 0$ case.

8.(b) Briefly explain the following terms: i) Drain induced barrier lowering, ii) Gate leakage. Marks: 5

1. Theory

1. The ideal switch and the real one (Core Idea)

In the ideal picture, a transistor has no current when off, its threshold is set by the gate alone, and its gate is a perfect insulator. Small transistors break all these rules. DIBL and gate leakage are two of the ways.

2. The barrier picture

Think of the electrons in the source as water behind a dam. The p-type region under the gate is a hill of potential energy between source and drain. The gate voltage sets the height of the hill. In a long channel the drain is too far away to affect the top of the hill.

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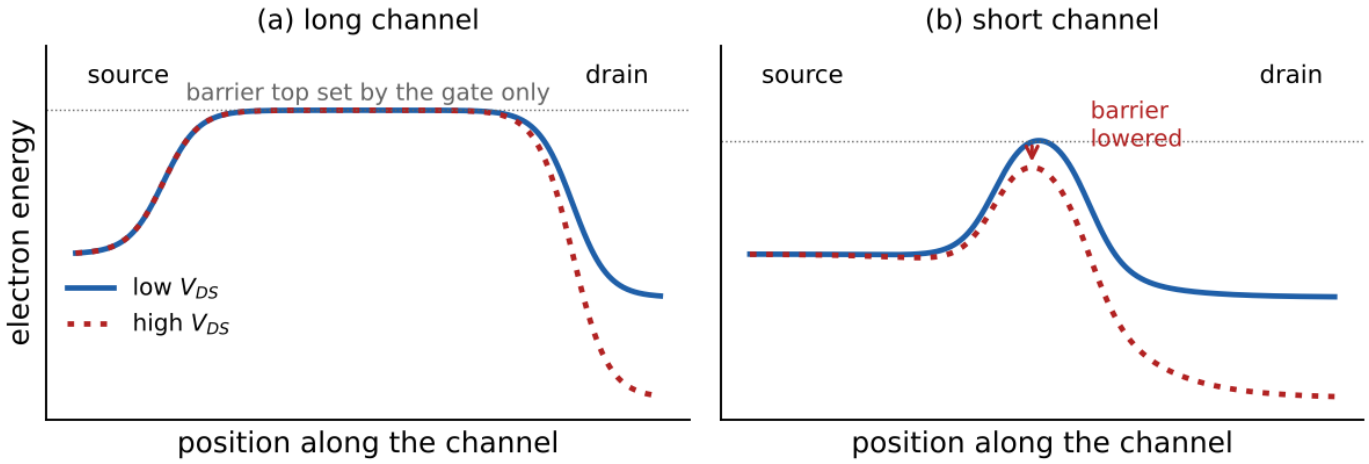


Figure 44: Electron energy along the channel: (a) in a long channel the drain does not change the barrier top; (b) in a short channel a high drain voltage lowers it (reused from the 2019-20 paper's solution)

Reading the figure. In (a), raising V_{DS} only pulls down the drain end. In (b), the channel is so short that the drain's field reaches the top of the barrier; at high V_{DS} the top is lower, so more electrons cross.

3. Tunnelling

A classical electron cannot cross an insulator whose barrier is above its energy. A quantum electron is a wave, and a wave does not stop dead at a barrier: it dies away exponentially inside. If the barrier is thin enough, some of the wave comes out the other side.

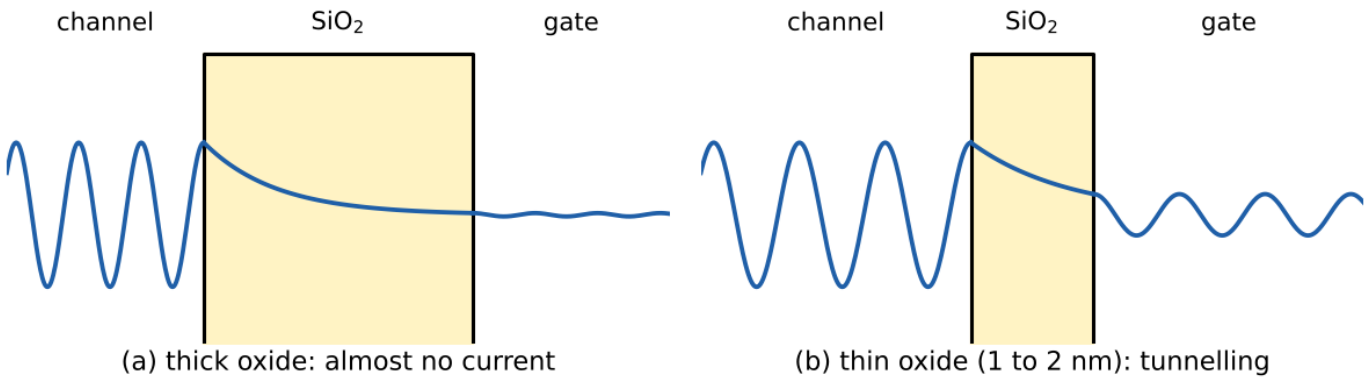


Figure 45: An electron wave at an oxide barrier: (a) in a thick oxide it dies away; (b) in a thin oxide part of it comes through

Reading the figure. The wave's size falls by the same rate inside each oxide. In (a) it has almost vanished by the far side. In (b) the oxide is thin, so a clear wave emerges. That wave is the tunnelling current.

2. Answer

i) Drain-induced barrier lowering (DIBL). In a short-channel MOSFET the drain field reaches the source end of the channel and lowers the potential barrier there, as the gate does. So the threshold falls as the drain voltage rises:

$$V_t = V_{t0} - \eta V_{ds}$$

where η is the DIBL coefficient (about 0.1, that is 100 mV per volt). Effects: more subthreshold current at high V_{ds} (the off current at $V_{ds} = V_{DD}$ is much larger), a saturation current that keeps rising with V_{ds}, and so low output resistance. It grows as the channel gets shorter.

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ii) Gate leakage. With gate oxides thinner than about 15–20 Å, electrons have a real chance of **quantum-mechanical tunnelling** through the oxide, between the channel and gate. A small current then flows into the gate even though it is meant to be insulated. It rises exponentially as the oxide thins (by a factor of 2.7 or more per ångström) and rises with the voltage across the oxide:

$$I_{gate} = WA\left(\frac{V_{DD}}{t_{ox}}\right)^2 e^{-B\,t_{ox}/V_{DD}}$$

where W is the width and A, B are constants of the technology. Consequences: static power, in ON transistors especially, and a limit on scaling t_{ox}. The cure is a high-k gate dielectric (such as HfO_2), which can be physically thicker for the same C_{ox}. In SiO_2, the tunnelling current of nMOS can be an order of magnitude higher than of pMOS, because holes see a higher barrier than electrons.

3. Keyword Explanation

1. High-k dielectric

An insulator with a larger dielectric constant than SiO_2. It gives the same C_{ox} at greater thickness, so tunnelling is less.

2. Short-channel effect

Any change of the device's behaviour because the source and drain are close together.

4. Notes and Remarks

- DIBL also appears in 1(d) of the 2019-20 paper, as extra reading.
- The gate leakage value used in 4(d) is 5 nA/µm.

8.(c) Implement the following expression using CMOS logic: $Y = AB + C(A + B)$. Marks: 5

1. Theory

1. Building a CMOS gate (Core Idea)

A static CMOS gate is a network of nMOS below the output and a network of pMOS above it. The nMOS network pulls the output down, so a single gate always gives an **inverted** function: $\overline{Y}$ of whatever the nMOS network computes.

Rules for the nMOS network: AND is series, OR is parallel. Rule for the pMOS network: its **dual**, with series and parallel swapped.

$Y = AB + C(A + B)$ is **not inverted**, so we build $\overline{Y}$ as a compound gate and add an inverter.

2. The two networks

- **nMOS:** $Y = AB + C(A + B)$ is an OR of two parts, so two branches in parallel: (A in series with B) and (C in series with (A parallel B)).
- **pMOS:** take the dual of each piece. The whole becomes series: [(A parallel B)] in series with [C parallel (A in series with B)].

To check the dual: $\overline{Y} = \overline{AB} \cdot \overline{C(A + B)} = (\overline{A} + \overline{B})(\overline{C} + \overline{A}\,\overline{B})$. The pMOS conduct on a 0 input, so this reads exactly as the network above.

3. The widths

For unit drive (resistance R), unit nMOS width is 1 and unit pMOS width is 2.

- **nMOS:** every path has 2 in series, so each has width 2.
- **pMOS:** the longest path is top (A or B) then the A-B series pair, 3 in series, so these are width 6 ($R/3$ each). The path through C has the top device, $R/3$, so C needs $2R/3$, which is width 3.

2. Answer

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Step 1: Y is not inverted, so implement $\overline{Y}$ with a compound gate, then an inverter.
Step 2: the pull-down network conducts when $Y = 1$: (A series B) parallel [C series (A parallel B)].
Step 3: the pull-up network is its dual: (A parallel B) series [C parallel (A series B)].
Step 4: an inverter gives Y from $\overline{Y}$.

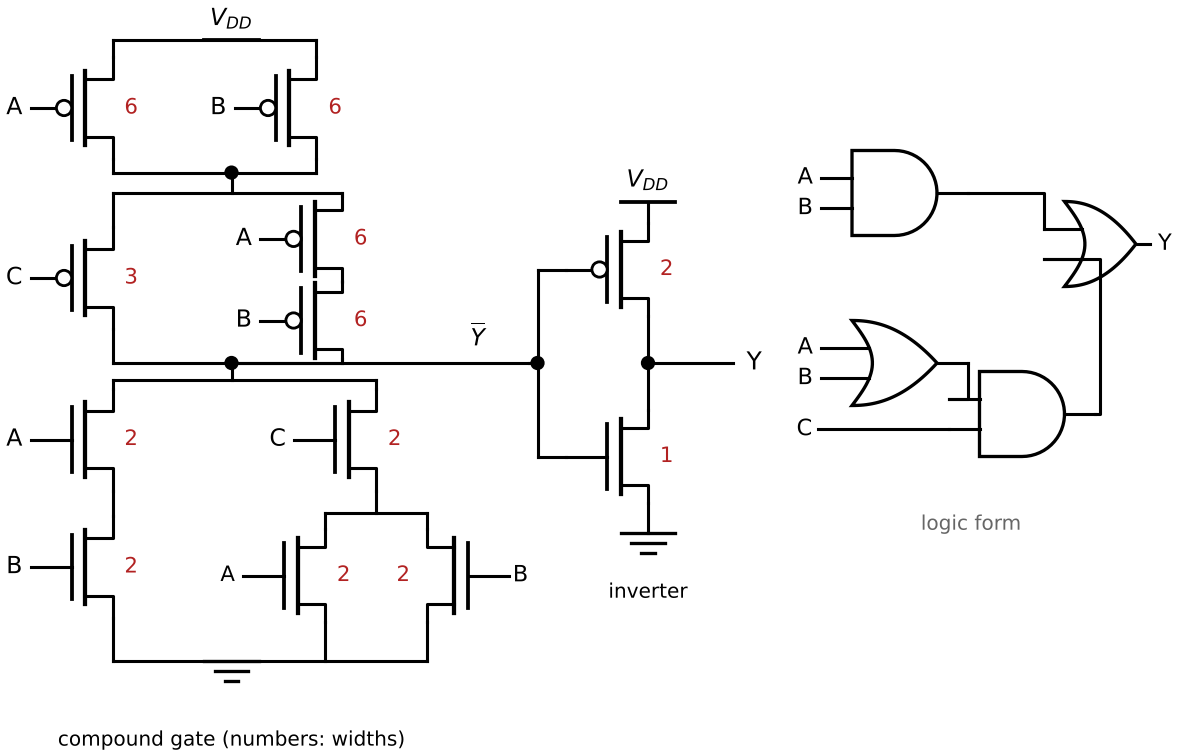


Figure 46: CMOS implementation of Y = AB + C(A + B): the compound gate with widths, the inverter and the logic form

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table 13: Truth table of Y

The gate uses 10 transistors, and the inverter 2, so 12 in all. Y is 1 when two or more inputs are 1.
Reading the figure. For A = B = 1, C = 0: the left nMOS branch conducts, so $\overline{Y} = 0$, and the inverter gives Y = 1. The top pMOS are off, so there is no path from V_{DD}. For A = 0, B = 0, C = 1: neither nMOS branch conducts. The top pMOS are ON and the lower A-B pair is ON, so $\overline{Y} = 1$ and Y = 0.

3. Keyword Explanation

1. Compound gate

A single-stage CMOS gate for a function such as AOI or OAI, with series and parallel branches in each network.

4. Notes and Remarks

- This Y is the **carry** of a full adder; see 6(b). There, the pull-up uses the mirror form, $AB + C(A + B)$ again, which also works, because the carry is self-dual.

8.(d) ⚠ Note on the paper: The paper prints "minimum number delay". The sentence goes on to "the gate size to achieve this delay", so it means the minimum delay.

1. Theory

1. Logical effort in brief (Core Idea)

A gate's delay, in units of $\tau = 3RC$, is $d = gh + p$, where g is the logical effort, $h = C_{out}/C_{in}$ the electrical effort, and p the parasitic delay.

Gate	g	p
Inverter	1	1
NAND2	4/3	2
NAND3	5/3	3
NOR2	5/3	2

Table 14: Logical effort and parasitic delay

The NAND3 value: width 3 series nMOS and width 2 parallel pMOS give $C_{in} = 5$, against 3 for an inverter.

2. The path

The path effort $F = GBH$ is the product of the logical efforts, the branching efforts and the electrical effort. Stage efforts equal to $F^{1/N}$ give the minimum delay, $D = NF^{1/N} + P$. Sizes come from working back from the load with $C_{in} = g\,C_{out}/f$.

2. Answer

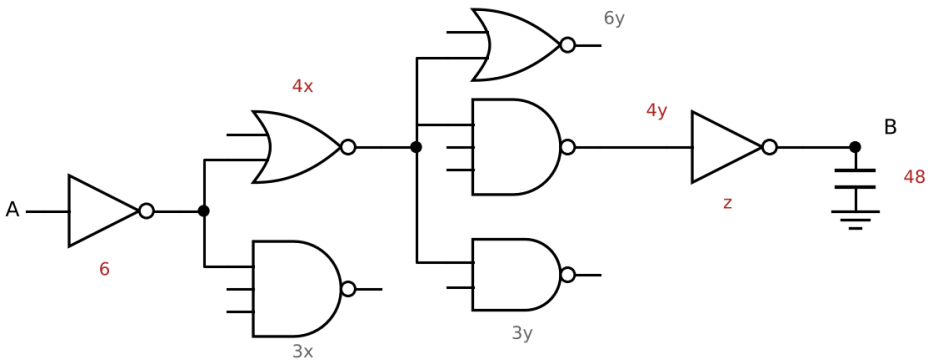


Figure 47: Repeated from the question: the circuit of Term Test 2, question 4 (redrawn)

Reading the circuit. The labels are input capacitances. Path A to B: inverter (6), NOR2 (4x), NAND3 (4y), inverter (z), load 48. There are 4 stages.

Path effort:

$$G = 1 \times \frac{5}{3} \times \frac{5}{3} \times 1 = \frac{25}{9} = 2.78$$

$$b_1 = \frac{4x + 3x}{4x} = \frac{7}{4}, \qquad b_2 = \frac{6y + 4y + 3y}{4y} = \frac{13}{4}, \qquad B = \frac{7}{4} \times \frac{13}{4} = 5.69$$

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$$H = \frac{48}{6} = 8$$

$$F = GBH = 2.78 \times 5.69 \times 8 = 126.4$$

Stage effort and delay:

$$\hat{f} = 126.4^{1/4} = 3.35$$

$$P = 1 + 2 + 3 + 1 = 7$$

$$D = 4(3.35) + 7 = 20.4\tau$$

$$\boxed{D = 20.4\ \tau \approx 4.1 \text{ FO4 delays}}$$

Sizes, from the load back:

$$z = \frac{1 \times 48}{3.35} = 14.3$$

$$4y = \frac{(5/3)(14.3)}{3.35} = 7.12 \ \Rightarrow \ y = 1.78$$

The NOR2 drives $13y = 23.1$:

$$4x = \frac{(5/3)(23.1)}{3.35} = 11.5 \ \Rightarrow \ x = 2.87$$

Check at the first inverter: it drives $7x = 20.1$, so $C_{in} = 20.1/3.35 = 6.0$. ✔

$$\boxed{x = 2.87, \quad y = 1.78, \quad z = 14.3}$$

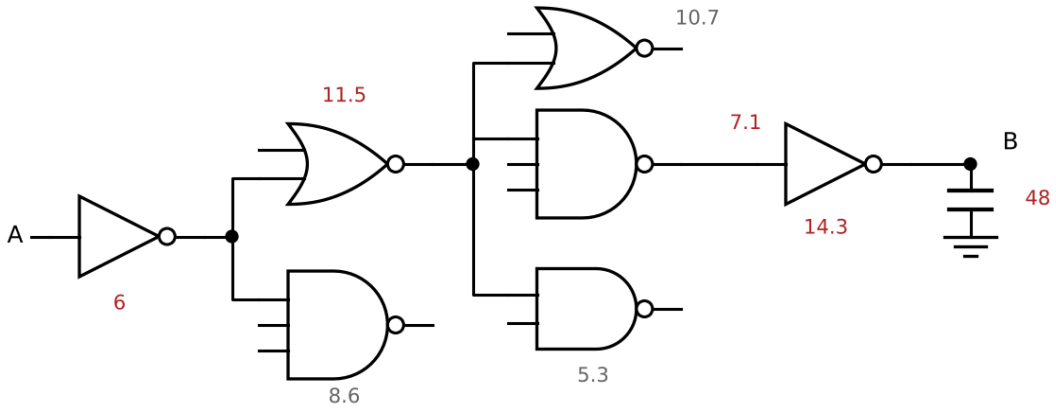


Figure 48: The circuit with the input capacitance of every gate

Reading the figure. On the path the sizes are 6, 11.5, 7.1 and 14.3; off the path they are 3x = 8.6, 6y = 10.7 and 3y = 5.3.

3. Keyword Explanation

1. Electrical effort

The ratio of the output to input capacitance of a stage or path.

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4. Notes and Remarks

- **Reading of the gates.** The paper's symbols show 4x as an OR shape with a bubble (NOR2), and 4y and 3x with three input leads (NAND3), 3y and 6y with two. The answer follows that drawing. If 4y were a NAND2: $G = 20/9$, $F = 101.1$, $f = 3.17$, $P = 6$, $D = 18.7\tau$. The very similar Figure 3.1 of the 2019-20 paper has NAND2 for both, with C_{in} 6 and 48 too: see 3(b) of that paper, as extra reading.
- The printed "48 F" is read as 48 units of capacitance, the same units as the other labels.
- The stage electrical efforts are 3.35, 2.01, 2.01 and 3.35; with g, every stage effort is 3.35.

8.(e) Find the stage effort for the ideal number of stages. Marks: 5

1. Theory

1. The trade-off (Core Idea)

For a path with effort F, more stages share the effort (less $NF^{1/N}$) but add parasitic delay. The ideal number of stages makes D smallest. The stage effort at that point is called ρ.

2. Deriving ρ, step by step

Suppose the path has n_1 logic stages and we add $N - n_1$ inverters, each with parasitic p_{inv}. F does not change:

$$D = NF^{1/N} + \sum_{i=1}^{n_1} p_i + (N - n_1)\, p_{inv}$$

Differentiate with respect to N. Since $F^{1/N} = e^{(\ln F)/N}$, we have $\frac{d}{dN}F^{1/N} = -F^{1/N} \ln F/N^2$, so by the product rule

$$\frac{dD}{dN} = F^{1/N} - F^{1/N}\,\frac{\ln F}{N} + p_{inv}$$

Set to zero with $\rho = F^{1/N}$, so that $\ln \rho = (\ln F)/N$:

$$\rho - \rho \ln \rho + p_{inv} = 0$$

$$p_{inv} + \rho(1 - \ln \rho) = 0$$

The equation has no closed form.

2. Answer

Minimising $D = NF^{1/N} + \sum p_i + (N - n_1)p_{inv}$ with respect to N gives

$$p_{inv} + \rho(1 - \ln \rho) = 0, \qquad \rho = F^{1/\hat{N}}$$

where ρ is the stage effort at the ideal number of stages. Solved numerically for $p_{inv} = 1$:

$$\boxed{\rho \approx 3.59 \ \text{(about 4 in practice)}}$$

Check: $1 + 3.59(1 - \ln 3.59) = 1 + 3.59(1 - 1.278) \approx 0$. ✔
The other values are $\rho = e = 2.72$ for $p_{inv} = 0$, and $\rho = 4.32$ for $p_{inv} = 2$.
The ideal number of stages is $\hat{N} = \log_\rho F$.
For the circuit of 8(d) ($F = 126.4$):

$$\hat{N} = \frac{\ln 126.4}{\ln 3.59} = \frac{4.84}{1.278} = 3.79 \ \Rightarrow \ N = 4$$

$$\hat{f} = 126.4^{1/4} = 3.35$$

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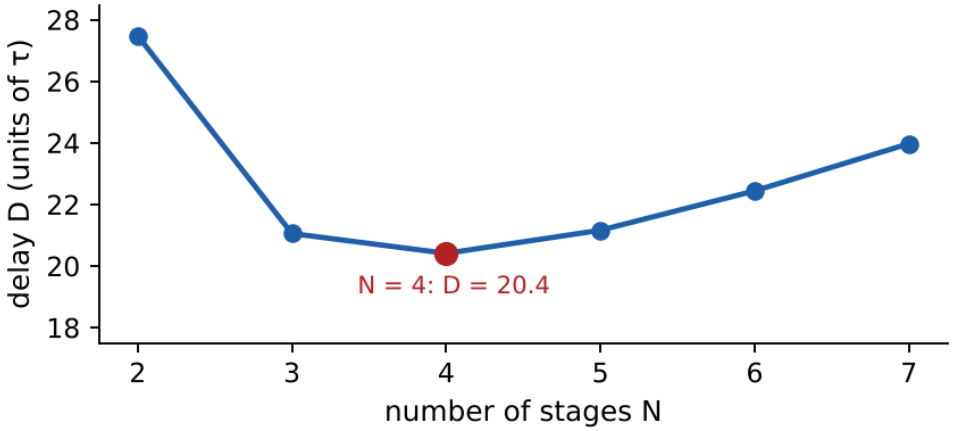


Figure 49: Path delay against the number of stages for F = 126.4

Reading the figure. The delay is smallest at $N = 4$, the number the circuit has, at 20.4τ. At that point the stage effort is 3.35, close to the ideal 3.59. The whole number of stages is why it is not exactly 3.59.

3. Keyword Explanation

1. Fanout-of-4 rule

A stage effort of about 4 gives close to the best delay; stage efforts of 2.4 to 6 stay within about 15 % of the minimum (Weste and Harris).

4. Notes and Remarks

- The paper names no circuit in this question, so the general result is the answer, and the number for 8(d) is given too.
- The final's 3(d) asks the same idea from the side of the number of stages.

5. Final Concise Answer

(a) $V_T = V_{T0} + \gamma\left(\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}\right)$, with $\gamma = \sqrt{2qN_A\varepsilon_{Si}}/C_{ox}$. Example: $\gamma = 0.82\ \mathrm{V}^{1/2}$, $V_T = 1.18$ V at $V_{SB} = 2.5$ V.
(b) DIBL: the drain field lowers the source barrier in short channels, $V_t = V_{t0} - \eta V_{ds}$, raising off current. Gate leakage: electrons tunnel through a thin oxide (below about 15–20 Å); high-k dielectrics reduce it.
(c) $\overline{Y}$ gate: nMOS (A–B series) parallel [C series (A parallel B)]; pMOS (A parallel B) series [C parallel (A–B series)]; then an inverter. 12 transistors.
(d) $G = 25/9$, $B = 91/16$, $H = 8$, $F = 126.4$, $f = 3.35$, $P = 7$, $D = 20.4\tau$; $x = 2.87$, $y = 1.78$, $z = 14.3$.
(e) $p_{inv} + \rho(1 - \ln\rho) = 0$ gives $\rho \approx 3.59$ for $p_{inv} = 1$. For $F = 126.4$: $\hat{N} = 3.79$, so $N = 4$ with $f = 3.35$.

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FORMULA SHEET

EEE455

VLSI Design and Testing

Formula Sheet

Session 2018–19 paper *Final Examination-2022 and Term Tests 1 and 2*

TOPICS

1 Fabrication — 11 formulas

2 MOS transistor — 6 formulas

3 Scaling — 3 formulas

4 Inverters — 6 formulas

5 Delay and logical effort — 9 formulas

6 Power — 4 formulas

7 Sequential, datapath and memory — 6 formulas

Each topic is one table: **Title** · **Equation** · **Notation** · **Reference**.

Prepared by Nur Mohammad

November 2026

BACK OF THE COVER

Before you use the sheet

How to read a row

Plasma dissociation	$CF_4 + e^- \rightarrow CF_3 + F + e^-$	F = fluorine radical, the main etchant	1(a)
↑ the name you search for	↑ the formula, exactly as derived	↑ every new symbol with its unit	↑ link: opens the derivation

In the exam

- Write the formula first, then put the numbers in: the formula alone often earns a mark.
- Read the symbols in the Notation column before you substitute, and check the units.
- Not sure where a formula comes from? Its Reference opens the derivation.
- The **Key formulas** page at the end gathers the rows marked ★.
- Lost a symbol? The **Symbol index** at the end links to the row where it first appears.

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1 Fabrication 4

2 MOS transistor 5

3 Scaling 5

4 Inverters 6

5 Delay and logical effort 6

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7 Sequential, datapath and memory 7

8 Key formulas 8

9 Symbol index 9

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Conventions: widths and capacitances in logical effort are in units of the unit inverter's nMOS (unit inverter: nMOS 1, pMOS 2). Delays are in units of $\tau = 3RC$; 1 FO4 $= 5\tau$. Threshold formulas use Kang's signs for an nMOS on a p-type body ($\phi_F < 0$). References point to the solution in this folder.

Columns: Title · Equation · Notation · Ref. (reference).

1. Fabrication:

Title	Equation	Notation	Ref.
Plasma dissociation	$CF_4 + e^- \rightarrow CF_3 + F + e^-$	F = fluorine radical, the main etchant	1(a)
Silicon etch	$Si + 4F \rightarrow SiF_4 \uparrow$	SiF_4 = volatile product	1(a)
★ **Adding O_2**	$O + CF_3 \rightarrow COF_2 + F$	more free F: Si rate peaks near 12 % O_2	1(a)
★ **Adding H_2**	$H + F \rightarrow HF$	less free F: SiO_2:Si selectivity rises	1(a)
Plasma ionisation	$e^- + A \rightarrow A^+ + 2e^-$	sustains the plasma, with secondary electrons from the electrodes	1(c)
Deal-Grove fluxes	$F_1 = h(C^* - C_O),\ F_2 = D\dfrac{C_O - C_I}{x_O},\ F_3 = k_S C_I$	C^* = oxidant solubility D = diffusivity in SiO_2 k_S = interface rate constant	1(b)
★ **Oxide growth law**	$x_O^2 + Ax_O = B(t + \tau)$	$A = 2D(1/k_S + 1/h)$ $B = 2DC^*/N_1$ $\tau = (x_i^2 + Ax_i)/B$	1(b)
Thin-oxide limit	$x_O \approx \dfrac{B}{A}(t + \tau)$	linear, reaction limited	7(b)
Thick-oxide limit	$x_O \approx \sqrt{B(t + \tau)}$	parabolic, diffusion limited	7(b)
Lithography resolution	$W_{min} = k_1 \dfrac{\lambda}{NA}$	λ = wavelength (m) NA = numerical aperture	7(a)
Epitaxy reactions	$SiH_4 \rightarrow Si + 2H_2$; $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$	1000–1250 °C	7(c)

Table 1: Fabrication

2. MOS transistor:

Title	Equation	Notation	Ref.
Fermi potential	$\phi_F = \frac{kT}{q} \ln \frac{n_i}{N_A}$	negative for p-type	2(a)
Depletion charge	$Q_{B0} = -\sqrt{2qN_A\varepsilon_{Si}\lvert 2\phi_F\rvert}$	C/m^2	2(a)
★ **Threshold, no body effect**	$V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$	ϕ_{GC} = work-function difference Q_{ox} = oxide charge $C_{ox} = \varepsilon_{ox}/t_{ox}$	2(a)
★ **Threshold with body effect**	$V_T = V_{T0} + \gamma\left(\sqrt{\lvert 2\phi_F\rvert + V_{SB}} - \sqrt{\lvert 2\phi_F\rvert}\right)$	$\gamma = \sqrt{2qN_A\varepsilon_{Si}}/C_{ox}$ (V$^{1/2}$)	8(a)
DIBL	$V_t = V_{t0} - \eta V_{ds}$	$\eta \approx 0.1$	8(b)
Gate leakage	$I_{gate} = WA\left(\frac{V_{DD}}{t_{ox}}\right)^2 e^{-Bt_{ox}/V_{DD}}$	A, B = technology constants	8(b)

Table 2: MOS transistor

3. Scaling:

Title	Equation	Notation	Ref.
Moore's law	count $\times 2$ every 2 years	Moore, 1965 and 1975	2(b)
★ **Constant-field scaling**	$L, W, t_{ox}, V : 1/S;\ N_A : S$	delay $1/S$; power per gate $1/S^2$; density S^2; power density 1	2(b)
Constant-voltage scaling	$L, W, t_{ox} : 1/S;\ V : 1;\ N_A : S^2$	delay $1/S^2$; power density S^3	2(b)

Table 3: Scaling

4. Inverters:

Title	Equation	Notation	Ref.
Saturated load, high level	$V_{OH} = V_{DD} - V_{TL}$	V_{TL} = load threshold, with body effect	2(c)
Saturated load, low level	$k_R\left[2(V_{OH} - V_T)V_{OL} - V_{OL}^2\right] = (V_{OH} - V_{OL})^2$	$k_R = k_D/k_L$	2(c)
CMOS, region B	$V_{out} = V_{in} + V_t + \sqrt{(V_{DD} - 2V_{in})(V_{DD} - 2V_t)}$	symmetric inverter, $V_t \leq V_{in} < V_{DD}/2$	2(d)
CMOS, region D	$V_{out} = V_{in} - V_t - \sqrt{(2V_{in} - V_{DD})(V_{DD} - 2V_t)}$	symmetric inverter	2(d)
★ **Switching threshold**	$V_M = \dfrac{V_{tn} + r(V_{DD} - \lvert V_{tp}\rvert)}{1 + r}$	$r = \sqrt{\beta_p/\beta_n}$	2(d)
Noise-margin points	$V_{IL} = \dfrac{3V_{DD} + 2V_t}{8}$, $V_{IH} = \dfrac{5V_{DD} - 2V_t}{8}$	$NM_L = V_{IL} - V_{OL}$ $NM_H = V_{OH} - V_{IH}$	2(d)

Table 4: Inverters

5. Delay and logical effort:

Title	Equation	Notation	Ref.
★ **Elmore delay**	$t_{pd} = \sum_i R_{i,\text{to source}}\, C_i$	each C times all R to the source	3(b)
NOR3 driving h NOR3	$t_{pdr} = (21 + 7h)RC$, $t_{pdf} = (9 + 7h)RC$	fully contacted diffusion	3(b)
Gate delay	$d = gh + p$	g: inverter 1, NAND2 4/3, NAND3 5/3, NOR2 5/3 p: inverter 1, n-input gate n	3(c)
Branching effort	$b = \dfrac{C_{on} + C_{off}}{C_{on}}$	$B = \prod b_i$	3(c)
★ **Path effort**	$F = GBH$	$G = \prod g_i$, $H = C_{out}/C_{in}$	3(c)
★ **Minimum delay**	$\hat{f} = F^{1/N}$, $D = N\hat{f} + P$	$P = \sum p_i$	8(d)
Sizing	$C_{in} = \dfrac{g\,C_{out}}{\hat{f}}$	from the load backwards	3(c)
★ **Best stage effort**	$p_{inv} + \rho(1 - \ln\rho) = 0$	$\rho = 3.59$ for $p_{inv} = 1$	8(e)

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Title	Equation	Notation	Ref.
Best number of stages	$\hat{N} = \log_\rho F$	about $\log_4 F$	3(d)

Table 5: Delay and logical effort

6. Power:

Title	Equation	Notation	Ref.
★ **Switching power**	$P = \alpha C V_{DD}^2 f$	α = activity factor f = clock frequency (Hz)	4(b)
Static power	$P_{static} = (I_{sub} + I_{gate} + I_{junct} + I_{contention})V_{DD}$	half the width counts for I_{sub}, half for I_{gate}	4(d)
★ **Activity factor**	$\alpha = P(1-P)$	P = probability the node is 1	4(c)
Output probabilities	NAND2: $1 - P_A P_B$; NOR2: $\overline{P_A}\,\overline{P_B}$	independent inputs	4(c)

Table 6: Power

7. Sequential, datapath and memory:

Title	Equation	Notation	Ref.
★ **Max-delay**	$T_c \geq t_{pcq} + t_{pd} + t_{setup}$	fixed by a slower clock	5(c)
★ **Min-delay**	$t_{cd} \geq t_{hold} - t_{ccq}$	not fixed by the clock	5(c)
Synchronous reset	$Q_{next} = D \cdot \overline{reset}$	at the rising edge	5(d)
★ **Full adder**	$S = A \oplus B \oplus C_{in}$, $C_{out} = AB + C_{in}(A \oplus B)$		6(b)
Mirror adder sum	$S = ABC_{in} + (A + B + C_{in})\overline{C_{out}}$	28 transistors	6(b)
★ **DRAM read swing**	$\Delta V = \pm\frac{V_{DD}}{2}\frac{C_{cell}}{C_{cell} + C_{bit}}$	$C_{cell} \ll C_{bit}$	6(d)

Table 7: Sequential, datapath and memory

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8. Key formulas:

The formulas marked ★ in the sheet, on one page. Each title links back to its row.

Title	Equation
Adding O_2	$\mathrm{O} + \mathrm{CF_3} \rightarrow \mathrm{COF_2} + \mathrm{F}$
Adding H_2	$\mathrm{H} + \mathrm{F} \rightarrow \mathrm{HF}$
Oxide growth law	$x_O^2 + Ax_O = B(t + \tau)$
Threshold, no body effect	$V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$
Threshold with body effect	$V_T = V_{T0} + \gamma\left(\sqrt{\lvert 2\phi_F\rvert + V_{SB}} - \sqrt{\lvert 2\phi_F\rvert}\right)$
Constant-field scaling	$L, W, t_{ox}, V : 1/S;\ N_A : S$
Switching threshold	$V_M = \frac{V_{tn} + r(V_{DD} - \lvert V_{tp}\rvert)}{1 + r}$
Elmore delay	$t_{pd} = \sum_i R_{i,\text{to source}}\, C_i$
Path effort	$F = GBH$
Minimum delay	$\hat{f} = F^{1/N},\ D = N\hat{f} + P$
Best stage effort	$p_{inv} + \rho(1 - \ln\rho) = 0$
Switching power	$P = \alpha C V_{DD}^2 f$
Activity factor	$\alpha = P(1 - P)$
Max-delay	$T_c \geq t_{pcq} + t_{pd} + t_{setup}$
Min-delay	$t_{cd} \geq t_{hold} - t_{ccq}$
Full adder	$S = A \oplus B \oplus C_{in},\ C_{out} = AB + C_{in}(A \oplus B)$
DRAM read swing	$\Delta V = \pm\frac{V_{DD}}{2}\frac{C_{cell}}{C_{cell} + C_{bit}}$

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9. Symbol index:

Every symbol the sheet defines, in the order it first appears. The last column links to that row.

Symbol	Meaning	First used in
F	fluorine radical, the main etchant	Plasma dissociation
SiF_4	volatile product	Silicon etch
C^*	oxidant solubility	Deal-Grove fluxes
D	diffusivity in SiO_2	Deal-Grove fluxes
k_S	interface rate constant	Deal-Grove fluxes
λ	wavelength (m)	Lithography resolution
NA	numerical aperture	Lithography resolution
ϕ_{GC}	work-function difference	Threshold, no body effect
Q_{ox}	oxide charge	Threshold, no body effect
A, B	technology constants	Gate leakage
V_{TL}	load threshold, with body effect	Saturated load, high level
α	activity factor	Switching power
f	clock frequency (Hz)	Switching power
P	probability the node is 1	Activity factor

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